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Draft NSP/004/046 – Specification for 33-132kV Single Circuit Wood Pole Lines with span lengths upto 150m

1. Purpose

The purpose of this document is to provide a specification detailing the design and constructional requirements for single circuit EHV overhead lines with the capability of operating at 132kV and with maximum span lengths of up to 150m.

Additionally this specification confirms the reference OHL 9/09 as the reference to be quoted to DEFRE for all new lines.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NSP/004/046	1.0	Specification for 33-132kV Single Circuit Wood Pole Lines with span lengths upto 150m
	1.0	OHL9-01 – Draft Specification for 132kV Overhead Lines on Wood Poles
CE/C/37(M1)	2.0	
O.311	Jan 1986	OHL4 Specification for the construction of 66kV Lines on Wood Poles

2. Scope

This specification covers the design and constructional requirements for 3-phase overhead lines designed to operate at voltages up to and including 132 kV. It has been designed to accommodate conductor sizes upto 200mm AAAC (Poplar) on single pole supports. The design is suitable for the modification, renovation or extension of existing 33-66kV wood pole lines and mast lines.

This design includes information to allow a wood pole line to be designed for eventual 132kv operating voltage, whilst being currently operated at 33kV or 66kV.

All angle and terminal structures have been selected to duplicate those used in and existing CE Electric UK specification NSP/004/045 "Specification for EHV Wood Pole Lines operating up to 132kV with span lengths between 140m and 220m".

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Where spanning is consistently in excess of 150m, the lines shall be designed to a companion specification NSP/004/045 "OHL 10 - Specification for EHV Single Circuit Wood Pole Lines operating up to 132kV with span lengths between 140m and 220m.

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3. Design Criteria

The structures within this specification have been prepared using a deterministic approach on a 130m basic span, such that with 200mm² AAAC "POPLAR", the maximum design tension shall not exceed 17.79 kN (1814.1 kgf) at -5.6°C coincident with wind pressures of 380(N/m²) and diametric ice thickness of 19mm. Additionally to reduce the risk of vibration, the design tension of the conductors at 5°C in still air shall not exceed 20% of their calculated breaking load.

This specification currently incorporates detail from ENATS 43-50 and a previous Northern Electric Specification known as NE/OHL 4/86.

This line design is based on the use of vertical and horizontal post type insulators of polymeric material on the intermediate positions and polymeric tension insulators for tension positions.

Data has also been provided to allow this specification to be used with alternative conductors or higher design tensions to match in with historical design standards

Conductor	Design Loadings	Used on historical specifications
175mm ² AAAC "Elm"	17.79 kN (1814.1 kgf)	OHL4 120m Basic
175mm ² ACSR (Lynx)	17.79 kN (1814.1 kgf)	OHL4 120m Basic
175mm ² ACSR (Lynx)	26.34 kN (2685.9 kgf)	OHL4 150m Basic
175mm ² m ACSR (Lynx)	25.740 kN (2624.7 kgf).	CE/C/37 150m Basic
200mm ² AAAC "Poplar"	23.517 kN (2397.76 kgf).	CE/C/37 (M1) 150m Basic

Whilst this design has been prepared on a 130m basic, extraordinary sections may be constructed using either 175 mm² ACSR conductor or 200 mm² AAAC with maximum working tensions up to 36 kN (3670 kgf).

Details of suitable support structures and appropriate sag charts will be provided by contacting the companies Overhead Line Standards Manager.

Where lines are erected with spans in excess of 130m or lines are erected using design tensions in excess of 17.79kN, the conductors shall be protected from vibration by the use of vibration dampers installed in accordance with drawing no. 1091010188.

3.1 Span Criteria

The structures detailed within this specification have been designed for the following span criteria:

Span Criteria	
Standard or Basic Span	130 m
Maximum Span Length	150 m *
Maximum Clashing Span	150 m
Extraordinary Spans	170m

Designers shall aim to construct new lines based on a typical span length of 130m +/-10m with the odd long span up to 150m (erected using a 130m sag chart as detailed in Appendix A1). Where due to

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wayleave or terrain issues the line requires a single or multiple span section of line to be in excess of these values, this is permissible upto a maximum span of 170 provided that the line is treated as an extraordinary line section and that the details are discussed with the overhead line standards manager.

To improve the clashing performance and reduce the sag on extraordinary line sections, the conductors shall be erected to higher design tensions (erected to 150m sag chart as detailed in Appendix A4)

3.2 Security Loads

A security loading condition (broken wire condition) is not normally considered for suspension or light angle timber pole structures, however since the previous NEDL specification NE/OHL 4/85 aimed to provide a factor of safety of at least 1.5 when supporting the static loads of single broken wire conditions.

To provide additional security, failure containment measures shall be designed into the line to limit damage in the event of cascade failure shall be incorporated. These should include, but not be limited to, the provision of longitudinal stays on selected angle/section structures which shall be designed to arrest cascades. Crossarms/insulators would then fail first on longitudinally stayed structures.

No section of line shall exceed 2000m or 10 spans without a section or angle structure which meets the requirements of failure containment as detailed above, being inserted into the line.

3.3 Electrical Design

Unless specified otherwise, the overhead line shall be designed to operate at 132 kV with a nominal system frequency of 50 Hz.

3.3.1 Current Ratings

Unless specified otherwise, the line shall be designed to operate with 200 mm² AAAC Poplar conductor with a maximum operating temperature of 75°C

This will provide the following thermal ratings (3% excursion values):

132 kV @ 75 deg C							
Conductor		Summer		Spring/Autumn		Winter	
Poplar		Amps	MVA	Amps	MVA	Amps	MVA
200	AAAC	624	143	678	155	708	162

Further information on alternative line ratings can be found by reference to IMP/001/011" Code of practice for Overhead Line ratings"

3.4 Electrical clearances

3.4.1 Internal Electrical Clearances

Wire clearances at the support shall be in accordance with ENA 43-125, that is the minimum clearance from live metal to structure steelwork and earthed fittings when tension insulators are used shall not be less than:

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1.2 m in still air

0.9 m for jumper loops with a maximum swing angle of 20°

To facilitate increased phase to earth clearances at section structures, all pilot insulators shall incorporate the support stool assembly detailed on drawing 1091010487 sht34

Minimum downlead phase-to-phase clearances shall be 2.5 m in still air.

3.4.2 External Electrical Clearances

External overhead line clearances shall fully meet the requirements of CE Electric UK code of practice document NSP/004/011, titled "Overhead Line Clearances", being designed for operation at 132 kV at the maximum operating temperature of 75°C.

3.5 Components

3.5.1 Conductors

Conductors shall be manufactured and supplied in accordance with CE Electric UK material specification NPS/001/007. All conductors shall be greased in accordance with Engineering Recommendation L38/1 and the weight of greased conductors shall be used in the design. Aluminium Alloy conductors shall be provided in an inert condition as detailed in BS EN 50182, clause 5.5.7 and have a maximum resistivity of 32.53 nΩm at 20°C.

Unless specified otherwise, conductors shall be 200 mm² (AAAC), (37/2.87mm) code-named Poplar. The minimum factor of safety shall be 2.0 on the conductor nominal breaking loads. The conductor Maximum Working Tension (MWT) shall be calculated at a minimum conductor temperature of -5.6°C.

3.5.2 Conductors Spacing to Avoid Clash

Phase separation shall comply with the requirements of EN 50423-3-9: 2005, Clause 5.4.3.1.

For the structures in this design, the assumed smoothed gust and lull wind pressures of 1.832 times the mean and 0.546 times the mean wind pressures respectively for a gust factor of 1.5 have been considered for a "2C" weather zone. That is a wind pressure of 380 N/m² and diametric ice thicknesses from 0 mm - 30 mm in increments of 10 mm. The worst combination of wind and ice loading is used for the calculation, with the actual wind and ice loads employed enhanced by a withstand factor of 1.1.

Maps of the UK showing the appropriate weather zones at varying altitudes from sea level up to 500 m can be found in EN50423-3-9:2005 and ENA TS 43-40

The clashing span in this design has been artificially limited to 150m, thus providing a reasonable match to the support capability of a single Extra Stout Pole. For span requirements in excess of this figure will be provided by contacting the companies Overhead Line Standards Manager.

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3.5.3 Conductor Terminations, Clamps and Joints

Conductor terminations and joints shall be designed for 200 mm² AAAC 'Poplar' conductor supplied in accordance with CE Electric UK material specification NPS/001/005 and the requirements of IEC 61284.

3.5.4 Conductor Terminations

Conductor terminations shall be full tension compression dead-end type supplied complete with jumper terminal flags. See drawing 1091010653 sht3 for details.

Jumper palms for connecting jumper loops to dead ends, shall be of the straight compression type. All jumper palms shall be supplied complete with the required bolt assemblies.

Bolt assemblies shall consist of bolts and nuts to grade 8.8/8.0 in accordance with BS 4190 complete with load spreading washers. Load spreading washers shall be 44mm diameter and 9.5mm thick for M16 bolts. To prevent loosening in service the minimum installation torque shall be 90Nm.

All conductor terminations shall comply with the performance and test requirements of IEC 61284.

3.5.5 Conductor - line post trunnion clamps

Conductors shall be secured at the line post insulator positions using pivot type trunnion clamps which are supplied with the insulators and have been designed to accommodate a conductor diameter of 20.1mm. The trunnion (pivot) shall be on the centre-line of the conductor.

Where alternative conductor types are used, the project engineer shall contact the standards sections for guidance on alternative clamp types.

Conductors shall be secured at intermediate pole positions by direct clamping onto the conductors, however all pilot insulator positions have been designed to accommodate armour grip suspension rods to improve the stability of long jumpers.

Prior to any conductor being secured into the trunnion clamps, it shall be wrapped with a layer of aluminium chafing tape (1.27mm x 7.62mm) to BS 1470 EN 573 Alloy ENAW-1050A Class 'A' annealed to 'O' in accordance with BSEN 515 and mechanical properties to BSEN 485-2 Table 4. The tape shall be applied as a continuous single layer such that it extends beyond the ends of the clamp by at least 3 complete turns.

3.5.6 Conductor - Tension Joints

The fullest possible use shall be made of the maximum conductor lengths in order to reduce to a minimum the number of mid-span tension joints. The number, span and location of any mid-span tension joint used on the line shall be approved by the supervising engineer managing the project. No more than one such joint per conductor in any one span shall be allowed.

Mid span Joints shall not be used

- a) Within 3 m of the line post insulator attachment or dead-end tension joint.

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- b) In spans over railways, navigable rivers, motorways, buildings or spans covered by special wayleave conditions.

Mid-span tension joints and compression dead ends shall be in accordance with the performance and test requirements of NPS/001/016 and IEC 61284. The tensile test shall be carried out to the requirements of Clause 11.5.1, alternative a.

3.5.7 Conductor - Non-tension Joints

Non-tension joints shall be in accordance with the performance and test requirements of partial tension fittings to IEC 61284 Conductor Terminations and Joints and NPS/001/016. The maximum resistance of all joints shall not exceed 75% of the resistance of an equivalent length of conductor. A record of joint resistance measurements shall be kept and stored with the line records.

3.6 Steelwork

3.6.1 Steelwork Assemblies

Steelwork assemblies on Intermediate structures utilise Mild steel and have been designed using the unit loadings detailed in the conductor data. The loads are compared to the ultimate tensile strength of steel divided by a factor of safety of 2.5 whilst incorporating the requirements of BS 449: Part 2.

Steelwork assemblies on the section and terminal structures utilise Mild steel and have been designed in accordance with BS5950 Part 1 except that a partial material factor γ_m of 0.64 shall be applied to the yield strength of grade S275 steel to EN 10025. For other grades of steel a partial factor equal to the ratio of yield strength to ultimate tensile strength shall be used.

All steelwork assemblies shall be supplied in accordance with CE Electric UK technical specification NPS/001/005.

Following the completion of all fabrication processes on the support steelwork, including nuts, bolts and washers, they shall be hot dip galvanised and tested in accordance with the requirements of BS EN ISO 1461.

Steelwork assemblies for intermediate positions are designed to support a conductor downpull on each side of the support, provided that the downpull is acting in a direction such that a tangent of the angle to the horizontal plane is not greater than 0.1.

Steelwork assemblies for section positions are designed to support a conductor downpull on each side of the support, provided that the downpull is acting in a direction such that a tangent of the angle to the horizontal plane is not greater than 0.2

3.6.2 Fasteners and Washers

All bolts shall be grade 4.6 in accordance with BS EN 20898

All nuts, bolts, fasteners and washers shall be supplied in accordance with CE Electric UK material specification NPS/001/010 and ENA TS 43-96.

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The length of a bolt or tie rod must be such that after installation there is a minimum of two full threads protruding from the nut. The excess length shall not exceed 25 mm. Where nuts or bolt heads come into contact with wood, or where slotted holes are involved, the use of washers is required and normally a washer will be used under both the bolt head and under the nut to adequately protect the galvanised skin of steelwork.

3.6.3 Galvanising

Galvanizing of all ferrous material except messenger cable wires used in Stockbridge type vibration dampers shall be in accordance with ISO 1461 and shall be applied by the hot dip process to provide a mass of zinc coating of not less than 610 gm of zinc per square metre of surface on steel bars, plates, sections and fittings. Threaded work shall have a coating mass of 305 gm of zinc per square metre.

3.7 Supports

Supports have been designed to resist the specified loads due to the climatic conditions, accidental-security conditions and construction and maintenance conditions taking into consideration the specified partial load factors and partial material strength factors. All supports have been designed to have a minimum factor of safety of 2.5.

The wind loading spans on intermediate supports are based on pole strengths in accordance with BS 1990 and the limiting case between the pole strength and the foundation capability.

Please note this design is still limited by the clashing performance which is currently selected as 150m. Increased clashing performance may be available to match the support capability by contacting the Standards section.

The stresses created in stayed supports are crippling stresses due to stay tension and vertical conductor loads acting at the pole top. The distribution of stresses in the limbs of 'H' supports on which all stays are adjusted to equal tension is considered to be as follows:

Stayed equally on each limb – 50% of stress in each limb.

The bending stresses induced in unstayed supports result from

Centre phase loadings applied 1755 mm above pole top

Outer phase loadings applied 802 mm below pole top.

This results in a combined pole loading point of 50mm below the pole top.

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3.7.1 Timber

The strength of timber supports shall be derived from formulae contained in BS 1990 part 1 and in respect of *Pinus Sylvestris* wooden products, the pole producers in the UK have confirmed the following parameters:

Mean ultimate extreme fibre stress (bending stress)	53.3 N/mm ²
Mean modulus of elasticity	10054 N/mm ²

These parameters are based on a *Pinus Sylvestris* population whose southern border occurs at 60° latitude. Where some other species and / or pole population is considered, other strength parameters may be applicable and in this respect, amended support capabilities may be used.

Poles shall be fabricated and supplied in accordance with CE Electric UK material specification NPS/001/001 and the fabrication drawings detailed within this specification.

3.7.2 Preferred Support Lengths

Preferred pole lengths are given for guidance to maximise available structural strength for differing loading conditions. Alternative pole lengths may be used to suit site conditions.

Unstayed Section or Intermediate	- Preferred Length 15m – E/Stout
Single Pole Stayed Section or Stayed Section Angle	- Preferred Length 14m – E/Stout
H Terminal or H Section Angle	- Preferred Length 14m – E/Stout

3.7.3 Supports Types

Drawing Number	Structure Type
1.09.138.0006 sht1	Single intermediate pole.
1.09.138.0006 sht2	Renovation of 'A' Pole Intermediate Structure.
1.09.139.0013 sht1	Single section/angle section pole. Dev 0-20°
1.09.139.0014 sht1	Single section/angle section pole. Dev 20°-40°
1.09.139.0018 sht1	'H' section angle support (2.9m centres). Max Angle Dev 60°
1.09.139.0019 sht2	Terminal/Out of Balance Support 2.93m centres.

3.7.4 Refurbishment of existing lines containing 'A' pole supports

Arrangement drawing 1.09.138.0006 sht 2 has been provided to allow existing 66kV lines containing 'A' pole supports to be refurbished for continued use with this specification. These supports are capable of supporting all the conductor types listed in this specification with windspans in excess of 168m. Where supports are re-fabricated to this assembly drawing, all unused holes must be plugged with wood plugs.

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3.7.5 Support Windspan Capability

This table has been created to select the limiting factor between the pole strength and the foundation capability. The Table has been formulated based on pole foundations with a 2500mm top block and a 1300mm lower block. The figures in brackets in the E/Stout column indicate the actual pole capability however this has been artificially limited to 150m to match the clashing design capability.

Other span/clash options may be available with different foundation arrangements or pole top special arrangements, contact the standards section for other options

* Preferred Intermediate pole height

Windspan Capability – Single poles

Pole Height (m)	Sinking Depth (m) Inter Poles	Sinking Depth (m) Single Section Poles	Sinking Depth (m) H Section Poles	Intermediate Supports and Single Pole Section Supports Available Windspan E/ Stout Grade Poles (m)		Intermediate Supports Available Windspan Stout Grade Poles (m)	H Section Supports Available Windspan E/ Stout Grade Poles (m)
				Ground Type "Good & Average 628 -471 kN/m ² /m	Ground Type Poor 314 kN/m ² /m	All Ground Types 628 - 314 N/m ² /m	All Ground Types 628 -314 kN/m ² /m
14.0	2.4	2.4	2.2	150 (159)	150 (159)	124	200+
15.0 *	2.4	2.4	2.2	150 (165)	150 (162)	128	200+
16.0	2.4	2.4	2.2	150 (172)	150 (150)	133	200+
17.0	2.4	2.4	2.4	150 (173)	139	132	200+
18.0	2.4	2.4	2.4	150 (180)	130	135	200+

Note :- All 'H' supports can provide spanning in excess of 200m irrespective of ground conditions

3.7.6 Guyed Structures

Where stays are used, the ability of the poles to resist the crippling loads due to the vertical component of the forces in the stays shall be calculated in accordance with BS 1990: Part 1: 1984.

Stay spreads shall normally be such that an angle of 45° is provided between stay and pole. In circumstances when tighter stay spreads are unavoidable due to physical obstructions, reduced angles may be used, subject to the approval of the company's Overhead Line Standards Manager. However the minimum allowable stay angle shall be 30°.

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At single pole angle supports where a single stay is used the stay shall be set so as to bisect the angle of deviation to counteract the transverse component of the conductor tension. When two or more stays are fitted, they shall be arranged in a splayed formation symmetrical about the angle of deviation bisector.

Triple stays shall comprise two splayed stays as above, set at the approved angle to the pole, the third stay being placed to bisect the two and entering the ground at least 2 metres behind them.

Where structures with multiple stays are required to provide resistance for failure containment purposes, the location of stays shall be chosen to provide adequate longitudinal resistance. The ideal location for the outer stays in such cases is in line with the conductors in the opposite span, but if this does not provide adequate transverse resistance, a compromise position may need to be selected.

Multiple stay baulks shall be installed so that the length of undisturbed ground between stay pits or anchors is not less than 3.0m.

At section positions where change of conductor type or sag chart give rise to the need for out of balance stays, single or double splayed stays per leg will normally be fitted to provide an angle of 45° between pole and stay. Normally, out of balance stays will be set under the line in each direction sufficient in number to be capable of terminating the line in each direction, unless by examination of the appropriate sag charts it can be shown that out of balance tensions do not reverse with change of temperatures, in which case stays may only be fitted against the higher tension conductors at the discretion of the Standards Manager.

All stay terminations attached to metal fittings shall be fitted with galvanised thimbles.

All stays supporting HV lines shall be effectively bonded at the top of the line crossarm or hamper. The method of bonding stays is to take one wire from the stay strand, or centre 'king' wire from a preformed pole top make off, and connect it to the designated bolt on the crossarm. Stays shall be considered to be effectively bonded where they are either terminated direct onto a stay plate which itself is bolted direct to pole top steelwork, or where one wire is taken from the pole top make off and connected to a designated bolt on the crossarm as stipulated in ENATS 43-91.

3.7.7 Loading limitations on Guyed Single Structures

To maximise the allowable strut load capability on single Extra Stout supports, this specification has amended the minimum acceptable pole diameters as detailed in BS 1990: Part 1 :1984. The tabulated "min" value has been increased to reflect a calculated mean value. The new effective min values are shown in the pole fabrication drawing 1091390003.

The following stay chart has been created to show the optimised stay arrangements for lines constructed using Poplar conductor which have been designed for operation at 17.79kN. The table details the minimum number of stays and minimum permissible stay angle (stay to pole) for varying angles of line deviation.

Additionally to ensure that no loading situation exceeds the maximum allowable design values, an artificial maximum windspan limit has been set at 150m.

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Variations in allowable pole diameters outside of these limits, stay spreads other than those detailed or maximum windspans in excess of these tabulated values may only be allowed with the approval of the CE Electric UK Overhead Standards Manager.

Optimised Stay Chart specific to Poplar Conductor operating @17.79kN

Structure Code	Pole Type	Length	No Stays and min angle to pole	Structure Drawing Ref
0 ⁰ - 10 ⁰	E/Stout	14m -17m	1 stays @ min 30 ⁰	1091390013 sht1
10 ⁰ - 20 ⁰	E/Stout	14m -17m	2 stays @ min 40 ⁰	1091390013 sht1
10 ⁰ - 20 ⁰	E/Stout	14m -17m	3 stays @ min 30 ⁰	1091390013 sht1
20 ⁰ - 25 ⁰	E/Stout	14m -17m	2 stays @ min 35 ⁰	1091390014 sht1
25 ⁰ - 30 ⁰	E/Stout	14m -17m	2 stays @ min 40 ⁰	
25 ⁰ - 30 ⁰	E/Stout	14m -17m	3 stays @ min 38 ⁰	
30 ⁰ - 40 ⁰	E/Stout	14m -16m	3 stays @ min 40 ⁰	
30 ⁰ - 40 ⁰	E/Stout	17m	3 stays @ min 45 ⁰	
40 ⁰ - 60 ⁰	E/Stout H Pole	14m -17m	4 stays @ min 30 ⁰	1091390018 sht1
Terminal	E/Stout H Pole	14m -17m	4 stays @ min 30 ⁰	1091390019 sht1

Generic Pole / Stay Selection chart suitable for the whole range of conductors and design tensions considered within this design document

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Structure Code	Pole Type	Length	Stays	Structure Drawing ref
0° - 2°	E/Stout	15m	None (**)	1091380006 sht1
0° - 10°	E/Stout	14m - 17m	1 stay @ 45°	1091390013 sht1
10°-20°	E/Stout	14m -17m	2 stay @ 45°	1091390014 sht1
20°-30°	E/Stout	14m	2 stay @ 45°	
20°-30°	E/Stout*	17m	3 stay @ 45°	
30°-40°	E/Stout	14m - 17m	3 stay @ 45°	
Large Angle H Pole 40° - 60°	E/Stout	14m -17m	4 stay @ 45°	1091390018 sht1
Terminal	E/Stout H - Pole	13m	4 stay @ 45°	1091390019 sht1
Terminal	E/Stout H - Pole	17m	4 stay @ 45°	

3.7.8 Use of Section Supports

Use of Section Type Supports - Section supports shall normally be used at the following locations:

- At positions of line deviation
- At positions where conductor out of balance produced by the use of alternative sag charts or conductors applies.
- At each of the supports adjacent to a tee-off arrangement.
- At the penultimate pole adjacent to a cable termination.
- At power line crossing beneath steel tower lines of similar or higher voltage
- Where a construction section is additionally justifiable.

For extreme angles of deviation (i.e. above 60°) the line will be terminated in each direction such that the line conductors cross one above the other, the centre of the cross being approximately 6m from each terminal structure, the conductors separated by approximately three metres in the vertical plane and connected by vertical jumpers at the point of cross of each similar phase.

3.7.9 Support Foundations

The minimum factor of safety on foundations is 2.5.

Structure foundations shall be designed for three soil conditions with ultimate lateral rupture capacities of 628 kN/m², 471 kN/m² and 314 kN/m² per metre depth for good, good / average and average / poor soils respectively. Unless specific soil strength information is available, all calculations shall be based on 471 kN/m² or "average" soil on the premise that soil additives will be used.

All un-stayed structures shall be fitted with two wood foundation blocks. As a default the poles shall be fitted with a (2500 x 250 x 125mm) top block as detailed on 1091010670 sht 11 and a (1300 x 250 x 125mm) lower block as shown in ENA TS 43-91 fig3 type 2. The blocks shall be placed as follows:

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Ground line to top block centres = 500 mm

Ground line to lower block centres = 800 mm.

Where use of the long top block creates wayleave / installation difficulties eg installation of poles adjacent to hedge lines or walls or the potential deep ploughing in fields, the top block may be substituted with a standard (1300 x 250 x 125mm) block. However this amended foundation is only permissible where the ground type has been classified as "good/average"

All pole foundations shall incorporate the use of the backfill "Perma-Soil" backfill additive as detailed below.

25kg of Perma-Soil shall be added per cubic metre of excavated spoil. The mixture shall be replaced in the excavation and fully compacted in layers not exceeding 300mm thick. Initially the bottom baulks only shall be fitted to unstayed structures to facilitate this compaction. The top baulks shall only be fitted when the compacted fill reaches the underside of these baulks.

Research work has shown that where poor foundations are identified, the Perma-Soil stabiliser will normally provide sufficient improvements in the foundation capability to improve the soil quality to that of a Average(Medium) backfill.

Ground Classification for Foundations

Ground Classification in kN/m ² /m of depth	Soil Description
314	Poor: Soft clay, clay loam, poorly compacted sand clays containing a large amount of silt and vegetable matter, and made ground. Poor soils will normally be wet and have poor drainage.
471	Average(Medium): Compact fine sand, medium clay, compact, well drained sandy loam, loose coarse sand and gravel. Average soils should drain sufficiently well that water does not stand on the surface.
628	Good: Compact well graded sand and gravel, hard clay, well-graded fine and coarse sand, decomposed granite rock and soil. Good material should be well drained and in locations where water will not stand.

3.8 Stays

3.8.1 Staywire

All stays shall be adjustable, utilise 7/4.00 mm grade 1150 galvanised steel stay strand and shall be in accordance with ENA TS 43-91.

The tensile strength of all components of the stay assembly shall be not less than the stay wire which has a MFL = 101kN.

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3.8.2 Stay Insulators

All stays shall be fitted with an approved type of stay insulator(s) and shall be positioned such that under a broken jumper or a broken stay condition the insulator shall prevent the stay becoming 'live'. The insulator shall therefore be placed below any likely point of contact with live metal, but must remain a minimum of 3.0 m vertically above the ground.

All stay insulators (except approach) shall be provided with insulator protective devices with an arc gap of 1120mm and comply with the electrical and mechanical requirements of ENA TS 43-91 as given below:

Nominal system voltage	132kV
Minimum wet power frequency withstand voltage	190kV
Minimum dry lightning impulse withstand voltage	200kV

Within the approach zone (up to 1.6km from the substation) all stays shall be fitted with insulators that comply with drawing 439107 Type 2 as detailed in ENA TS 43-9.

3.8.3 Stay Assemblies

Stays assemblies shall be constructed and attached to the structures as shown on drawing 1091010429 sht 3 and shall be installed in accordance with CE Electric UK guidance note NSP/004/104.

3.8.4 Stay Foundations

Stay blocks shall be to ENA TS 43-91 drawing no. 439103 type 2 and installed at a depth of 1.8 m below ground level in accordance with CE Electric UK guidance note NSP/004/104.

Alternatively screw anchors of an approved design may be used.

3.9 Insulators

Unless specified otherwise, all insulators shall be manufactured from composite materials and be fully compliant with the CE Electric UK material specification NPS/001/006 and ENA TS 43-93: issue 4: 2004, line Insulators.

Insulator Types	Insulator Drawing No.	Structure type used
Vertical Post	1091010487 sht 18	Intermediate
Horizontal Post *	1091010487 sht 16	
Tension Insulator	1091010487 sht 30 or 1091010487 sht 31	Section / Terminal
Vertical post/pilot + Support Stool	1091010487 sht 18	Section
Stay insulators	1091010372 sht 2	All Unearthed stayed supports
Stay insulators	1000439107 sht 2	All Earthed stayed supports

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All lines shall be assumed to be in areas designated as “heavy pollution”. Creepage distances for 132kV insulators shall therefore be **3625mm** and **2538mm** for vertical and horizontal composite insulators respectively.

The insulation co-ordination requirements for the replacement lines shall be as follows:

Specific Requirements	Nominal System Voltages		
	33 kV	66 kV	132 kV
Wet power frequency withstand voltage (60s)	90	140	275
Dry lightning impulse withstand voltage (1.2/50µs wave)	200	325	650

3.9.1 Post Insulators

All post insulators shall be supplied in accordance with CE Electric UK technical specification NSP/001/006 and the requirements of ENA TS 43-93, in particular Section 6, Quality Assurance and Section 8, Mechanical Performance. Design, type, sample and routine tests shall be carried out as required. Tracking and Erosion Tests, RIV Requirements and Corona Extinction Voltage Tests shall be conducted as specified.

Unless agreed otherwise, the specified cantilever load (SCL) shall comply with Clause 8.2 of ENA TS 43-93. The SCL shall therefore be at least 21kN.

It should be noted that suppliers generally use **Routine Cantilever Load (RCL)** or **Maximum Design Cantilever Load (MDCL)** when specifying the mechanical strength of post insulators. This is defined as the maximum recommended working load and is usually specified by suppliers as being 50% of the ultimate cantilever strength rating of the insulator defined as the **Specified Cantilever Load (SCL)** or the **Specified Mechanical Load (SML)**

If the SCL is not specified by the supplier, then to comply with the specification, the RCL or MDCL of the insulator shall not be less than 50% of specified cantilever load

3.9.2 Tension Insulators

Unless otherwise specified composite tension insulators shall be ball and socket and have a mechanical rating of 120 kN with a standard coupling size of 20 mm. End fittings shall be in accordance with BS EN 61466-1. Sockets shall have “W” type phosphor bronze security clips in accordance with IEC 60372.

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3.9.3 Arc horns

Arc horns shall only be fitted to intermediate type structures fitted with post insulators in the approach zone (up to 1.6km from the substation)

Arc gap requirements on tension sets shall be as tabulated below:

Arc gaps	Nominal system Voltage		
	33kV	66kV	132kV
Normal (beyond initial 1.6km from substation on unearthed wood poles)	1120mm	1120mm	1120mm
Approach – (Wood poles assumed to be earthed structures within 1.6km of substation)	453mm*	453mm	995mm

*normal 268mm arc gap increased to 453mm to reduce bird strike risks,

Where replacement lines are installed with 132kV insulator arrangements but are required to be initially operated at a reduced nominal system voltage, adjustable arcing horns shall be provided on tension sets such that the arc gaps for the appropriate system voltage can be provided whilst providing a simple upgrade option.

3.9.4 Insulator Fittings

All ball, socket, tongue, clevis and eye couplings shall be supplied in accordance with the CE Electric material specification NPS/001/005, have a mechanical rating of 120 kN and be in accordance with the requirements of BS 3288. All ball and socket couplings shall also comply with the requirements of IEC 60372.

3.10 Lightning Protection and Earthing

With the exception of structures located in the substation approach zone, all structures have been designed to be of unearthed construction.

To avoid a capacitive induced potential on unearthed supports, all insulator support steelwork shall be bonded together using flexible insulated earthwire bonds. Similarly, all stays shall be bonded at the top of the pole steelwork.

3.10.1 Earthing Structures – Substation Approach Zone

Over the last 1.6km of route into any substation or where a cable section is inserted into the route, the last 5 supports in both directions shall be converted to earthed structures. Pole crossarms shall be earthed by means of a 70mm PVC (Green) insulated copper cable down the pole, fixed to the pole at intervals of not more than 600 mm and connected to earth rods. Rods shall be driven into the ground as necessary to obtain a maximum earth resistance of 10 ohms. Earth rods and their connectors shall comply with the requirements of ENA TS 43-94 and CE Electric UK material specification NPS/002/001. Where earthed structures are installed, all insulators shall be fitted with arcing horns

3.11 Anti Climbing Devices

Anti-climbing devices (ACD) shall be designed in accordance with material specification NPS/001/005 and be installed in accordance with NSP/004/109.

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They shall consist of pre-wrapped barbed wire Anti-climbing guards (Enhanced Design) and shall as a default be installed on all poles..

3.12 Safety Signs, Labels and Notices

All supports shall be fitted with at least one Safety Sign. The sign shall be mounted approximately 3 m above the ground level, above any anti-climbing guard and clearly visible to an observer on the ground. Safety Signs, labels & notices shall be designed in accordance with material specification NPS/001/010 and shall be installed in accordance with CE Electric UK guidance note NSP/004/109.

3.13 Conductor Sag Charts and Design DataErection Guidance

See Appendix A contains details of applicable conductor data and sag charts allowed for use with this specification.

3.13.1 Conductor Creep Compensation

Suspended conductors are subject to longitudinal stresses that cause permanent long-term elongation or conductor creep; this results in an increase in sag in the conductors. In order to ensure satisfactory ground clearance exists throughout the life of the line, a combination of compensation techniques have been employed within this Specification.

(i) Over-Tensioning

The reduction of sag due to the increased tension compensates for the increase in sag due to conductor creep during the life of the line. The over tension can be applied either as a temperature shift or as a percentage increase over the design tension. The following over-tension values should be used.

Conductor Type	Spans less than 100m	Spans greater than 100m
AAAC	10% increase in tension	-20°C temperature shift
ACSR	10% increase in tension	-20°C temperature shift

(ii) Ordinate Shift

Longitudinal profiles for overhead lines commonly use the "Optimal, PoleCad Software" to design overhead lines and to ensure that all statutory clearance requirements have been met. The calculations used in the production of ground clearance curves apply a technique of modifying the clearance ordinates by reducing the conductor tension by a specific percentage. This process builds in a small safety factor into the clearances. All percentage reduction values are based on an application temperature of 15°C. Lines to this specification shall utilise a 10 % reduction

3.14 Survey and profiling

Survey and profiling shall be carried out in accordance with the company's guidance document NSP/004/024. All proposed routes shall be surveyed to ensure conformance with company's design

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and clearance criteria.

The survey data together with the proposed pole type and position shall be presented back to the Company in an electronic design file format for inspection and approval.

The company uses the Optimal Software PoleCad package and will provide any necessary Cell or Conductor Libraries for use with the system.

In addition way-leaves shall be obtained and forwarded together with plans to a 1 : 10,000 scale to enable Form B, Section 37 DTI consents to be applied for. Upon completion of a satisfactory design and receipt of way-leaves and consents, line schedules shall be produced for construction and line record purposes. All CAD designs shall be given a unique file reference, related to the line feeder route number and archived.

When the line is ready for connection to the Companies distribution system, all network diagrams and asset databases shall be updated.

3.14.1 Conductor Design Checks

- a) Proposed lines shall be checked for possible conductor uplift conditions through the use of an unloaded, -6.0°C conductor catenary curve. The check shall ensure that 0.5m exists between the top of the insulator and the cold curve position.
- b) Where uplift conditions are designed out of the line, this process shall not result in excessive pole heights (ie under normal circumstances due to ground clearance requirements only it is envisaged that poles will not exceed 16m, poles in excess of this value shall be sectioned).
- c) Lines shall be checked for conformance with the clearance requirements of CE Electric UK guidance document NSP/004/011.

3.15 Climbing systems and work positioning attachment points

In accordance with the requirements of the Working at Height Regulations 2005, these structures have been designed to accommodate the installation of an assisted temporary climbing system. This system involves the permanent installation of fixing brackets at specific points up the structures during the initial construction process which allow the later installation of a lightweight temporary ladder system. (See Appendix E, item E3 for details). The ladder system provides a combined access and fall arrest system via its internal track system. The internal track system allows a travelling device (See Appendix E, Item E5 for Details) located in the track to follow the direction of the climber and in the event of a fall, the travelling device locks onto the track.

The ladder system is constructed from individual small ladder sections approx 2m in length (See Appendix E, item E4 and E4.1 for details) allowing the climber to provide an access system from a point above the structure ACD to a point above the pole top. A selection of ladders will be provided at a central point within CE Electric UK to provide access to structures when required.

Additionally all poles have been designed to incorporate fall arrest / work positioning attachment points. These are provided via the installation of M20 eyebolts at strategic points around the pole top working area.

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Although this climbing system has been designed into the structures, it is envisaged that in most cases pole top access is likely to be achieved through independent means ie mobile access platforms.

Ladders are available in 5 standard sizes. To accommodate the varying pole heights and arrangements the following matrix has been produced to indicate the combinations required in each situation.

Component parts

Type	Description	Cat Number	No Rungs	Weight in Kgs
A	Common base ladder section 2240mm in length	313080	8	9.2
B	Ladder section 1960mm in length	313081	7	8.3
C	Ladder section 1680mm in length	313082	6	7.4
D	Ladder section 1400mm in length	313083	5	6.5
E	Shaft Entry Top Section 1120mm in length	313084	1	4.5
-	Permanent Female Mounting Brackets	313086	-	-
-	Glidelock Traveller	313085	-	-

Ladder Selection table

Structure Type	(Rutter Pole)	Bkts Req	Single section poles & H section / terminal poles	Bkts Req
12m	(1 x A) + (1 x B) + (2 x E)	8	(1 x A) + (3 x D)	8
13m	(1 x A) + (1 x C) + (1 x D) + (2 x E)	10	(1 x A) + (1 x B) + (1 x C) + (1 x D)	8
14m	(1 x A) + (3 x D) + (2 x E)	12	(1 x A) + (1 x C) + (3 x D)	10
15m	(1 x A) + (3 x C) + (2 x E)	12	(1 x A) + (5 x D)	12
16m	(1 x A) + (2 x B) + (2 x D) + (2 x E)	14	(1 x A) + (3 x C) + (2 x D)	14
17m	(1 x A) + (5 x D) + (2 x E)	16	(1 x A) + (2 x C) + (4 x D)	16

Note : H pole require the number of brackets and ladders to be doubled (ladders fitted to each leg)

3.16 Specialist Pole top working Equipment

3.16.1 Pole Platforms

The pole top has been designed to allow installation of the standard 2m long pole top temporary access platform. Once in position the platform is capable of providing full access to the outer insulators.

3.16.2 Insulator Mounted Running blocks

Specialist insulator mounted running blocks have been designed to allow the conductors to be installed and manipulated into the insulator conductor clamps in a safe controlled manner without the need for any other additional equipment. (See Appendix E, Item E1 and E2 for details)

3.17 Inspection

All lines shall be inspected before commissioning.

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3.18 Renovation Strategy

The following table details the proposed design strategy to be utilised when existing lines are identified for replacement

Voltage	Historical YEDL Specs	Specification Summary	Replacement Harmonised Specs
33kV	CE/C/36	Typically constructed using 175mm ACSR conductor, with a basic span of 76.2m and based on a 23kN MWT. Poles were single Stout or E/Stout with Rutter poles at Section angle points.	OHL 5 (NSP/004/042) – Due to the higher conductor tensions all section/terminal structures must utilise 125kN insulator assemblies. A series of design tables are available to detail the equivalent renovation structures to utilise. Where cable terminal supports are inserted to facilitate U/G diversions, this will be accommodated on OHL5 type structures designed to drawing 1000434011 sht 4 or sht 6
33kV / 66kV	CE/C/37 (DSS/004/040)	Typically constructed using 175mm ACSR, 100/125mm HDBC or 200mm AAAC conductors to 150m basic spans. Poles are Stout grade H poles supporting a portal type suspension arrangement. Conductor tensions are based on a 23-25kN MWT.	Where lines are diverted/extended or replaced as long sections (determined by proportion of rotten poles) the line shall be replaced with OHL 9 construction (NSP/004/046). Where lines are refurbished or only require conductor replacement and the majority of the existing supports are suitable for continued use, any defective supports shall be replaced on a like for like basis. Where cable terminal supports are inserted to facilitate U/G diversions, this will be accommodated on structures designed to drawing 1091231173 sht4
33kV	OHL 2/69	Typically constructed using 150/175mm ACSR or 100/125mm HDBC conductors to 120m basic spans. Poles are single Stout Intermediate poles supporting pin insulators. Conductor tensions were based on a 17.79kN MWT.	OHL 5 (NSP/004/042) Where cable terminal supports are inserted to facilitate U/G diversions, this will be accommodated on single pole terminals to drawing 1000434010 sht 6 or 'H' pole structures to 1000434011 sht 4 or sht 6
33kV / 66kV	WoodHouse Mast	Typically constructed using 175mm ACSR or 100/125mm HDBC conductors to 200m basic spans. Supports are	All supports to be replaced - Where typical span lengths are up to 150m they shall be replaced with OHL 9 construction (NSP/004/046) supports

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		steel masts with stays on angles. Conductor tensions are based on a 23-25kN MWT.	(ie single Extra Stout Supports). Where typical span lengths are in excess of 150m upto 220m they shall be replaced with OHL 10 construction (NSP/004/046) – ie Extra Stout Rutter supports. Where cable terminal supports are inserted to facilitate U/G diversions, this will be accommodated on structures designed to drawing 1091231173 sht4. Where individual supports require replacement eg. Fault repair, then H poles incorporating a suitable earthwire arrangement shall be used
66kV	O.311 (OHL 4)	Typically constructed using 150/175mm ACSR, 100/175 AAAC or 100/125mm HDBC conductors to 120m basic spans. Occasional spans upto 170m. Poles are typically single Stout or E/stout Intermediate poles or Medium/Stout 'A' poles with 66kV porcelain post insulators. Conductor tensions are based on a 17.79kN MWT.	Where lines are diverted/extended or replaced as long sections (determined by proportion of rotten poles) then OHL9 (NSP/004/046) construction shall be utilised. The design utilises single e/stout grade structures with composite post insulators. It is designed for 130m basic spans with options up to 170m. Where lines are refurbished or only require conductor replacement and the majority of the existing supports are suitable for continued use, all steelwork shall be replaced with OHL9 steelwork and any defective supports replaced with OHL9 equivalents. Where cable terminal supports are inserted to facilitate U/G diversions, this will be accommodated on structures designed to drawing 1091231173 sht4
66kV	Riley & Neate Mast Lines	Typically constructed using 150/175mm ACSR, 175 AAAC or 100/125mm HDBC conductors to 150m basic spans. Occasional spans upto 170m. Supports are narrow based lattice type steel masts supporting suspension strings . Conductor tensions are based on a 23-25kN MWT	As for the renovation of O.311 (OHL 4)
66kV	M&M A Pole	Typically constructed using 150/175mm ACSR, 175 AAAC	As for the renovation of O.311 (OHL 4)

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	and Portal Construction.	or 100/125mm HDBC conductors to 150m basic spans. Occasional spans upto 170m. Supports are 'A' poles or 'H' portal suspension structures. Conductor tensions are based on 17.79kN MWT	
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4 References

4.1 External Documentation

Reference	Title
ENA TS 43-8	Overhead Line Clearances
ESQCR	The Electricity Safety, Quality and continuity Regulations 2002, SI 2665
BS EN 50341-1	Overhead electrical lines exceeding AC 45kV Part 1: General Requirements - Common Specifications
BS EN 50423-3-9	Overhead electrical lines exceeding 1 kV, Part 3: National Normative Aspects for Great Britain and Northern Ireland
BS 1990-1	Wood Poles for Overhead Power and Telecommunication Lines
ENA TS 43-88	Selection and treatment of wood poles and associated timber for overhead lines.
ENA TS 43-90	Anti-Climbing devices and Safety signs for HV lines up to and including 400kV.
ENA TS 43-91	Stay strands and stay fittings for overhead lines
ENA TS 43-93	Line insulators
ENA TS 43-94	Specification for Earth Rods and Connectors
ENA TS 43-95	Steelwork for overhead lines
ENA TS 43-96	Fasteners and washers for wood pole overhead lines
IEC 61109	Composite Insulators for a.c. overhead lines with a nominal voltage greater than 1000V – definitions, test methods and acceptance criteria

4.2 Internal Documentation

Reference	Title
NSP/004/011	Guidance on Overhead Line Clearances
NPS/001/001	Technical Specification for Wood Poles and Associated Products for Overhead Lines
NPS/001/002	Technical Specification for Helical Products
NPS/001/005	Technical Specification for Steelwork, Conductor/Insulator Fittings and Stay Fittings for Overhead Lines and General Construction Works
NPS/001/006	Technical Specification for Insulators for Overhead Lines Up to And Including 132kV
NPS/001/007	Technical Specification for Conductors for Overhead Lines
NPS/001/010	Technical Specification for Notice Plates and Signs for Use on NEDL and YEDL Distribution Systems
NPS/001/011	Technical Specification for Fasteners And Fixings For Wood Pole Overhead Lines and General Construction Works
NPS/001/013	Technical Specification for Stay wire for Overhead Lines
NPS/001/015	Technical Specification for Barbed Wire
NPS/001/016	Technical Specification for Compression Fittings and Mechanical Conductor Fittings for Overhead Lines
NSP/002/001	Technical Specification for Earthing Materials

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Reference	Title
NSP/004/011	Overhead Line Clearances
NSP/004/101	(OHI 1) Guidance on Pole and Stay Holes
NSP/004/102	(OHI 2) Guidance on Erecting Single or 'H' poles
NSP/004/103	(OHI 3) Guidance on the type and fabrication of overhead line supports
NSP/004/104	(OHI 4) Guidance on the types and installation requirements for stays
NSP/004/105	(OHI 5) Guidance on the selection, erection and sagging of O/H line conductors
NSP/004/106	(OHI 6) Guidance on the selection and application of conductor joints, termination & binders
NSP/004/107	(OHI 7) Guidance on the selection of conductor jumpers and non-tension joints
NSP/004/108	(OHI 8) Guidance on the installation of compression joints
NSP/004/109	(OHI 9) Guidance on anti-climbing devices, safety signs and labels required on O/H lines
NSP/004/127	(OHI 27) Guidance on the selection and application of insulators and insulator assemblies.

5 Definitions

Term	Definition
Aeolian Vibration	Conductor Oscillation caused by low winds (1 m/s to 7 m/s) blowing steadily across the conductor.
Average Span	The arithmetic average of a number of spans in a line or section of line
Basic Span	The span length adopted for sag/tension calculations
Clashing	The electrical contact between two dissimilar line conductors of an overhead line displaced from their normal position by environmental forces such that the conductors are likely to touch.
Conductor Creep	Permanent long term elongation of the conductor
Conductor Downpull	The vertical loading imposed by conductors corresponding to a gradient measured between adjacent pole tops
Conductor Uplift	The upward vertical loading imposed by the conductors resulting from a gradient between adjacent points of conductor support
Everyday Tension (EDT)	The design stress in an unloaded conductor at 5°C. Used in sag / tension calculations to limit harmful conductor vibrations
Extraordinary Section	A section of line of one or more spans operating within, but at a higher set of tensions than the normal line and requiring out of balance staying of the section supports at both ends of the section
Freezing Point Tension (FPT)	The design tension of an unloaded conductor at 0°C in still air.
Ice Co-ordinate	The intercept on the ice axis of the weather incidence load line (See ENA TR 111) whose value is given for differing height and UK locations in the associated weather maps
Intermediate Support	A support in a straight run of line on which the conductors are supported

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Term	Definition
	on post insulators.
Maximum Conductor Pressure (MCP)	The maximum transverse component of applied conductor load when subjected to a wind load.
Maximum Conductor Weight (MCW)	The maximum vertical component of applied conductor load, including the weight of accreted ice, if present.
Maximum Span	The maximum permitted length of any span
Maximum Working Tension (MWT)	The absolute maximum conductor tension assessed at -5.6°C with wind and ice loading.
Minimum Failing Load	The minimum applied mechanical load that will cause failure by breakage of a component.
Over-tensioning	Excess tension applied above normal tension at time of erection to compensate for conductor creep.
Pre- tensioning	The tension treatment applied to a conductor for a short duration before final erection tension is established to remove a proportion of conductor creep.
Sag	The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured at mid-span
Section Angle Support	A support at which a line deviates and the conductors are made off on either side of the crossarm on tension insulator sets
Section Support	A support in a straight line on which the conductors are made off on either side of the crossarm on tension insulator sets
Span	The horizontal distance between adjacent supports. Individual spans will normally be within 20% of the chosen basic span
Spatial Ratio	The ratio between the sag of the conductor and the midspan clearance between this and an adjacent conductor under a given set of loading conditions.
Wind Co-ordinate	The intercept on the wind axis of the weather incidence load line (See ENA TR 111) whose value is given for differing height and UK locations in the associated weather maps
Weather Zone	A geographical area in which the likely mean wind pressure and absolute maximum ice accretion thickness may be described by a numeral and letter respectively. The wind co-ordinate is described in 190N/m ² increments, whilst the ice co-ordinate is measured in 10mm diametric thickness increments.
Wind Loading Span	Half the sum of the spans adjacent to the support.

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6 Authority for Issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G.Hammel	Overhead Standards Manager	<<Signature>>	<<Date>>

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M.Nicholson	Head Of System Strategy	<<Signature>>	<<Date>>

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C.Holdsworth	Standards & Assessment Manager	<<Signature>>	<<Date>>

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	<<Signature>>	<<Date>>

6.5 Approval

Approval is given for the content of this document

		Sign	Date
Sean Caskey	Network Construction Manager	<<Signature>>	<<Date>>

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Neil Denley	Primary Engineering Projects Manager	<<Signature>>	<<Date>>

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Appendix A – Conductor Design Data

Appendix A1 - 200 mm² AAAC (Poplar) – Based on 17.29Kn (1814 kgf) – 130m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	POPLAR
Greased Conductor Weight (kg/m)	0.68992
Cross Sectional Area of Conductor (mm ²)	239.4
Conductor Overall Diameter (mm)	20.1
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm ²)	5700
Rated Breaking Strength of Conductor (kgf)	7200.2
Basic / Recommended Span (m)	130
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1814.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1440.04
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1814.4
Maximum Conductor Weight (MCW) (kg/m)			1.496
Maximum Conductor Pressure (MCP) (kg/m)			1.515
Freezing Point Tension (FPT) (kgf)	at	0°C	891.0

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	POPLAR
Basic / Recommended Span (m)	130
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		70	80	90	100	110	120	130	140	150
-5.6	973.4	0.43	0.57	0.72	0.89	1.07	1.28	1.50	1.74	1.99
0	891.0	0.47	0.62	0.78	0.97	1.17	1.39	1.64	1.90	2.18
5	826.9	0.51	0.67	0.84	1.04	1.26	1.50	1.76	2.04	2.35
10	770.8	0.55	0.72	0.91	1.12	1.35	1.61	1.89	2.19	2.52
15	721.9	0.59	0.76	0.97	1.19	1.45	1.72	2.02	2.34	2.69
20	679.1	0.62	0.81	1.03	1.27	1.54	1.83	2.15	2.49	2.86
25	641.6	0.66	0.86	1.09	1.34	1.63	1.94	2.27	2.63	3.02
30	608.5	0.69	0.91	1.15	1.42	1.71	2.04	2.40	2.78	3.19
35	579.3	0.73	0.95	1.21	1.49	1.80	2.14	2.52	2.92	3.35
40	553.3	0.76	1.00	1.26	1.56	1.89	2.24	2.63	3.05	3.51
45	530.1	0.80	1.04	1.32	1.63	1.97	2.34	2.75	3.19	3.66
50	509.3	0.83	1.08	1.37	1.69	2.05	2.44	2.86	3.32	3.81
55	490.4	0.86	1.13	1.42	1.76	2.13	2.53	2.97	3.45	3.96
60	473.3	0.89	1.17	1.48	1.82	2.20	2.62	3.08	3.57	4.10
65	457.7	0.92	1.21	1.53	1.88	2.28	2.71	3.18	3.69	4.24
70	443.5	0.95	1.24	1.58	1.94	2.35	2.80	3.29	3.81	4.38
75	430.4	0.98	1.28	1.62	2.00	2.42	2.89	3.39	3.93	4.51
80	418.3	1.01	1.32	1.67	2.06	2.49	2.97	3.48	4.04	4.64

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Appendix A2 - 175mm² ACSR (Lynx) – Based on 17.29Kn (1814 kgf) – 130m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	LYNX	
Greased Conductor Weight (kg/m)	0.8626	
Cross Sectional Area of Conductor (mm²)	226.2	
Conductor Overall Diameter (mm)	19.5	
Coefficient of Linear Expansion (/Degree C)	1.8E-05	
Modulus of Elasticity (kg/mm²)	8157.73	
Rated Breaking Strength of Conductor (kgf)	8154.7	
Basic / Recommended Span (m)	130	
Wind Pressure on Conductor (N/m²)	380	
Radial Ice Thickness (mm)	9.5	
Ice Density (kg/m³)	913	
Absolute Maximum Working Tension (MWT) Limit (kgf)	1814.4	
Temperature at MWT Limit (Degrees C)	-5.6	
Maximum "Everyday" Tension (EDT) Limit (kgf)	1630.9	
Temperature at EDT Limit (Degrees C)	5	
Maximum Conductor Tension (MCT) (kgf) at	-5.6°C	1814.4
Maximum Conductor Weight (MCW) (kg/m)		1.653
Maximum Conductor Pressure (MCP) (kg/m)		1.492
Freezing Point Tension (FPT) (kgf) at	0°C	891.2

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	LYNX
Basic / Recommended Span (m)	130
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		70	80	90	100	110	120	130	140	150
-5.6	943.6	0.56	0.73	0.93	1.14	1.38	1.65	1.93	2.24	2.57
0	891.2	0.59	0.77	0.98	1.21	1.46	1.74	2.04	2.37	2.72
5	849.5	0.62	0.81	1.03	1.27	1.54	1.83	2.15	2.49	2.86
10	811.9	0.65	0.85	1.08	1.33	1.61	1.91	2.24	2.60	2.99
15	778.1	0.68	0.89	1.12	1.39	1.68	2.00	2.34	2.72	3.12
20	747.4	0.71	0.92	1.17	1.44	1.75	2.08	2.44	2.83	3.25
25	719.6	0.73	0.96	1.21	1.50	1.81	2.16	2.53	2.94	3.37
30	694.3	0.76	0.99	1.26	1.55	1.88	2.24	2.62	3.04	3.49
35	671.1	0.79	1.03	1.30	1.61	1.94	2.31	2.72	3.15	3.62
40	649.8	0.81	1.06	1.34	1.66	2.01	2.39	2.80	3.25	3.73
45	630.2	0.84	1.10	1.39	1.71	2.07	2.46	2.89	3.35	3.85
50	612.1	0.86	1.13	1.43	1.76	2.13	2.54	2.98	3.45	3.96
55	595.3	0.89	1.16	1.47	1.81	2.19	2.61	3.06	3.55	4.08
60	579.7	0.91	1.19	1.51	1.86	2.25	2.68	3.14	3.65	4.18
65	565.2	0.93	1.22	1.55	1.91	2.31	2.75	3.22	3.74	4.29
70	551.6	0.96	1.25	1.58	1.95	2.37	2.81	3.30	3.83	4.40
75	538.9	0.98	1.28	1.62	2.00	2.42	2.88	3.38	3.92	4.50
80	527.0	1.00	1.31	1.66	2.05	2.48	2.95	3.46	4.01	4.60

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Appendix A3 - 175 mm² AAAC (Elm) – Based on 17.29Kn (1814 kgf) – 150m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	ELM
Greased Conductor Weight (kg/m)	0.5922
Cross Sectional Area of Conductor (mm²)	211
Conductor Overall Diameter (mm)	18.8
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	5710.41
Rated Breaking Strength of Conductor (kgf)	6346.7
Basic / Recommended Span (m)	130
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1814.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1269.3
Temperature at EDT Limit (Degrees C)	5
Maximum Conductor Tension (MCT) (kgf) at	-5.6°C
Maximum Conductor Weight (MCW) (kg/m)	1.363
Maximum Conductor Pressure (MCP) (kg/m)	1.465
Freezing Point Tension (FPT) (kgf) at	0°C

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	ELM
Basic / Recommended Span (m)	130
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		70	80	90	100	110	120	130	140	150
-5.6	1051.5	0.34	0.45	0.57	0.70	0.85	1.01	1.19	1.38	1.58
0	954.1	0.38	0.50	0.63	0.78	0.94	1.12	1.31	1.52	1.75
5	876.2	0.41	0.54	0.68	0.84	1.02	1.22	1.43	1.66	1.90
10	807.0	0.45	0.59	0.74	0.92	1.11	1.32	1.55	1.80	2.06
15	746.1	0.49	0.63	0.80	0.99	1.20	1.43	1.68	1.94	2.23
20	692.8	0.52	0.68	0.87	1.07	1.29	1.54	1.81	2.09	2.40
25	646.5	0.56	0.73	0.93	1.15	1.39	1.65	1.94	2.24	2.58
30	606.0	0.60	0.78	0.99	1.22	1.48	1.76	2.06	2.39	2.75
35	570.7	0.64	0.83	1.05	1.30	1.57	1.87	2.19	2.54	2.92
40	539.8	0.67	0.88	1.11	1.37	1.66	1.97	2.32	2.69	3.09
45	512.6	0.71	0.92	1.17	1.44	1.75	2.08	2.44	2.83	3.25
50	488.5	0.74	0.97	1.23	1.52	1.83	2.18	2.56	2.97	3.41
55	467.1	0.78	1.01	1.28	1.58	1.92	2.28	2.68	3.11	3.57
60	447.9	0.81	1.06	1.34	1.65	2.00	2.38	2.79	3.24	3.72
65	430.6	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.37	3.87
70	415.0	0.87	1.14	1.44	1.78	2.16	2.57	3.01	3.50	4.01
75	400.8	0.90	1.18	1.50	1.85	2.23	2.66	3.12	3.62	4.16
80	387.9	0.94	1.22	1.55	1.91	2.31	2.75	3.23	3.74	4.29

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Appendix A4 - 200 mm² AAAC (Poplar) – Based on 23.5Kn (2397.7kgf) 150m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	POPLAR
Greased Conductor Weight (kg/m)	0.68992
Cross Sectional Area of Conductor (mm²)	239.4
Conductor Overall Diameter (mm)	20.1
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	5700
Rated Breaking Strength of Conductor (kgf)	7200.2
Basic / Recommended Span (m)	150
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	2397.8
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1440.04
Temperature at EDT Limit (Degrees C)	5
Maximum Conductor Tension (MCT) (kgf) at	-5.6°C
Maximum Conductor Weight (MCW) (kg/m)	1.496
Maximum Conductor Pressure (MCP) (kg/m)	1.515
Freezing Point Tension (FPT) (kgf) at	0°C

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	POPLAR
Basic / Recommended Span (m)	150
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		70	80	90	100	110	120	130	140	150
-5.6	1620.5	0.26	0.34	0.43	0.53	0.64	0.77	0.90	1.04	1.20
0	1487.9	0.28	0.37	0.47	0.58	0.70	0.83	0.98	1.14	1.30
5	1377.1	0.31	0.40	0.51	0.63	0.76	0.90	1.06	1.23	1.41
10	1274.1	0.33	0.43	0.55	0.68	0.82	0.97	1.14	1.33	1.52
15	1179.7	0.36	0.47	0.59	0.73	0.88	1.05	1.24	1.43	1.64
20	1094.0	0.39	0.50	0.64	0.79	0.95	1.14	1.33	1.55	1.77
25	1017.0	0.42	0.54	0.69	0.85	1.03	1.22	1.43	1.66	1.91
30	948.4	0.45	0.58	0.74	0.91	1.10	1.31	1.54	1.78	2.05
35	887.5	0.48	0.62	0.79	0.97	1.18	1.40	1.64	1.90	2.19
40	833.7	0.51	0.66	0.84	1.03	1.25	1.49	1.75	2.03	2.33
45	786.1	0.54	0.70	0.89	1.10	1.33	1.58	1.85	2.15	2.47
50	743.9	0.57	0.74	0.94	1.16	1.40	1.67	1.96	2.27	2.61
55	706.5	0.60	0.78	0.99	1.22	1.48	1.76	2.06	2.39	2.75
60	673.2	0.63	0.82	1.04	1.28	1.55	1.84	2.16	2.51	2.88
65	643.5	0.66	0.86	1.09	1.34	1.62	1.93	2.26	2.63	3.02
70	616.8	0.69	0.89	1.13	1.40	1.69	2.01	2.36	2.74	3.15
75	592.7	0.71	0.93	1.18	1.46	1.76	2.10	2.46	2.85	3.27
80	570.8	0.74	0.97	1.22	1.51	1.83	2.18	2.55	2.96	3.40

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Appendix A5 - 175 mm² ACSR (Lynx) – Based on 25.73Kn (2624.7kgf) – 150m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	LYNX
Greased Conductor Weight (kg/m)	0.8626
Cross Sectional Area of Conductor (mm²)	226.2
Conductor Overall Diameter (mm)	19.5
Coefficient of Linear Expansion (/Degree C)	1.8E-05
Modulus of Elasticity (kg/mm²)	8157.73
Rated Breaking Strength of Conductor (kgf)	8154.7
Basic / Recommended Span (m)	150
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	2624.7
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1630.9
Temperature at EDT Limit (Degrees C)	5
Maximum Conductor Tension (MCT) (kgf) at -5.6°C	2624.7
Maximum Conductor Weight (MCW) (kg/m)	1.653
Maximum Conductor Pressure (MCP) (kg/m)	1.492
Freezing Point Tension (FPT) (kgf) at 0°C	1661.9

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	LYNX
Basic / Recommended Span (m)	150
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		70	80	90	100	110	120	130	140	150
-5.6	1784.2	0.30	0.39	0.49	0.60	0.73	0.87	1.02	1.18	1.36
0	1661.9	0.32	0.42	0.53	0.65	0.79	0.93	1.10	1.27	1.46
5	1560.4	0.34	0.44	0.56	0.69	0.84	1.00	1.17	1.35	1.55
10	1466.2	0.36	0.47	0.60	0.74	0.89	1.06	1.24	1.44	1.65
15	1379.6	0.38	0.50	0.63	0.78	0.95	1.13	1.32	1.53	1.76
20	1300.3	0.41	0.53	0.67	0.83	1.00	1.19	1.40	1.63	1.87
25	1228.1	0.43	0.56	0.71	0.88	1.06	1.26	1.48	1.72	1.98
30	1162.7	0.45	0.59	0.75	0.93	1.12	1.34	1.57	1.82	2.09
35	1103.5	0.48	0.63	0.79	0.98	1.18	1.41	1.65	1.92	2.20
40	1049.9	0.50	0.66	0.83	1.03	1.24	1.48	1.74	2.01	2.31
45	1001.4	0.53	0.69	0.87	1.08	1.30	1.55	1.82	2.11	2.42
50	957.5	0.55	0.72	0.91	1.13	1.36	1.62	1.90	2.21	2.53
55	917.7	0.58	0.75	0.95	1.17	1.42	1.69	1.99	2.30	2.64
60	881.5	0.60	0.78	0.99	1.22	1.48	1.76	2.07	2.40	2.75
65	848.6	0.62	0.81	1.03	1.27	1.54	1.83	2.15	2.49	2.86
70	818.4	0.65	0.84	1.07	1.32	1.59	1.90	2.23	2.58	2.96
75	790.8	0.67	0.87	1.10	1.36	1.65	1.96	2.30	2.67	3.07
80	765.4	0.69	0.90	1.14	1.41	1.70	2.03	2.38	2.76	3.17

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Appendix B – Drawing List

Drawing Number	Sheet No	Title
Pole Arrangement Drawings		
1 09 138 0006	1	132kV OVERHEAD LINE SINGLE POLE INTERMEDIATE STRUCTURE
1 09 138 0006	3	132KV OVERHEAD LINE REFURBISHMENT OF EXISTING 'A' POLE STRUCTURE
1 09 139 0013	1	132kV OVER HEAD LINES ASSEMBLY OF SINGLE POLE 0°-20° DEVIATION
1 09 139 0014	1	132kV OVER HEAD LINE SINGLE POLE 20-40 DEGREE DEVIATION
1 09 139 0018	1	132kV SINGLE CIRCUIT OVERHEAD LINES LARGE ANGLE STRUCTURE 2.9m CENTRES
1 09 139 0019	1	132kV SINGLE CIRCUIT OVERHEAD LINES TERMINAL STRUCTURE 2.9m CENTRES
1 09 123 1173	4	33 & 66KV SINGLE CIRCUIT OVERHEAD LINES TERMINAL SUPPORT WITH SURGE ARRESTORS AND CABLE TERMINATION
ENA TS 43-50 FIG 12		132KV SINGLE CIRCUIT OVERHEAD LINES TERMINAL SUPPORT WITH SURGE ARRESTORS AND CABLE TERMINATION (4 POLE ARRANGEMENT)
Pole Fabrication Drawings		
1 09 138 0006	2	132kV OVERHEAD LINES DRILLING, SCARFING & MARKINGS FOR SINGLE POLE INTERMEDIATE STRUCTURE
1 09 139 0003	2	132kV OVERHEAD LINES DRILLING, SCARFING & MARKINGS FOR SINGLE ANGLE POLE STRUCTURE
1 09 139 0003	3	ADDITIONAL FABRICATION DETAILS FOR THE ATTACHEMENT OF THE HCL CLIMBING ACCESS SYSTEM TO DRAWING 109139003 SHT 2, 1091390018 sht 2 & 1091390019 sht2
1 09 139 0018	2	132kV SINGLE CIRCUIT OVERHEAD LINES DRILLING, SCARFING & MARKING FOR LARGE ANGLE H-POLE
1 09 139 0019	2	132kV SINGLE CIRCUIT OVERHEAD LINES DRILLING, SCARFING & MARKING FOR TERMINAL H-POLE
1 09 123 1173	5	132kV SINGLE CIRCUIT OVERHEAD LINES DRILLING, SCARFING & MARKING FOR 33/66KV CABLE TERMINAL H-POLE
ENA TS 43-50 FIG 4		132KV SINGLE CIRCUIT OVERHEAD LINES DRILLING, SCARFING & MARKING FOR 132KV TERMINAL SUPPORT WITH SURGE ARRESTORS AND CABLE TERMINATION (MAIN POLE)
ENA TS 43-50 FIG 5		132KV SINGLE CIRCUIT OVERHEAD LINES DRILLING, SCARFING & MARKING FOR 132KV TERMINAL SUPPORT WITH SURGE ARRESTORS AND CABLE TERMINATION (AUXILLARY POLE)
Insulators & Insulator fittings		
1 09 101 0487	16	132KV COMPOSITE HORIZONTAL POST INSULATOR – TECHNICAL PARAMETERS
1 09 101 0487	18	132KV COMPOSITE VERTICAL LINE/PILOT POST POLYMER INSULATOR – TECHNICAL PARAMETERS
1 09 101 0487	29	132KV COMPOSITE TENSION / STAY INSULATOR (120KN WITH 20MM BALL / SOCKET COUPLING) – TECHNICAL PARAMETERS

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Drawing Number	Sheet No	Title
1 09 101 0487	30	132KV COMPOSITE TENSION SET INSULATOR ASSEMBLY DRAWING – (NORMAL UNEARTHED SITUATIONS) FOR USE WITH DRAWING 1091010487 SHT 29
1 09 101 0487	31	132KV TENSION SET INSULATOR ASSEMBLY DRAWING – (APPROACH / EARTHED SITUATIONS) FOR USE WITH DRAWING 1091010487 SHT 29
1 09 101 0487	32	132KV COMPOSITE VERTICAL LINE/PILOT POST INSULATOR - ARCING HORN ASSEMBLY FOR (APPROACH / EARTHED SITUATIONS) FOR USE ON DRAWING 1091010487 SHT18
1 09 101 0487	33	132KV COMPOSITE HORIZONTAL POST INSULATOR - ARCING HORN ASSEMBLY FOR (APPROACH/EARTHED SITUATIONS) FOR USE ON DRAWING 1091010487 SHT16
1 09 101 0487	34	SUPPORT STOOL FOR USE WITH COMPOSITE LINE POST INSULATORS ON DRAWING 1091010487 SHT 18 AT PILOT POSITIONS
Stays and Stay Fittings		
1 09 101 0372	2	132KV COMPOSITE STAY INSULATOR ASSEMBLY DRAWING (UNEARTHED ARRANGEMENTS C/W ARCING HORN ASSEMBLIES)
1 09 101 0429	8	132kv SINGLE CIRCUIT METHOD OF FIXING STAYS
1 09 101 0675	16	132kv OVERHEAD LINES STAY YOKE LEG DETAIL (1000 LONG x 65 x 8)
1 09 101 0675	24	132kv OVERHEAD LINES STAY YOKE ARRAY FOR 1, 2 & 3 STAY OPTIONS
1 09 101 0675	46	132kv OVERHEAD LINES 10° CRANKED STAY STRAP
1 09 101 0675	47	132kv OVERHEAD LINES 45° CRANKED STAY STRAP
1 09 101 0675	48	132kv OVERHEAD LINES 40° CRANKED STAY STAP FOR TERMINAL H POLE STAYS
Pole top steelwork		
1 09 101 0675	1	132kv OVERHEAD LINES POST INSULATOR SUPPORT / TENSION PLATE (565 LONG X 220 X 15)
1 09 101 0670	6	Post Insulator Base Plate
1 09 101 0652 sht1	Item 1	Angles (Post)
1 09 101 0652 Sht1	Item 2	Channels (Post)
1 09 101 0652 Sht 1	Item 7	Horizontal Lock
1 09 101 0652 Sht 1	Item 6	Vertical Locks
1 09 101 0675	2	132kv OVERHEAD LINES CROSSARM 40-60 DEGREES (5260 LONG x 150 x 90 x 15)
1 09 101 0675	4	132kv OVERHEAD LINES CROSSARM BRACE (518 LONG x 50 x 50)
1 09 101 0675	5	132kv OVERHEAD LINES CROSSARM SUPPORT CHANNEL (1118 LONG x 230 x 75 CHANNEL)
1 09 101 0675	6	132kv OVERHEAD LINES CROSSARM 0-40 DEGREES (4860 LONG x 150 x 90 x 15)
1 09 101 0675	7	132kv OVERHEAD LINES SHORT CROSSARM FOR IN-LINE POLES (650 LONG x 150 x 90 x 15)

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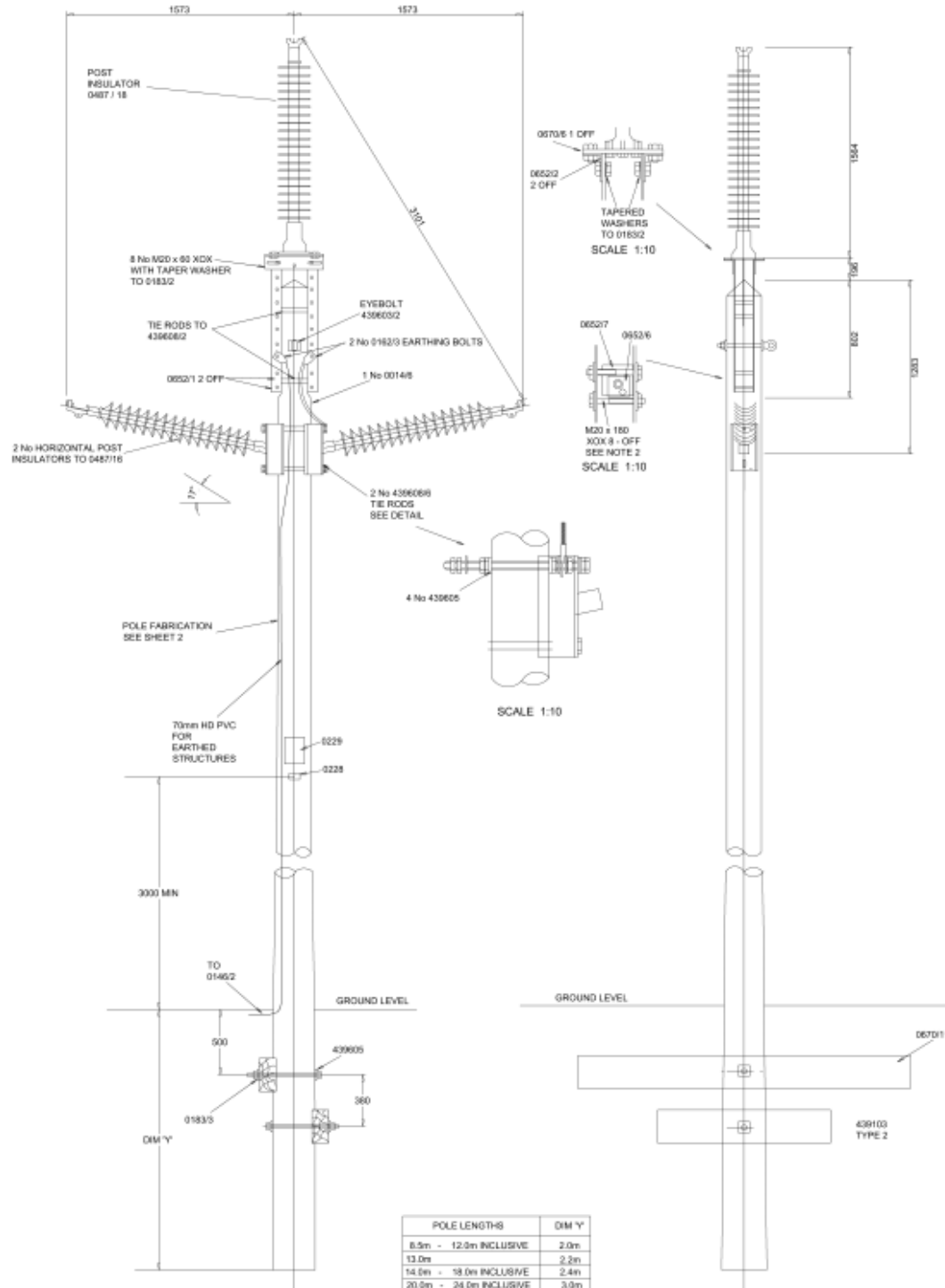
Drawing Number	Sheet No	Title
1 09 101 0675	8	132kV OVERHEAD LINES SIDE INSULATOR SUPPORT PLATE FOR IN-LINE POLES (409 LONG x 250 x 12)
1 09 101 0675	9	132kV OVERHEAD LINES INSULATOR SUPPORT PLATE BRACKET FOR IN-LINE POLES (250 LONG x 150 x 90 x 15)
1 09 101 0675	10	132kV OVERHEAD LINES SHORT XARM SUPPORT CHANNEL FOR IN-LINE POLES (1263 LONG x 230 x 74 CHANNEL)
1 09 101 0675	29	132KV OVERHEAD LINES M20 TIE ROD SCHEDULE
1 09 101 0675	30	132KV OVERHEAD LINES "RUTTER" TYPE POLE SHEAR BLOCK SCHEDULE
1 09 101 0675	31	132KV OVERHEAD LINES "RUTTER" POLE CROSSARM 0° - 2° DEVIATION (4860 X 150 X 90 X 15mm UNEQUAL ANGLE)
1 09 101 0675	32	132KV OVERHEAD LINES HORIZONTAL INSULATOR SUPPORT ANGLE (250 LONG x 100 x 75 x 12)
1 09 101 0675	34	ENHANCED ANTI-CLIMBING DEVICE FOR EXTRA STOUT POLE ASSEMBLY
1 09 101 0675	39	132kV OVERHEAD LINES RUTTER TYPE POLE HEAVY DUTY WELDED SHEAR BLOCK
1 09 101 0675	40	132kV OVERHEAD LINES CROSSARM STRUT (1460 LONG x 50 x 50 x 5)
1 09 101 0675	50	132kV OVERHEAD LINES H POLE CROSSARM COMPONENTS
Miscellaneous Items		
1 09 101 0653	3	COMPRESSION ANCHOR CLAMP – POPLAR CONDUCTOR
1 09 101 0461	1	125kN Shackle

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Appendix C - Pole Assembly drawings



NOTES

1. ALL STEEL WORK BOLTS TO BE COMPLETED WITH ONE ROUND WASHER UNDER NUT
1 ROUND WASHER UNDER BOLT HEAD EXCEPT WHERE SPECIFIED OTHERWISE.
2. LOCK 0652/7 ROTATED / FITTED TO LOCK TIE ROD BETWEEN CHANNEL SIDES WHILST MAINTAINING CHANNEL PLUMB AND CENTRAL ON POLE FACE. LOCK 0652/6 TO BE ROTATED FITTED TO LOCK TIE ROD BETWEEN 2 No M20 BOLTS WHILST PROVIDING LEVEL MOUNTING OF POLE ANGLES.
3. DETAIL DRAWINGS PREFIXED 43 REFER TO ESI DRAWINGS
OTHER DETAIL DRAWINGS REFER TO 1.09.101. SERIES
4. POST INSULATOR FIXED WITH 4 No 58" x 1-10" UNC HIGH TENSILE BOLTS.

		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF 132KV SINGLE CIRCUIT LINES SINGLE POLE INTERMEDIATE STRUCTURE	
Manufacturer Details Sheet No. 1 Scale 1:20 Prepared By Andrew Turnbull Revised 27/11/09 Date Issued 18/03/09		Type OVERHEAD Ref No. C102522 Historic Orig. No. 1091380006	
Checked By G. HARRIS Revision C Notes			

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MATERIAL LIST FOR 1 09 138.0006 SHEET No.1 (Intermediate)

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Wood Pole	109.138.0006/2	1	-
Anti Split Bolts / Climbing Attachment Point			
Eyebolts, M20 x 300, Galvd	439603/2	1	368059
Washers, Square Curved	439605	2	368078
Foundations			
Bolts, M20 x 530 Galvd	-	2	372815
Washers, Square Curved	439605	2	368078
Washer, Square Flat	0183/3	2	368063
Foundation Baulk 2500x225x125	0670/11	1	346581
Foundation Baulk 1300x225x125	439103/2	1	235124
Vertical Insulators			
132kv Polymeric Post Insulator -	0487/18	1	251550
Post insulator fixing bolts – 5/8" Unc, Grade 8.8	0487/6	4	239892
Post Insulator Base Plate	0670/6	1	346539
Bolts, M20 x 60, Galvd	-	8	378703
Square M20 Tapered Washer, Galvd	0183/2	8	374473
Angles (Post)	0652/2	2	252702
Channels (Post)	0652/1	2	252685
Tie Rod, M20 x 450, Galvd	439608/2	2	375832
Earthing Bolts, M20 x 60	0162/3	2	370504
Bolts, M20 x 180, Galvd	-	8	374098
Washers, M20, Round, Galvd	-	8	375616
Horizontal Lock	0652/7	4	252755
Vertical Locks	0652/6	4	252740
Earthing Strap –22mm lugs 750mm long	0014/6	1	251786
Horizontal Insulators			
132kv Horizontal Post Insulator	0487/16	2	251549
Tie Rod, M20 x 600, Galvd	439608/6	4	371225
Washers, Square Curved	439605	4	368078
Washers, M20, Round, Galvd	-	6	375616
Nuts M20	-	8	378915
Notices			
Safety Sign	0229	1	363318
Fibre Washer, 14mm dia x 6mm thick		6	374581
Fibre Washer, 14mm dia x 2mm thick	-	6	374577

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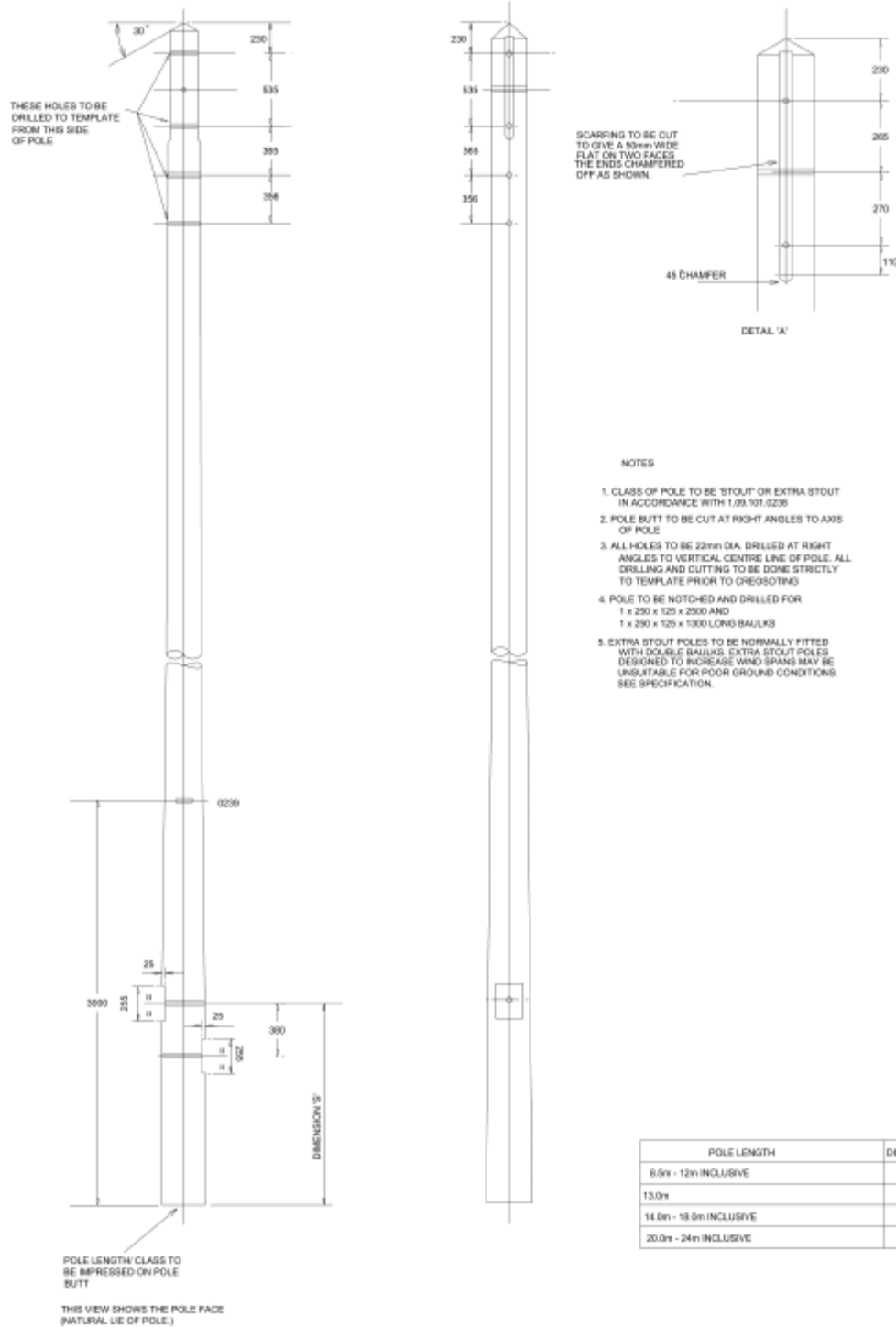
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Screw Nails, Hardened, B2p, 2in x 10G	-	6	371738
Number Base plate	0028	1	243258
Pole Numbers	0028	To Suit	-

Eathed Structures Only

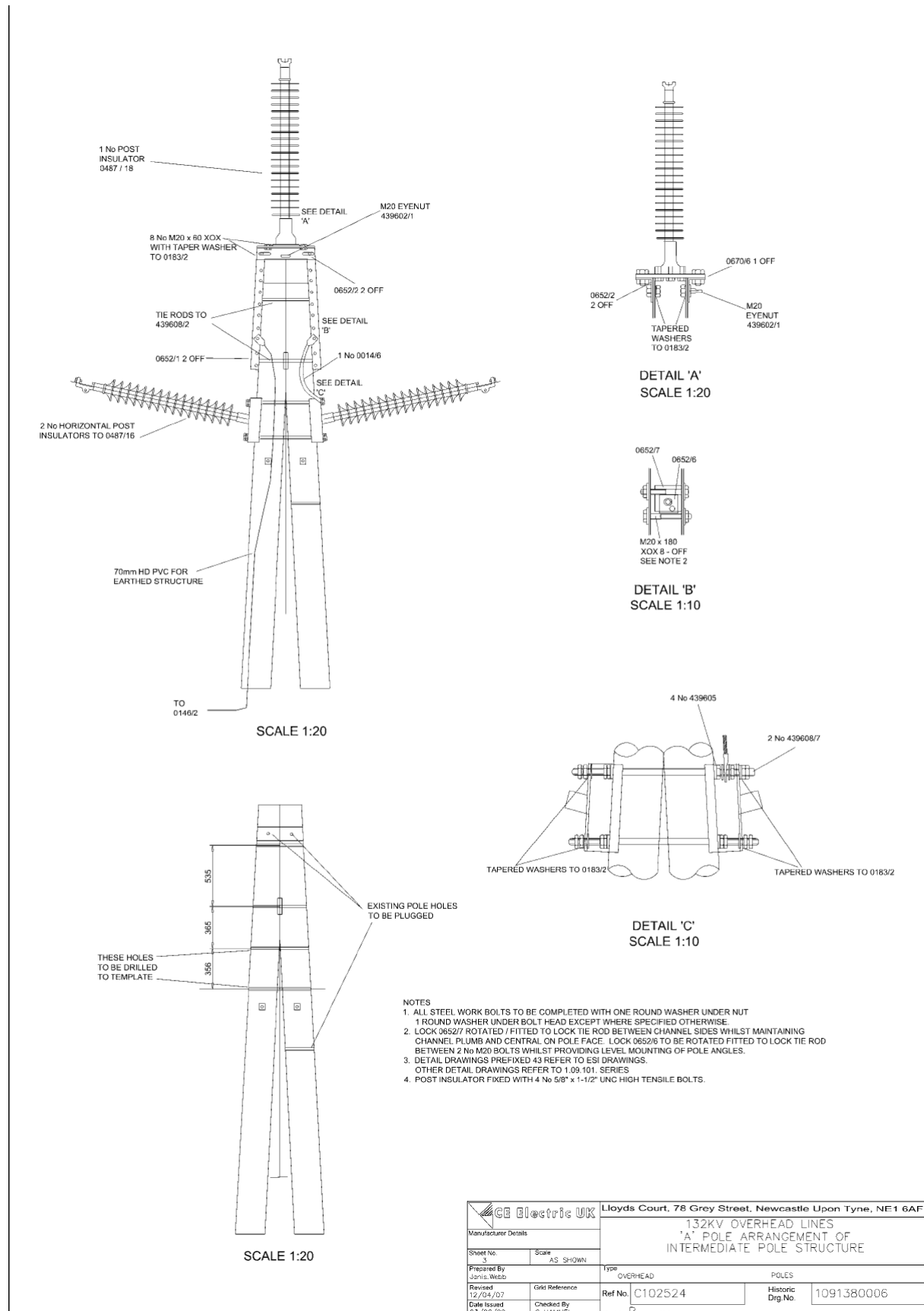
Comp Lug – M22 Hole (70mm PVC Cu)	0237	1	268819
70mm (7/3.55)HD CU Cond. PVC Covered type 8	-	20m Approx.	224774
Staples Copper 1½ x 8G	-	As Req.	374399
Earthing Lead casing	0149/3	1	235548
Earth Rod	3143	As Req.	354249
Rod Coupling	3143	As Req.	354361
Driving Bolt	3143	As Req.	354408
Saddle Clamp	3143/2	As Req.	354338

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CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AP	
Manufacturer Details		132KV SINGLE CIRCUIT OVERHEAD LINES DRILLING, SCARFING AND MARKING FOR SINGLE INTERMEDIATE POLE ARRANGEMENT	
Drawn No. 2	Scale 1:20	Type OVERHEAD	POLES
Prepared By Andrew Tarnhill	Qref Reference	Ref No. C102523	Historic Dwg No. 1091380006
Revised 21/11/09	Checked By M. 100307	Revision 03	Notes

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RENOVATION OF EXISTING 'A' POLE INTERMEDIATE STRUCTURE

Based on Drg Nos 1.09.138.0006/3

COMPONENT	DRAWING NO	QUANTITY	CAT NO
Vertical Insulator			
132kv Polymeric Post Insulator -	0487/18	1	251550
Reduced Creep Set Addition (Approach Situations)	0487/32	1	346397
Post insulator fixing bolts – 5/8" Unc, Grade 8.8	0487/6	4	239892
Composite Insulator Cond. Clamp	0487/6	1	346632
Chaffing Tape	-	1	339446
Post Insulator Base Plate	0670/6	1	346539
Bolts, M20 x 60, Galvd	-	8	378703
Square M20 Tapered Washer, Galvd	0183/2	8	374473
Angles (Post)	0652/2	2	252702
Channels (Post)	0652/1	2	252685
Tie Rod, M20 x 450, Galvd	439608/2	2	375832
Earthing Bolts, M20 x 60	0162/3	2	370504
Bolts, M20 x 180, Galvd	-	8	374098
Washers, M20, Round, Galvd	-	8	375616
Horizontal Lock	0652/7	4	252755
Vertical Locks	0652/6	4	252740
Earthing Strap –22mm lugs 750mm long	0014/6	1	251786
Climbing Attachment Point			
Eyenuit, M20	439602	1	368025
Bolt, M20 x 60, Galvd	-	1	378703
Washers, M20, Round, Galvd	-	2	375616
Horizontal Insulator			
132kv Horizontal Post Insulator	0487/16	2	251549
Reduced Creep Set Addition (Approach Situations)	0487/33	2	346398
Composite Insulator Cond. Clamp	0487/6	2	346632
Chaffing Tape	-	1	339446
Tie Rod, M20 x 600, Galvd	439608/6	4	371225
Washers, Square Curved	439605	4	368078
Square M20 Tapered Washer, Galvd	0183/2	4	374473
Washers, M20, Round, Galvd	-	2	375616
Nuts M20	-	8	378915
Notices			
Safety Sign	0229	1	363318

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Fibre Washer, 14mm dia x 6mm thick		6	374581
Fibre Washer, 14mm dia x 2mm thick	-	6	374577
Screw Nails, Hardened, B2p, 2in x 10G	-	6	371738
Number Base plate	0028	1	243258
Pole Numbers	0028	To Suit	-
Wood Plugs		5	263069

Earthed Structures Only

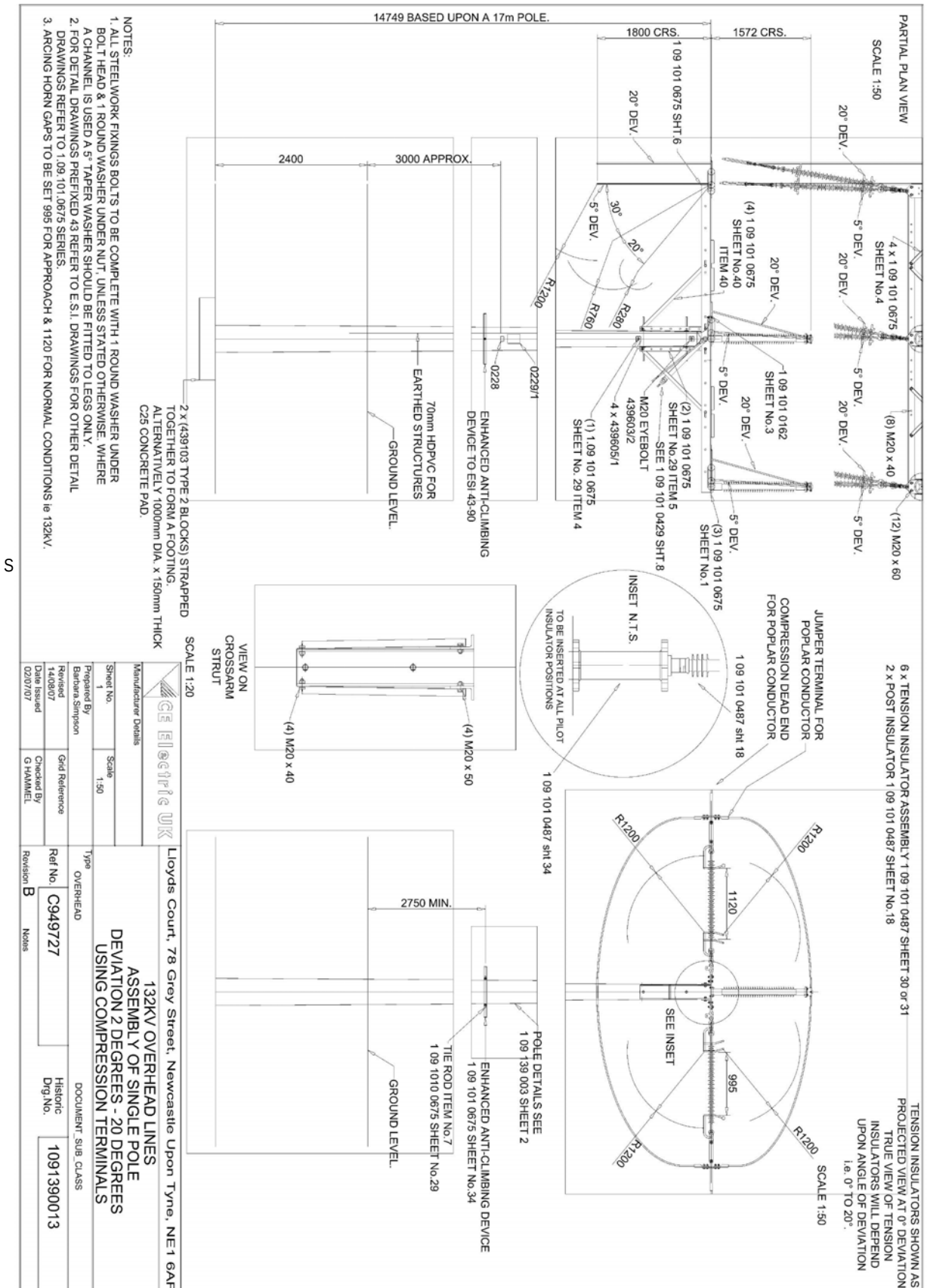
Comp Lug – M22 Hole (70mm PVC Cu)	0237	1	268819
70mm (7/3.55)HD CU Cond. PVC Covered type 8	-	20m Approx.	224774
Staples Copper 1½ x 8G	-	As Req.	374399
Earthing Lead casing	0149/3	1	235548
Earth Rod	3143	As Req.	354249
Rod Coupling	3143	As Req.	354361
Driving Bolt	3143	As Req.	354408
Saddle Clamp	3143/2	As Req.	354338

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MATERIAL LIST FOR 1 09 139 0013 SHEET No.1 (0° - 20° Deviation)

Pole fabrication Details

233999	12m Extra Stout Pole	1 09 139 0003 Sht2 Item 'A'
234000	13m Extra Stout Pole	1 09 101 0003 Sht2 Item 'B'
234001	14m Extra Stout Pole	1 09 101 0003 Sht2 Item 'C'
234002	15m Extra Stout Pole	1 09 101 0003 Sht2 Item 'D'
234003	16m Extra Stout Pole	1 09 101 0003 Sht2 Item 'E'
234004	17m Extra Stout Pole	1 09 101 0003 Sht2 Item 'F'

Poles supplied complete with foundation blocks and fixings

2	235124	439103 Type 2 Blocks Or 1 X 1000 Dia. X 150 Thk. C25 Concrete Pad.
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Materials Not supplied with Poles

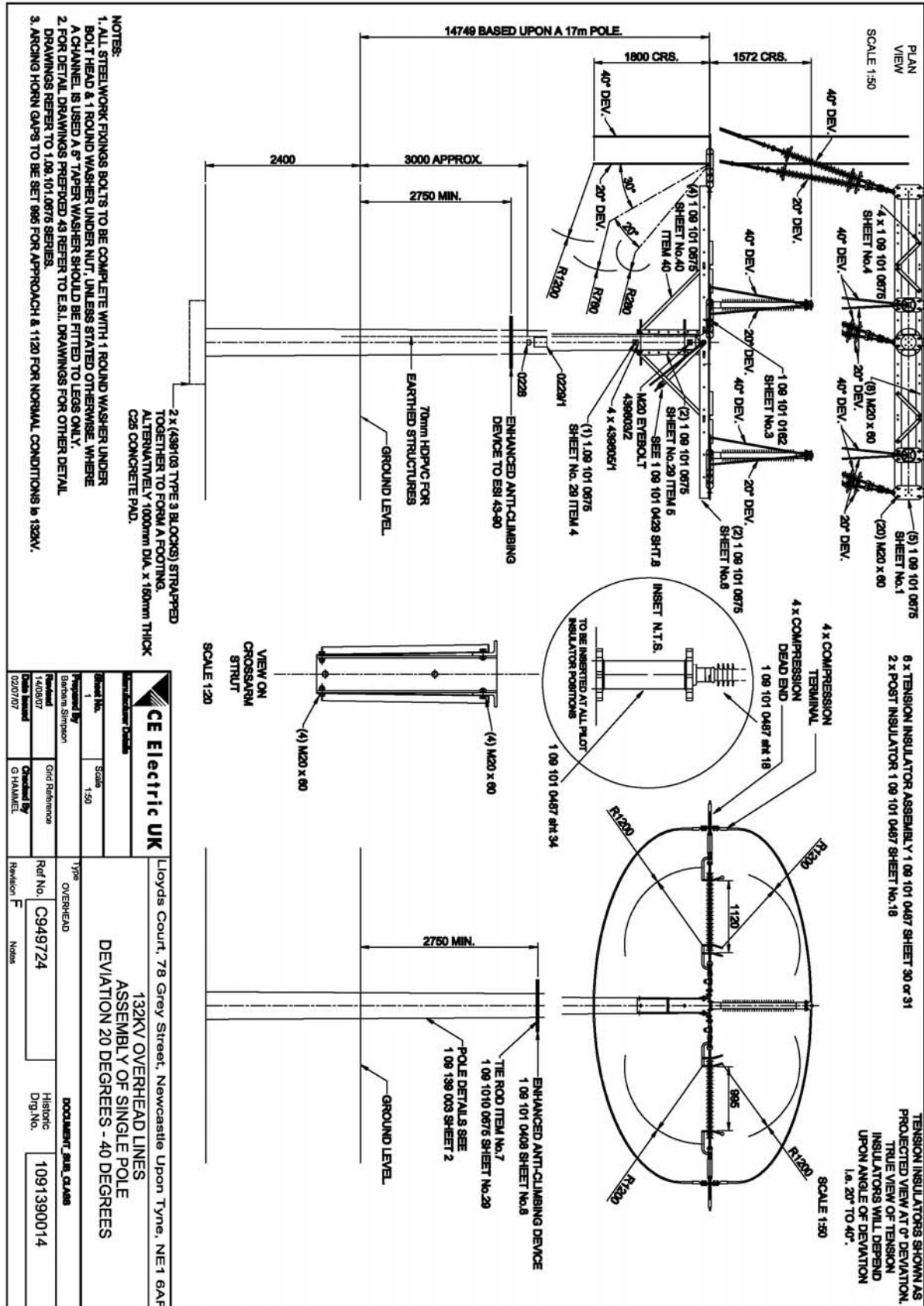
<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>	
2	251550	Pilot Post Insulator + (support stool)	1 09 101 0487 sht 18	
2	inc above	Pilot Insulator Support Stool (See above)	1 09 101 0487 sht 34	
6	251553	120kn Ten. Ins. Assembly 132kV Approach)	1 09 101 0487 sht 31	or
6	251554	120kn Ten. Ins. Assembly 33 & 66kV Approach)	1 09 101 0487 sht 31	or
6	251552	120kn Ten. Ins. Assembly (33-132kV Normal)	1 09 101 0487 sht 30	
6	243240	Poplar Comp. dead end c/w straight lug	1 09 101 0653 sht 3	
5	346474	Insulator/ Tension Plate	1 09 101 0675 Sht1	
2	346478	0-40° Crossarm	1 09 101 0675 Sht6	
4	346502	Crossarm Strut	1 09 101 0675 Sht40 Item 40	
4	346476	Crossarm Brace	1 09 101 0675 Sht4	
2	346477	Crossarm Support	1 09 101 0675 Sht5	
1	To Suit	Pole Number Plate	1 09 101 0228 Sht 3	
1	363318	Pole Safety Sign	1 09 101 0229 Sht1	
1	251362	Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8 (If Required)	
1	346487	M20 Tie Rod (360 Lg)	1 09 101 0675 Sht29 Item 4 (Anti-	
Split)				
1	36 8059	Eye Bolt Type 2	1 09 043 9603 Sht1	
4	368078	Square Curved Washer	1 00 043 9605 Sht1	
1	370504	Earthing Fastenings	1 09 101 0162 Sht1	
36	378703	Bolt Ms Hex Galv M20 X 60 Lg		
60	375616	Plain Washer Ms Galv M20		
2	346493	M20 Tie Rod (700 Lg)	1 09 101 0675 Sht29 item 5	
8	337630	Nut Ms Hex Galv M16		
8	378987	Plain Washer Ms Galv M16		
20m	220146	70mm HD Green PVC Insulated earthing Conductor –		Earthed Poles
1	075234	70mm compression Lug (m20 clearance hole	1091010237	Earthed Poles

For details on stay requirements see the following detailed stay drawings:

* For 1 To 3 Stays (1 Per Stay) Refer To Drawing 1 09 101 0429 Sht8

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* For 1 To 3 Stays Refer To Stay Yoke Array Drawing 1 09 101 0675 Sht24



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MATERIAL LIST FOR 1 09 139 0014 SHEET No.1 (20° - 40° Deviation)

Pole fabrication Details

233999	12m Extra Stout Pole	1 09 139 0003 Sht2 Item 'A'
234000	13m Extra Stout Pole	1 09 101 0003 Sht2 Item 'B'
234001	14m Extra Stout Pole	1 09 101 0003 Sht2 Item 'C'
234002	15m Extra Stout Pole	1 09 101 0003 Sht2 Item 'D'
234003	16m Extra Stout Pole	1 09 101 0003 Sht2 Item 'E'
234004	17m Extra Stout Pole	1 09 101 0003 Sht2 Item 'F'

Poles supplied complete with foundation blocks and fixings

2	235124	439103 Type 2 Blocks Or 1 X 1000 Dia. X 150 Thk. C25 Concrete Pad.
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Materials Not supplied with Poles

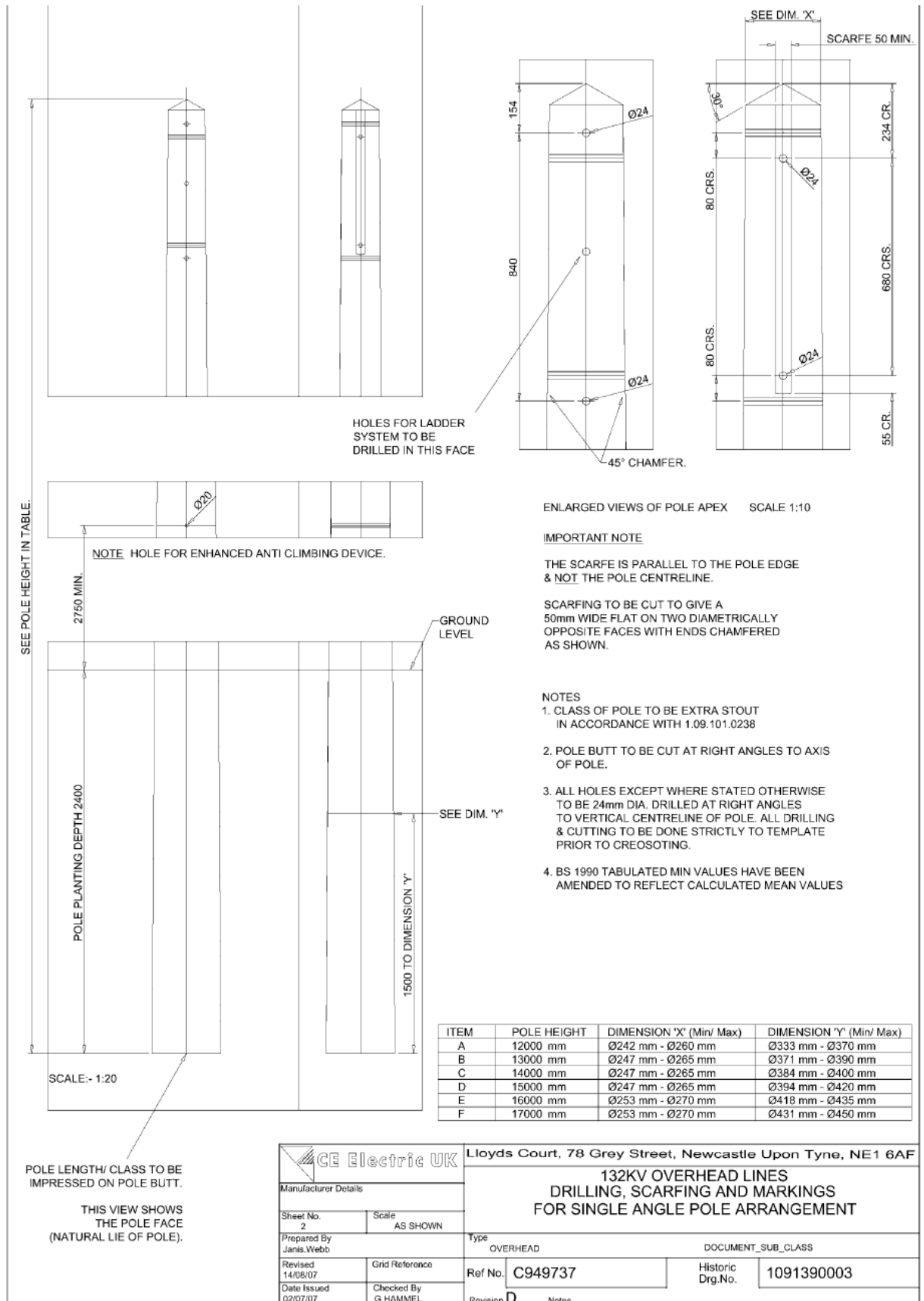
<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
2	251550	Pilot Post Insulator + (support stool)	1 09 101 0487 sht 18
2	inc above	Pilot Insulator Support Stool (See above)	1 09 101 0487 sht 34
6	251553	120kn Ten. Ins. Assembly (132kV Approachl)	1 09 101 0487 sht 31 or
6	251554	120kn Ten. Ins. Assembly (33 & 66kV Approachl)	1 09 101 0487 sht 31 or
6	251552	120kn Ten. Ins. Assembly (33 - 132kV Normal)	1 09 101 0487 sht 30
6	243240	Poplar Comp. dead end c/w straight lug	1 09 101 0653 sht 3
5	346474	Insulator/ Tension Plate	1 09 101 0675 Sht1
2	346478	0-40° Crossarm	1 09 101 0675 Sht6
4	346502	Crossarm Strut	1 09 101 0675 Sht40 Item 40
4	346476	Crossarm Brace	1 09 101 0675 Sht4
2	346477	Crossarm Support	1 09 101 0675 Sht5
1	To Suit	Pole Number Plate	1 09 101 0228 sht3
1	363318	Pole Safety Sign	1 09 101 0229 Sht1
1	251362	Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8 (If Required)
1	346487	M20 Tie Rod (360 Lg) –(Anti split)	1 09 101 0675 Sht29 Item4
1	36 8059	Eye Bolt Type 2	1 09 043 9603 Sht1
4	368078	Square Curved Washer	1 00 043 9605 Sht1
1	370504	Earthing Fastenings	1 09 101 0162 Sht1
36	378703	Bolt Ms Hex Galv M20 X 60 Lg	
60	375616	Plain Washer Ms Galv M20	
2	346493	M20 Tie Rod (700 Lg)	1 09 101 0675 Sht29 item 5
8	337630	Nut Ms Hex Galv M16	
8	378987	Plain Washer Ms Galv M16	
20m	220146	70mm HD Green PVC Insulated earthing Conductor –	Earthed Poles
1	075234	70mm compression Lug (m20 clearance hole	1091010237 Earthed Poles

For details on stay requirments see the following detailed stay drawings:

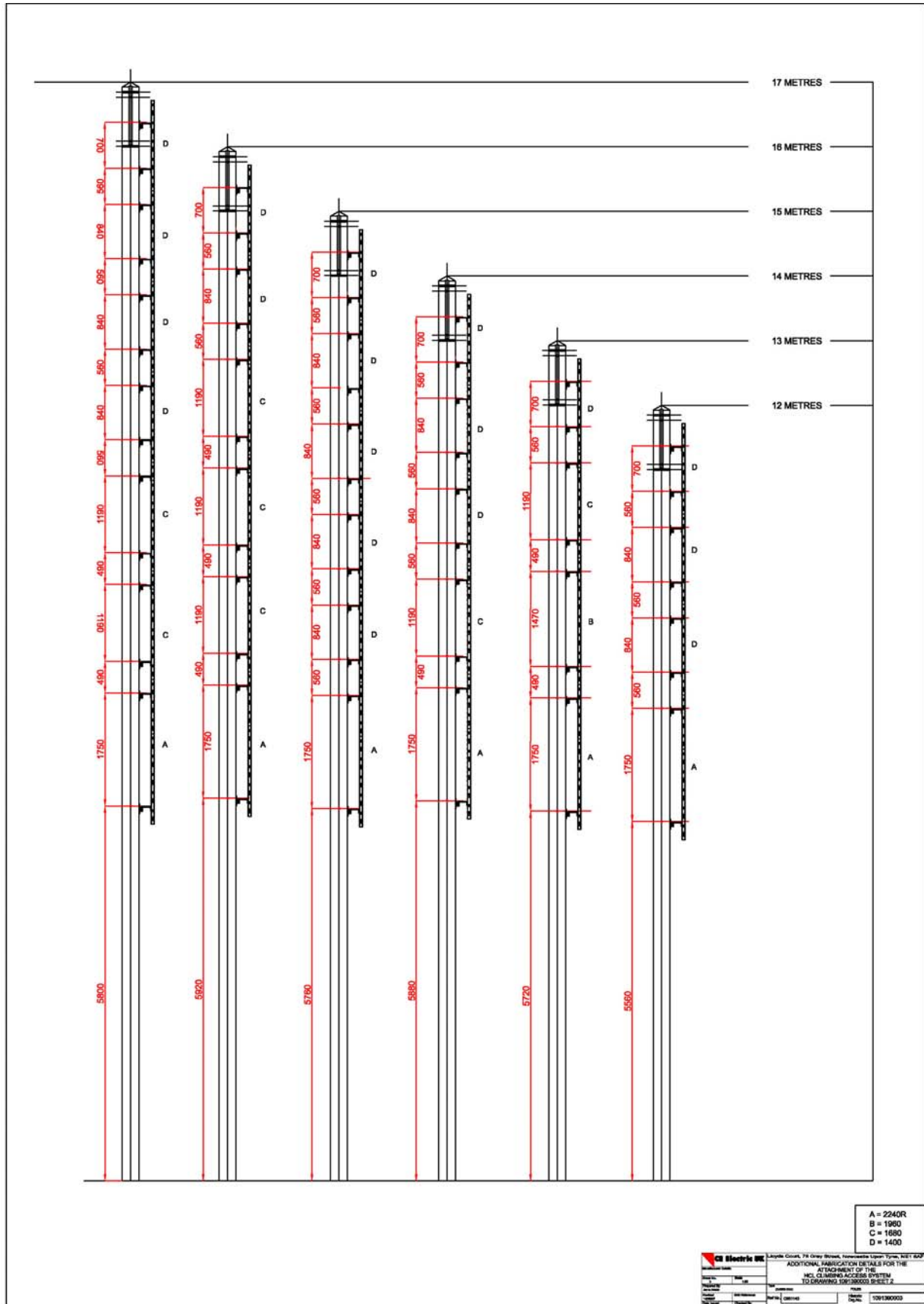
* For 1 To 3 Stays (1 Per Stay) Refer To Drawing 1 09 101 0429 Sht8

* For 1 To 3 Stays Refer To Stay Yoke Array Drawing 1 09 101 0675 Sht24

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MATERIAL LIST FOR 1 09 139 0018 SHEET No.1(40° - 60° Deviation)

Pole fabrication Details (H Pole Assembly select pole)

234005	12m E/Stout Pole	1 09 139 0018 Sht2
234006	13m E/Stout Pole	1 09 101 0018 sht2'
234007	14m E/Stout Pole	1 09 101 0018 sht2'
234008	15m E/Stout Pole	1 09 101 0018 sht2'
234009	16m E/Stout Pole	1 09 101 0018 Sht2
234010	17m E/Stout Pole	1 09 101 0018 Sht2 '

Note

Poles are supplied fabricated into structures using the materials listed below

<u>QTY</u>		<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
8	372815	Bolt Ms Hex Galv M20 X 530 Long	
2	375616	Plain Washer Ms Galv M20	
5	368078	Square Curved Washer	1 00 043 9605 Sht1
5	368063	Square Flat washers	1091010183 sht 3 item 6
2	240315	Foundation Braces	1091010650 sht 3 item 9
1	234507	Wood Block (3600 x 250 x 125mm)	1091010650 sht3 item 8
4	235124	Wood Blocks 439103 type 2	1000439103 type2

Materials Not supplied with Poles

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
2	251550	Pilot Post Insulator (+ support stool)	1 09 101 0487 Sht 18
2	inc above	Pilot Insulator Support Stool (See above)	1 09 101 0487 sht 34
6	251553	120kn Ten. Ins. Assembly (132kV Approach)	1 09 101 0487 Sht 31
6	251554	120kn Ten. Ins. Assembly (33 & 66kV Approach)	1 09 101 0487 Sht 31
6	251552	120kn Ten. Ins. Assembly (33-132kV Normal)	1 09 101 0487 Sht 30
6	243240	Poplar Comp. Dead End C/W Straight Lug	1 09 101 0653 sht 3
2	346507	Crossarm	1 09 101 0675 Sht 50 item 1
6	346508	Post Insulator & Tension Plate	1 09 101 0675 Sht 50 item 2
4	252670	Pole Cheek Plate	1 09 101 0675 Sht 50 item 4
2	372904	Bolt Ms Hex Galv M20 X 400 Long	
4	368078	Square Flat Washer	1 09 101 0183 Sht 3 item 6
4	240739	Square Plate For Item 4	1 09 101 0675 Sht 50 item 5
4	346510	Crossarm Spacer Channel	1 09 101 0675 Sht 50 item 6
4	368078	Square Curved Washer	1 00 043 9605 Sht1
2	368059	Eyebolt Type 2	1 00 043 9303 Sht 1
84	378703	Bolt Ms Hex Galv M20 X 60 Long	
88	374488	Taper Washer Galv M20	1 09 101 0183 item4
86	378915	Nut Ms Hex Galv M20	
88	375616	Plain Washer Galv M20	
4	346504	45 Degree Stay Strap	1 09 101 0675 Sht 47

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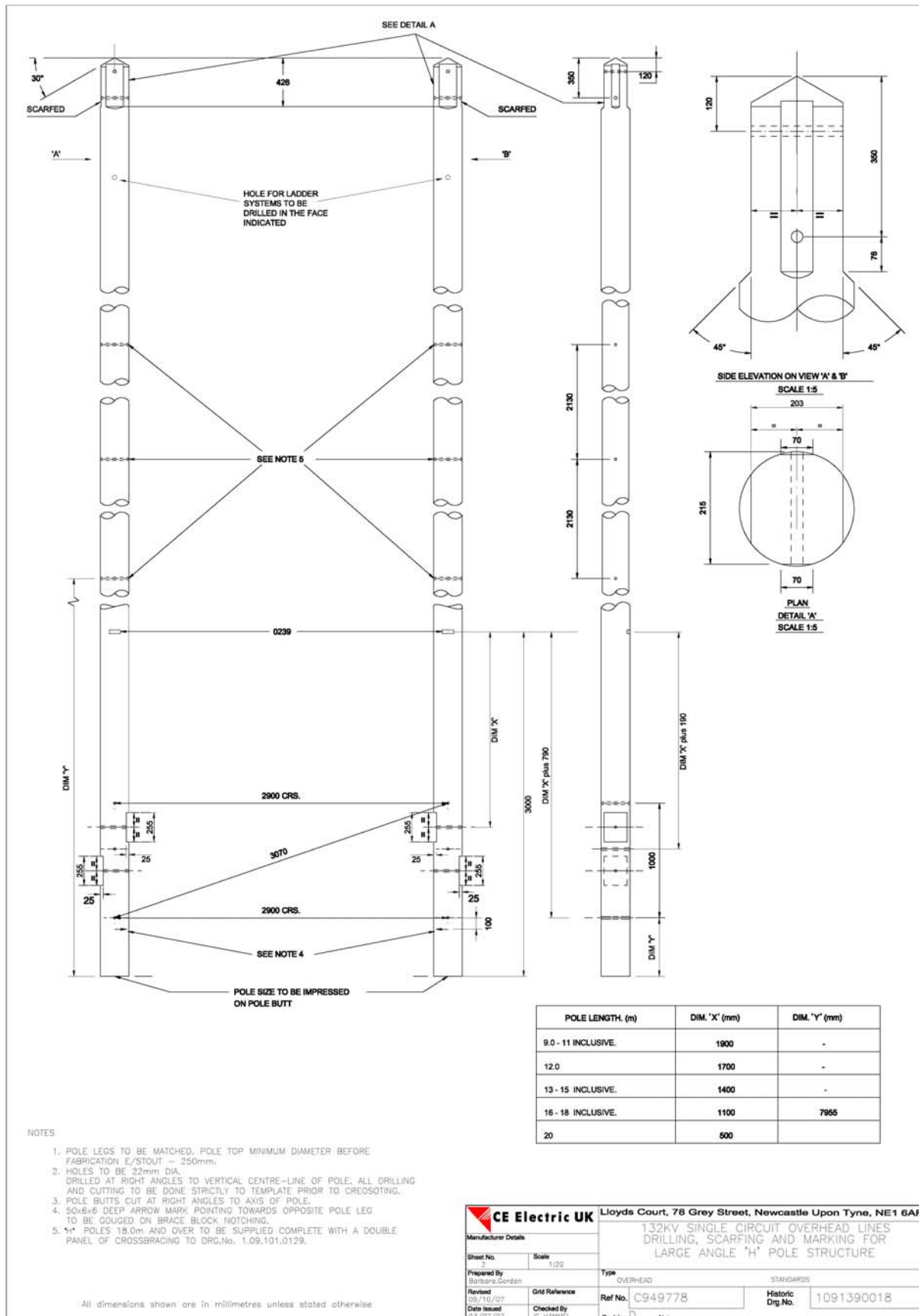
Stay Note:

4 STAYS GENERALLY TO DRAWING 1 09 101 0675 SHT8

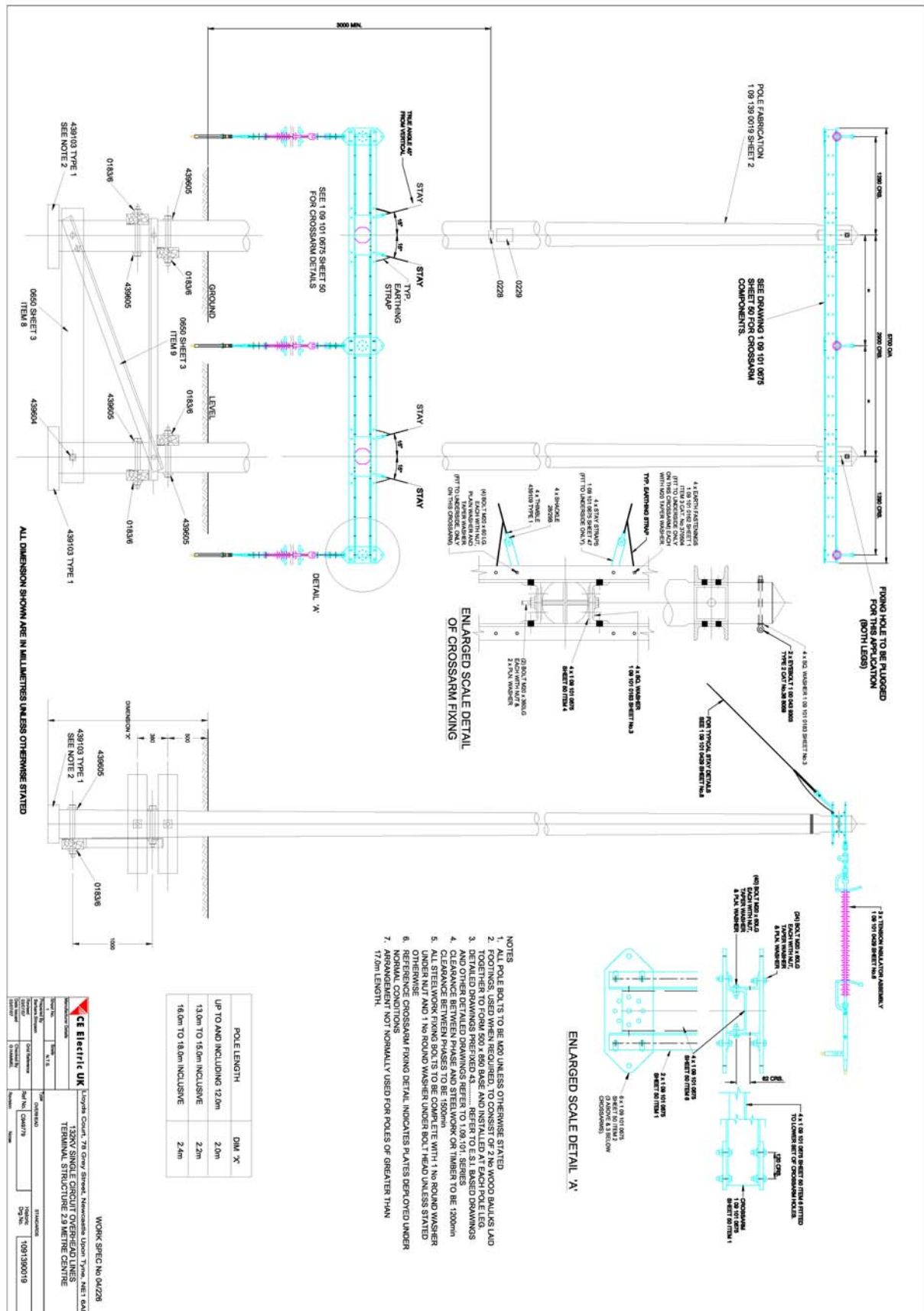
Continued

4	370504	Earthing Fastenings	1 09 101 0162 Sht1	
1	243258	Pole Number Plate	1 09 101 0228 sht 3 (number to suit)	
1	(number to suit)	Pole Numbers	1 09 101 0228 sht 3	
2	363318	Pole Safety Sign	1 09 101 0229 Sht1	
1	251362	Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8	(If Required)
4	375777	Brass Slotted Head screws 1" No8		
4	374581	Fibre washers (Thin)		
4	374577	Fibre washers (Thick)		
20m	220146	70mm HD Green PVC Insulated earthing Conductor –		Earthed Poles
1	075234	70mm compression Lug (m20 clearance hole	1091010237	Earthed Poles

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MATERIAL LIST FOR 1 09 139 0019 SHEET No.1(Terminal H)

Pole fabrication Details (H Pole Assembly select pole)

234005	12m Stout Pole	1 09 139 0019 Sht2
234006	13m Stout Pole	1 09 101 0019 sht2'
234007	14m Stout Pole	1 09 101 0019 sht2'
234008	15m Stout Pole	1 09 101 0019 sht2'
234009	16m Stout Pole	1 09 101 0019 Sht2'
234010	17m Stout Pole	1 09 101 0019 Sht2 '

Note

Poles are supplied fabricated into structures using the materials listed below

<u>QTY</u>		<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
8	372815	Bolt Ms Hex Galv M20 X 530 Long	
2	375616	Plain Washer Ms Galv M20	
5	368078	Square Curved Washer	1 00 043 9605 Sht1
5	368063	Square Flat washers	1091010183 sht 3 item 6
2	240315	Foundation Braces	1091010650 sht 3 item 9
1	234507	Wood Block (3600 x 250 x 125mm)	1091010650 sht3 item 8
4	235124	Wood Blocks 439103 type 2	1000439103 type2

Materials Not supplied with Poles

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>	
6	251553	120kn Ten. Ins. Assembly (132kV Approach)	1 09 101 0487 Sht 31	Or
6	251554	120kn Ten. Ins. Assembly (33 & 66kV Approach)	1 09 101 0487 Sht 31	Or
6	251552	120kn Ten. Ins. Assembly (33 – 132kV Normal)	1 09 101 0487 Sht 30	
6	243240	Poplar Comp. Dead End C/W Straight Lug	1 09 101 0653 sht 3	
2	346507	Crossarm	1 09 101 0675 Sht 50 item 1	
6	346508	Post Insulator & Tension Plate	1 09 101 0675 Sht 50 item 2	
4	252670	Pole Cheek Plate	1 09 101 0675 Sht 50 item 4	
2	372904	Bolt Ms Hex Galv M20 X 400 Long		
4	368078	Square Flat Washer	1 09 101 0183 Sht 3 item 6	
4	240739	Square Plate For Item 4	1 09 101 0675 Sht 50 item 5	
4	346510	Crossarm Spacer Channel	1 09 101 0675 Sht 50 item 6	
4	368078	Square Curved Washer	1 00 043 9605 Sht1	
2	368059	Eyebolt Type 2	1 00 043 9303 Sht 1	
84	378703	Bolt Ms Hex Galv M20 X 60 Long		
88	374488	Taper Washer Galv M20	1 09 101 0183 item4	
86	378915	Nut Ms Hex Galv M20		
88	375616	Plain Washer Galv M20		
4	346504	40 Degree Stay Strap	1 09 101 0675 Sht 47	

Stay Note:

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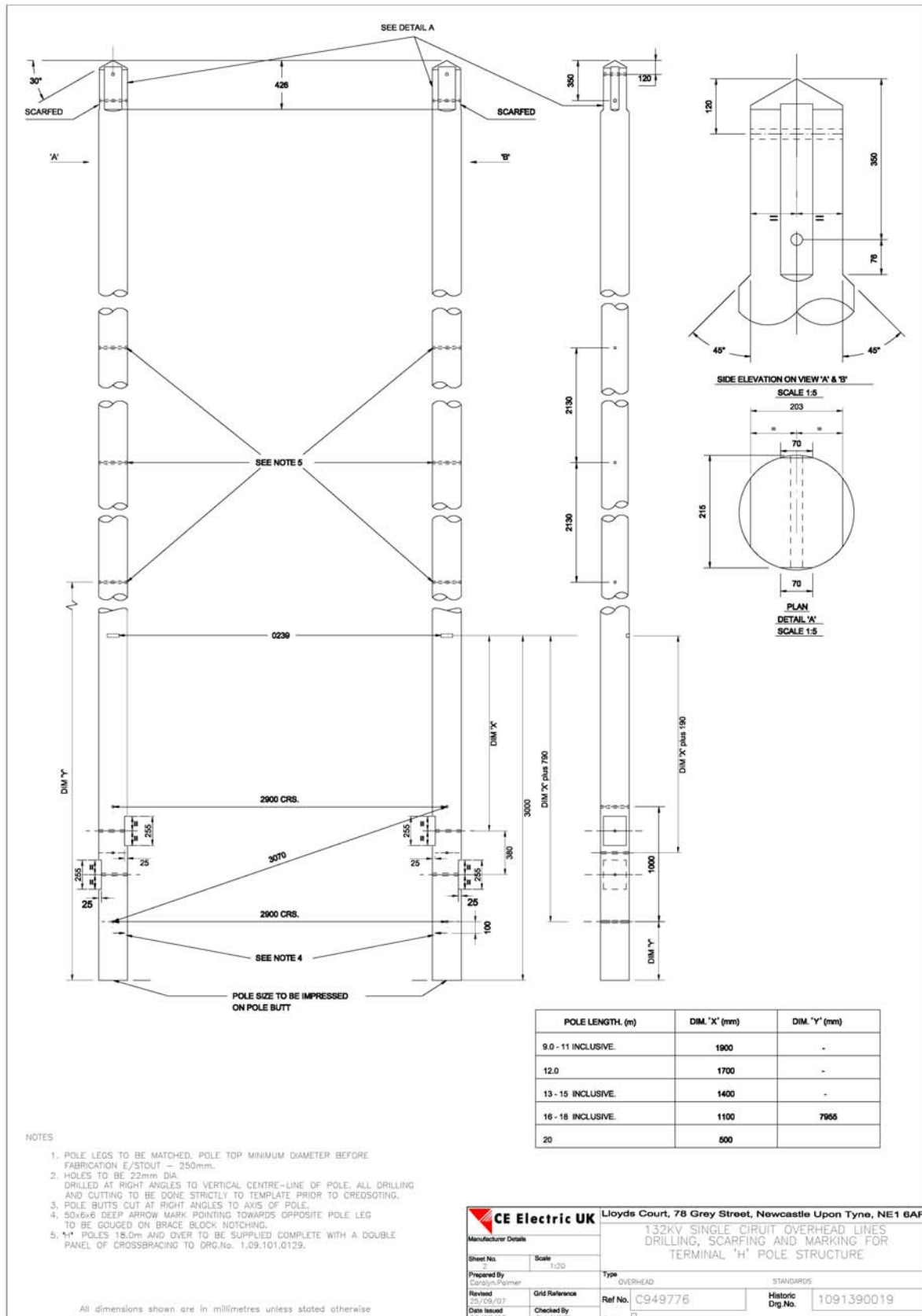
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4 STAYS GENERALLY TO DRAWING 1 09 101 0675 SHT8

Continued

4	370504	Earthing Fastenings	1 09 101 0162 Sht1	
1	243258	Pole Number Plate	1 09 101 0228 sht 3 (number to suit)	
1	(number to suit)	Pole Numbers	1 09 101 0228 sht 3	
2	363318	Pole Safety Sign	1 09 101 0229 Sht1	
1	251362	Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8	(If Required)
4	375777	Brass Slotted Head screws 1" No8		
4	374581	Fibre washers (Thin)		
4	374577	Fibre washers (Thick)		
20m	220146	70mm HD Green PVC Insulated earthing Conductor –		Earthed Poles
1	075234	70mm compression Lug (m20 clearance hole	1091010237	Earthed Poles

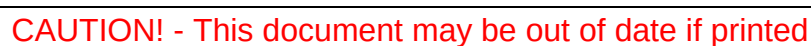
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Cable Termination Structures

- 33 or 66kV Cable Terminations shall be provided in accordance with drawing 1091231173 sht 4.
- 132kV Cable Terminations shall be provided in accordance with the drawing and material lists shown in ENA TS 43-50 fig 12 and the poles fabricated in accordance with figs 4 & 5



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33 - 66 KV 'H' TERMINAL

Based on Drg Nos 1.09.123.1173 Sheet 4

	DRAWING NO	QUANTITY	CAT NO
Poles			
12m Stout H Pole	1.09.138.0019 Sheet 2	1	234005
13m Stout H Pole	1.09.101.0019 Sheet 2	1	234006
14m Stout H Pole	1.09.101.0019 Sheet 2	1	234007
15m Stout H Pole	1.09.101.0019 Sheet 2	1	234008
16m Stout H Pole	1.09.101.0019 Sheet 2	1	234009
17m Stout H Pole	1.09.101.0019 Sheet 2	1	234010
Foundation Components			
Bolt Ms Hex Galv M20 X 530 Long	-	8	372815
Plain Washer Ms Galv M20	-	2	375616
Square Curved Washer	1 00 043 9605 Sht1	5	368078
Square Flat washers	1091010183 sht 3 item 6	5	368063
Foundation Braces	1091010650 sht 3 item 9	2	240315
Wood Block (3600 x 250 x 125mm)	1091010650 sht3 item 8	1	234507
Wood Blocks 439103 type 2	1000439103 type2	4	235124
Pole, Insulator and Conductor Fittings			
120kn Ten. Ins. Assembly (132kV Approach)	1 09 101 0487 Sht 31 <u>or</u>	6	251553
120kn Ten. Ins. Assembly (33 & 66kV Approach)	1 09 101 0487 Sht 31	6	251554
120kn Ten. Ins. Assembly (33 – 132kV Normal)	1 09 101 0487 Sht 30	6	251552
Poplar Comp. Dead End C/W Straight Lug	1 09 101 0653 sht 3	6	243240
Crossarm	1 09 101 0675 Sht 50 item 1	2	346507
Post Insulator & Tension Plate	1 09 101 0675 Sht 50 item 2	6	346508
Pole Cheek Plate	1 09 101 0675 Sht 50 item 4	4	252670
Bolt Ms Hex Galv M20 X 400 Long		2	372904
Square Flat Washer	1 09 101 0183 Sht 3 item 6		368078
Square Plate For Item 4	1 09 101 0675 Sht 50 item 5	4	240739
Crossarm Spacer Channel	1 09 101 0675 Sht 50 item 6	4	346510
Square Curved Washer	1 00 043 9605 Sht1	4	368078
Eyebolt Type 2	1 00 043 9303 Sht 1		368059
Bolt Ms Hex Galv M20 X 60 Long		84	378703
Taper Washer Galv M20	1 09 101 0183 item4	88	374488
Nut Ms Hex Galv M20		86	378915
Plain Washer Galv M20		88	375616
40 Degree Stay Strap	1 09 101 0675 Sht 47	4	346504
Earthing Fastenings	1 09 101 0162 Sht1	4	370504
Pole Number Plate	1 09 101 0228 sht 3	1	243258
Pole Numbers	1 09 101 0228 sht 3	1	(number to suit)
Pole Safety Sign	1 09 101 0229 Sht1	2	363318
Enhanced Anti-Climbing Device	1 09 101 0408 Sht 8	1	251362

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Brass Slotted Head screws 1" No8	-	4	375777
Fibre washers (Thin)	-	4	374581
Fibre washers (Thick)	-	4	374577
70mm HD Green PVC Insulated earthing Conductor –	-	20m	220146
70mm compression Lug (m20 clearance hole)	1091010237	1	075234
Stays (2 per leg)			
Pole Top Stay Dead End	0541/2	4	250209
7/4.00mm Grade 1150 Staywire	-	10	231481
Stay Dead End	0541/1	12	255217
Stay thimble	9109	12	244922
Polymeric Stay Insulator	0372/2	4	346596
66/132kv Stay Insulator Earthed	9107/2	2	253743
Type 2 Stay Rod	9101	4	231477
Wood Stay Block	9103/2	4	235124
Barbed wire	-	2	366829
66/132kv Stay Insulator Arcing Horn	0490/9	8	238014
Cable and Diverter Support			
Channel	0650 item 11	2	240508
Channel	0650 item 12	1	240512
Channel	0650 item 13	2	240289
Channel	0551/10	1	243421
Channel	0551/11	1	243493
Angle	0551/3	2	243578
Angle	0551/4	2	243582
SB Support Plates	0308 sheet 7 or 8	3	
Arrester Plates	0308/45	3	243987
Cleat Support	0309/7	3	245268
Bolt, M20 x 360		8	374609
Bolt, M16 x 50		46	378544
Bolt, M12 x 55 Earthing	0162/1	1	370487
Bolt, M12 x 40 Earthing	0162/2	3	370491
Washer Sq Curved	9605	8	368078
Washer, M16 Round		46	378987
Washer, M20 Round		8	375616
Washer, M16 Galvanised Tap		22	379852
Surge Arrester to suit Voltage		3	
Cable Support Cleat	0650 item 14	3	244640
Cable Sealing Bell Support Bars	0208 item 47	6	232732
Cable Sealing Bell Support Bars	0308 item 49	6	232747
Bolt, M12 x 40		24	378421
Cable Ladder			
Ladder Assy. To suit Pole Height	0550 sheet 2	1	To suit

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Bolts, M16 x 50		2	378544
Coach Screws ½ x 3"		2	372891
Baulk Type 2	9103	1	235124
Bolts, M20 x 530		1	372815
Washer, Flat Sq	0183/3	2	368063
Washer, M20 Round		2	375616
Flat Cleat Support	0308/22	5	242895
Flat Cleat Support	0309/37	1	245272
Bolts, M12 x 40		38	378421
Washer, M12 Round		38	375601

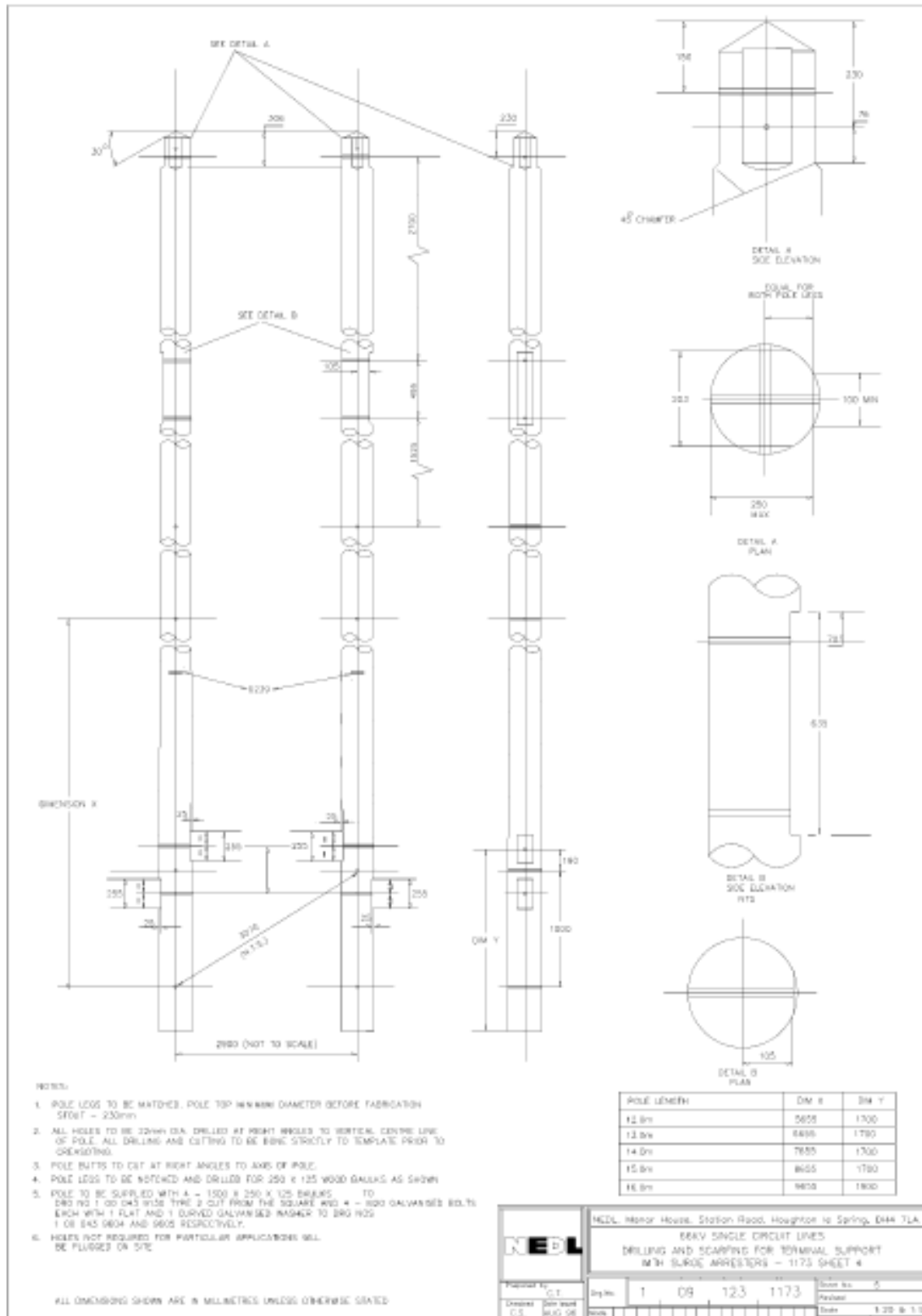
Protection

Cable Casing	0308/40	1	243084
Bolts, M16 x 40		18	378525
Bolts, M12 x 40		6	378421
Washer, M12 Round		6	375601
ACG	0408/12	1 pair	251362
ACG	0408/7	1 pair	243192
Bolt, M16 x 400		2	378671
Washer, M16 Round		20	378987
Steel Wire Galvanised		2m	330013
Barbed Wire		50m	366829
Staples, ¼ x 9G Galvanised		20	375599

Earth Materials

Comp Lug – M22 Hole (70mm PVC Cu)	0237	1	268819
70mm (7/3.55)HD CU Cond. PVC Covered type 8	-	20m	224774
		Approx.	
Staples Copper 1½ x 8G	-	As Req.	374399
Earthing Lead casing	0149/3	1	235548
Earth Rod	3143	As Req.	354249
Rod Coupling	3143	As Req.	354361
Driving Bolt	3143	As Req.	354408
Saddle Clamp	3143/2	As Req.	354338

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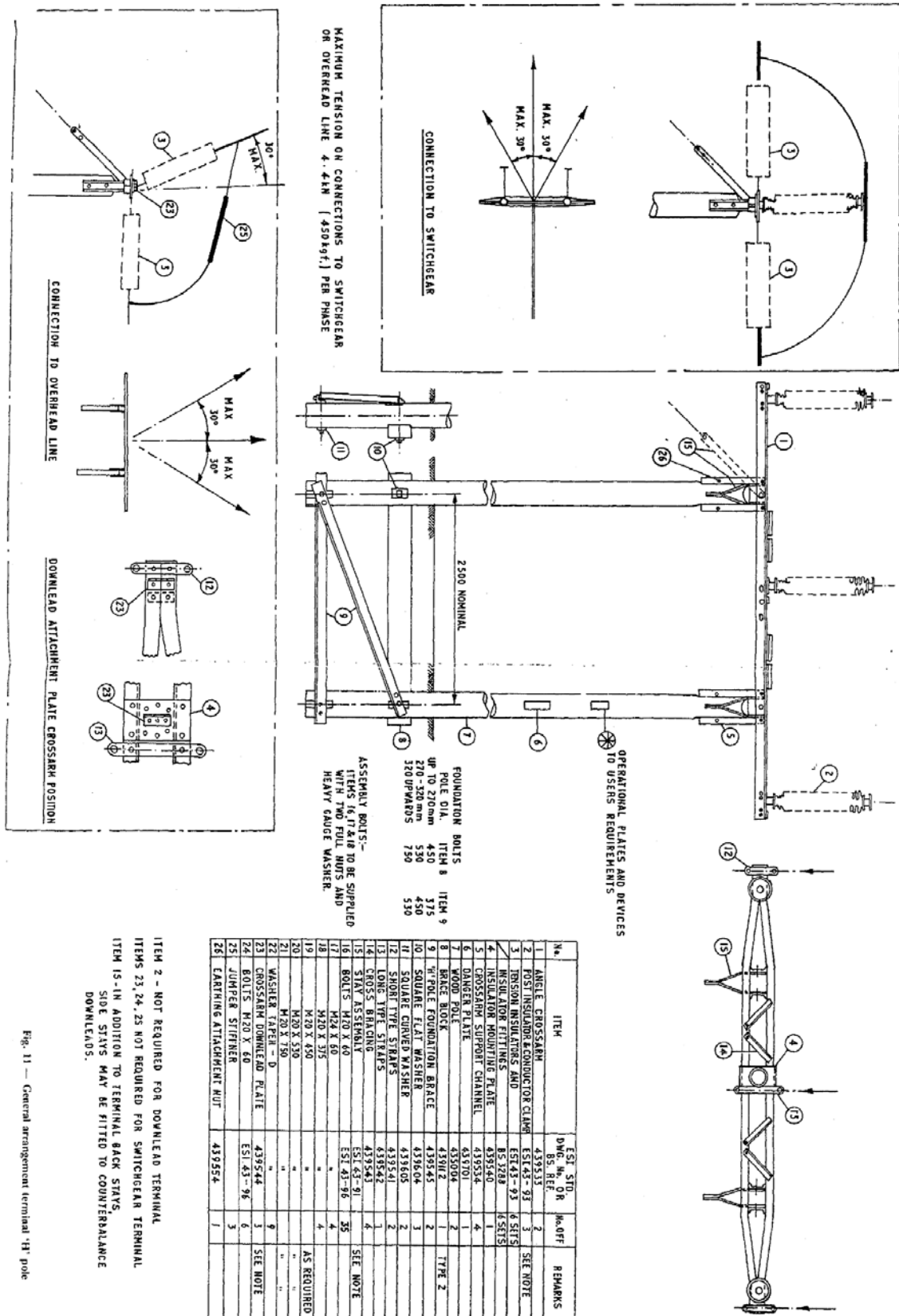
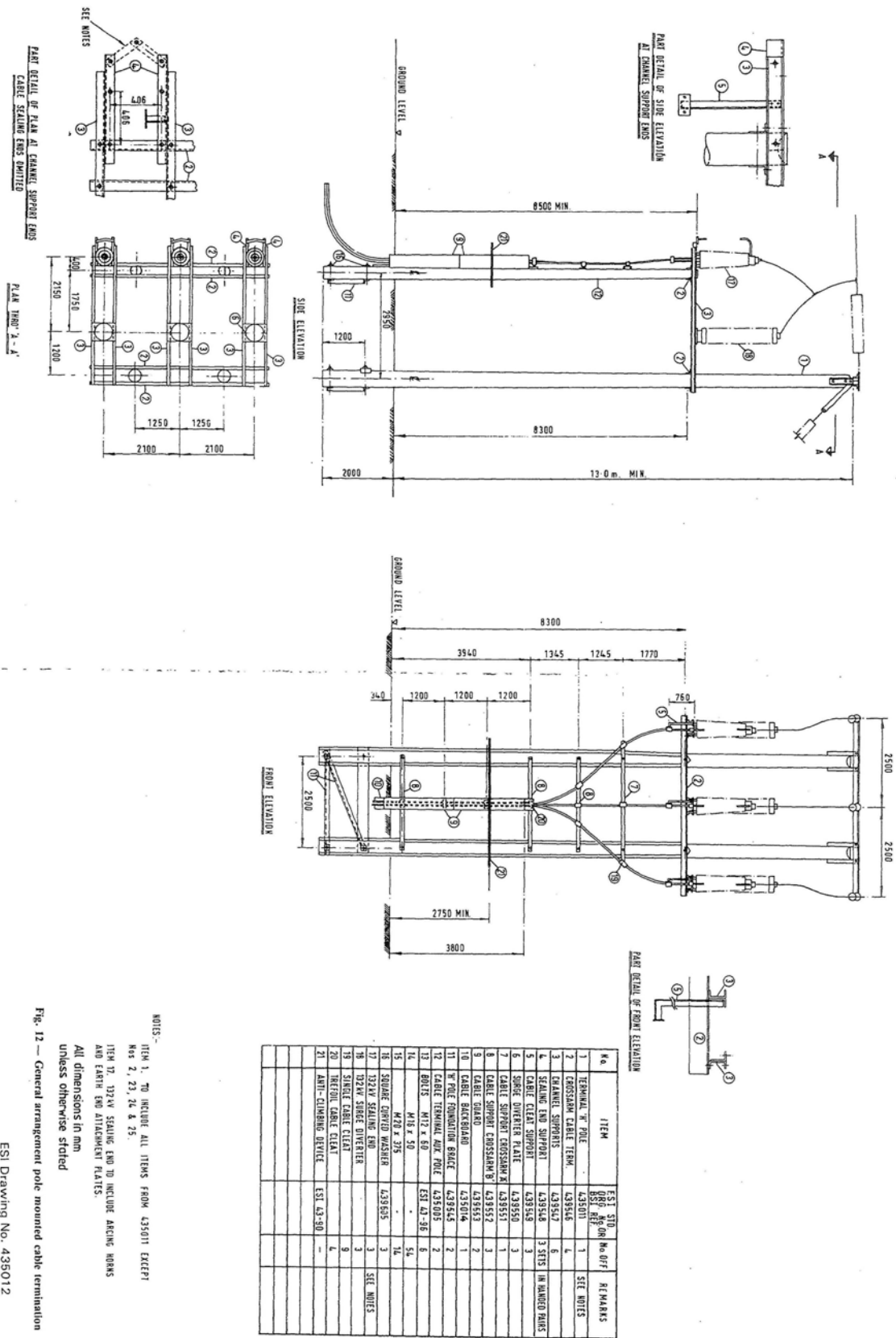


Fig. 11 - General arrangement terminal 'H' pole

66/630 ESI Drawing No. 435011

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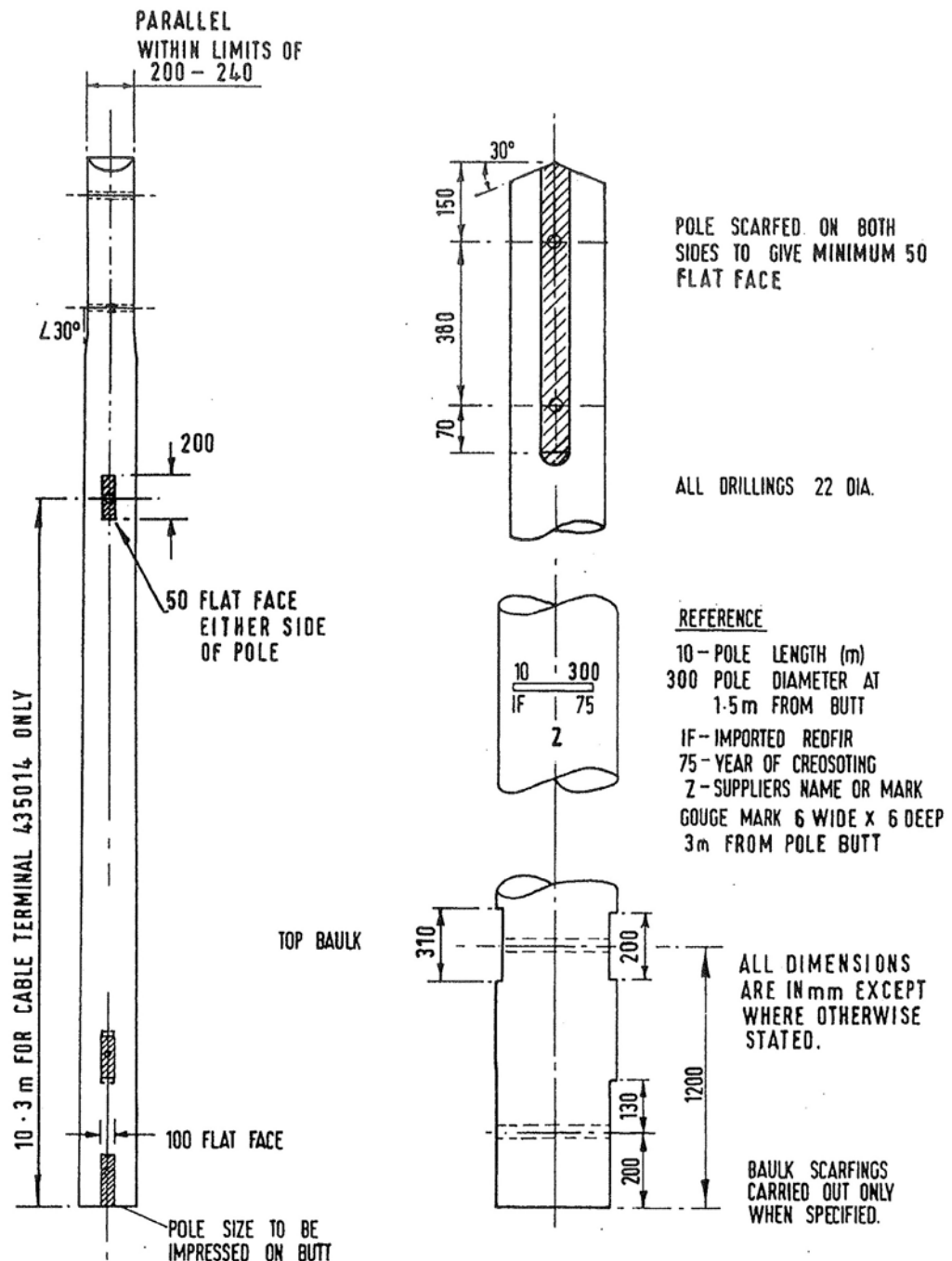
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ENA TS 43-50 Fig 4

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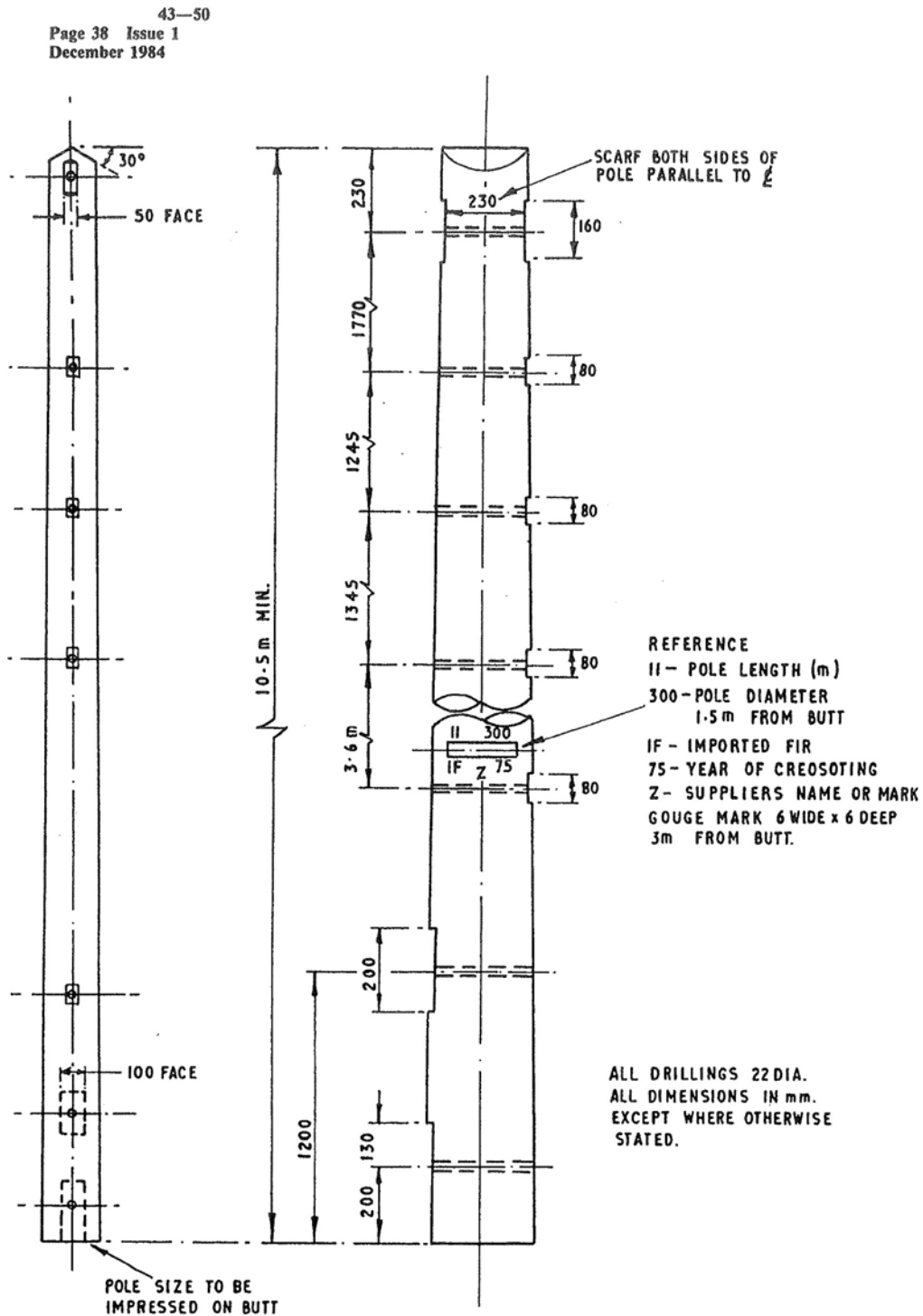


ESI Drawing No. 435004

Fig. 4 — Scarfing and drilling for terminal and section angle 'H' type supports

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ENA TS 43-50 Fig 5



ESI Drawing No. 435005

Fig. 5 — Scarfing and drilling — cable terminal structure auxiliary pole

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132 KV 'H' Cable Terminal

Based on Drg Nos ENA TS 43-50 sht 12

Item Description	Item Ref	Drawing No	Quantity	Cat No
Materials for Terminal Support leg				
Wood pole	435011/7	ENA TS 435004		To Suit
Foundation Components				
Bolt Ms Hex Galv M20 X 530 Long	435011/18 – 435011/20	ENA TS 43-96	8	372815
Square Curved Washer	435011/11	1 00 043 9605 Sht1	5	368078
Square Flat washers	435011/10	1091010183 sht 3 item 6	5	368063
Foundation Braces	435011/9	ENA TS 439545	2	TBA
Wood Block (3600 x 250 x 125mm)	435011/8	ENA TS 439112 (Type 2)	1	TBA
Pole, Insulator and Conductor Fittings				
Angle Crossarm	435011/1	ENA TS 439533	2	TBA
Short Type Straps	435011/12	ENA TS 439541	2	TBA
Long Type Straps	435011/13	ENA TS 439542	1	TBA
Cross bracing	435011/14	ENA TS 439543	4	TBA
Insulator Mounting Plate	435011/4	ENA TS 439540	1	TBA
Crossarm Support Channel	435011/5	ENA TS 439534	4	TBA
Crossarm Downlead Plate	435011/23	ENA TS 439544	3	TBA
120kn Ten. Ins. Assembly (132kV Approach)	435011/3	1 09 101 0487 Sht 31	6	251553
Poplar Comp. Dead End C/W Straight Lug	435011/3	1 09 101 0653 sht 3	6	243240
Bolt Ms Hex Galv M20 X 60 Long	435011/16	ENA TS 43-96	36	378703
Bolt Ms Hex Galv M24 X 60 Long	435011/17	ENA TS 43-96	4	TBA
Washer Taper - D	435011/22	ENA TS 43-96	9	TBA
Bolt Ms Hex Galv M20 X 60 Long	435011/24	ENA TS 43-96	6	378703
Earthing & Bonding				
Earthing Fastenings	435011/26	1 09 101 0162 Sht1	4	370504
70mm HD Green PVC Insulated earthing Conductor –	-	-	20m	220146
70mm compression Lug (m20 clearance hole)	-	1091010237	1	075234
Signs and Notices				
Pole Safety Sign	435011/6	1 09 101 0229 Sht1	2	363318
Pole Number Plate	435011/6	1 09 101 0228 sht 3	1	243258
Pole Numbers	435011/6	1 09 101 0228 sht 3	1	to suit)
Brass Slotted Head screws 1" No8	435011/6	-	4	375777
Fibre washers (Thin)	435011/6	-	4	374581
Fibre washers (Thick)	435011/6	-	4	374577
Enhanced Anti-Climbing Device		1 09 101 0408 Sht 8	1	251362

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Item Description	Item Ref	Drawing No	Quantity	Cat No
Stays (2 per leg)	435011/15			
190kN Heavy Duty Stay Straps (pair)	435011/15	1000439113	2	252647
Cranked Links (190kN) - 1091010469 sht 2	435011/15	BS 3288 ref 42/87	3	
Bolt Ms Hex Galv M24 X 60 Long Thread length 36mm with two round washers	435011/15	ENA TS 43-96	4	TBA
Bolt Ms Hex Galv M20 X 45 Long Thread length 36mm with two round washers	435011/15	ENA TS 43-96	2	378703
190KN Shackle	435011/15	BS 3288 ref 42/103	4	250938
190kN Yoke Plate	435011/15	BS 3288 ref 42/42	2	TBA
120kN Insulator Stay Assembly	435011/15	1 09 123 0372 Sht2	4	251547
Stay thimble	435011/15	1000439109 type 1	8	244922
7/4.00mm Grade 1150 Staywire	435011/15	-	20	231481
Stay Dead End	435011/15	0541/1	8	255217
Type 2 Stay Rod	435011/15	9101	4	231477
Wood Stay Block	435011/15	9103/2	4	235124
notes items 16, 17 & 18 to be supplied with two full nuts and a washer				
item 2 not required for download terminal				
items 23,24 and 25 not req for switchgear terminals				
item 15 - in addition to terminal back stays, side stays may be fitted to counterbalance downloads				
Cable Support Pole				
Cable Terminal Aux Pole Fabrication Drawing	435012/12	ENA TS 435005	2	TBA
Materials for Cable Support leg				
Crossarm - cable termination	435012/2	ENA TS 439546	4	TBA
Channel Supports	435012/3	ENA TS 439547	6	TBA
Sealing End Support	435012/4	ENA TS 439548	3 sets	TBA
Cable Cleat Support	435012/5	ENA TS 439549	3	TBA
Surge Diverter plate	435012/6	ENA TS 439550	3	TBA
Cable suport Crossarm (a)	435012/7	ENA TS 439551	1	TBA
Cable support crossarm (b)	435012/8	ENA TS 439552	3	TBA
Cable Guard	435012/9	ENA TS 439553	1	TBA
Cable backboard	435012/10	ENA TS 435014	1	TBA
H Pole Foundation Brace	435012/11	ENA TS 439545	2	TBA
Bolt Ms Hex Galv M12 X 60 Long	435012/13	ENA TS 43-96	6	TBA
Bolt Ms Hex Galv M16 X 50 Long	435012/14	ENA TS 43-96	54	TBA
Bolt Ms Hex Galv M20 X 375 Long	435012/15	ENA TS 43-96	14	TBA
Square Curved Washer	435012/16	1 00 043 9605 Sht1	3	TBA

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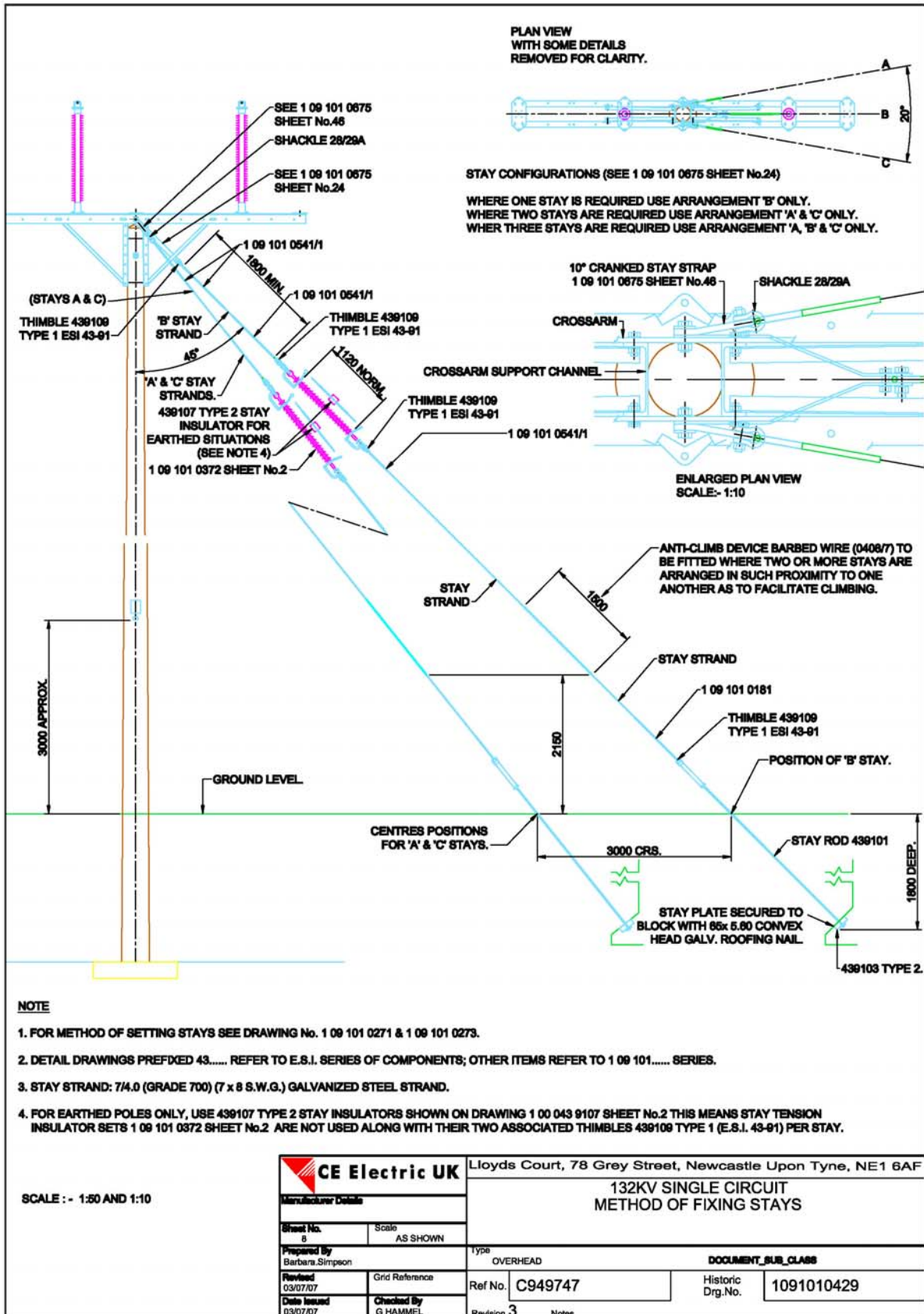
Item Description	Item Ref	Drawing No	Quantity	Cat No
132kV sealing End	435012/17		3	TBA
132kV Surge Diverter	435012/18		3	368078
Single cable Cleat	435012/19	1091010309 item 1	9	TBA
Trfoil cable cleat	435012/20	1091010309 item 5	4	TBA
Earth Materials				
Comp Lug – M22 Hole (70mm PVC Cu)		1091010237	1	075234
70mm (7/3.55)HD CU Cond. PVC Covered type 8		-	20m Approx.	220146
Polyprop cleats		-	As Req.	248815
Earthing Lead casing		1091190149/3	1	235548
Earth Rod		3143	As Req.	354249
Rod Coupling		3143	As Req.	354361
Driving Bolt		3143	As Req.	354408
Saddle Clamp		1091193143/2	As Req.	354338
Item 17 . 132kV sealing end to include arcing horns and earth end attachment plates				
Protection				
Anti Climbing Device - cable trunking	435012/21	1091010408 sht 11 item 7	2	243192
Anti Climbing Device - Cable Pole leg	435012/21	1090100408 sht 8	2	251362
Barbed Wire			50m	366829
Signs and Notices				
Pole Safety Sign		1 09 101 0229 Sht1	2	363318
Brass Slotted Head screws 1" No8		-	4	375777
Fibre washers (Thin)		-	4	374581
Fibre washers (Thick)		-	4	374577

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MATERIAL LIST FOR 1 09 101 0429 SHEET No.8 REV 1

IF ONE STAY IS USED THE FOLLOWING ITEMS ARE REQUIRED: -

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
1	253743	Single Stay Insulator Type 2 Earthed Poles	1 00 043 9107 Sht2 Or
1	346596	Stay Insulator Assembly Unearthed Poles	1 09 123 0372 Sht2
4	244922	Stay Thimble	1000439109 Type 1
4	255217	Helical Stay Dead End	1091010541
1	231477	Stay Rod	1000439101 Type 2 Item 3
3	337632	Bolt Ms Hex Galv M24 X 80 Long c/w full nut	
3	229195	Plain Washer Galv M24	
1	378703	Bolts MS Hex Galv M20 x 60 long	
1	375616	Washers, M20, round gald.	
Notes for more details See		STAY YOKE ARRAY (SEE DRAWING)	1 09 101 0675 SHT24
2	346483	Stay Yoke leg	1091010675 sht 16
1	235124	Stay Block 439103 type 2	1000439103 type 2

IF TWO STAYS ARE USED THE FOLLOWING ITEMS ARE REQUIRED: -

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
2	346503	10° Cranked Stay Strap	1091010675 Sht46
2	226365	Shackle BS3288 28/29a	1091010461 sht1
2	253743	Single Stay Insulator Type 2 Earthed poles	1 00 043 9107 Sht2 Or
1	346596	Stay Insulator Assembly Unearthed Poles	1 09 123 0372 Sht2
8	244922	Thimble	1000439109 Type 1
8	255217	Helical Stay Dead End	1091010541
2	231477	Stay Rod	1000439101 Type 2 Item 3
2	337632	Bolt Ms Hex Galv M24 X 80 Long c/w full nut	
2	229195	Plain Washer Galv M24	
1	235124	Stay Block 439103 type 2	1000439103 type 2

IF THREE STAYS ARE USED THE FOLLOWING ITEMS ARE REQUIRED: -

<u>QTY</u>	<u>CAT. No.</u>	<u>DESCRIPTION</u>	<u>CE DETAIL DRG.</u>
3	253743	Single Stay Insulator Type 2 Earthed Poles	1 00 043 9107 Sht2 Or
1	346596	Stay Insulator Assembly Unearthed Poles	1 09 123 0372 Sht2
2	346503	10° Cranked Stay Strap	1091010675 Sht46
2	226363	Shackle BS3288 28/29a	1091010461 sht1
2	346483	Stay Yoke leg	1091010675 sht 16
12	244922	Stay Thimble	1000439109 Type 1
12	255217	Helical Stay Dead End	1091010541
3	231477	Stay Rod	1000439101 Type 2 Item 3
3	337632	Bolt Ms Hex Galv M24 X 80 Long c/w full nut	
3	229195	Plain Washer Galv M24	
1	378703	Bolts MS Hex Galv M20 x 60 long	
1	375616	Washers, M20, round gald.	
1	235124	Stay Block 439103 type 2	1000439103 type 2

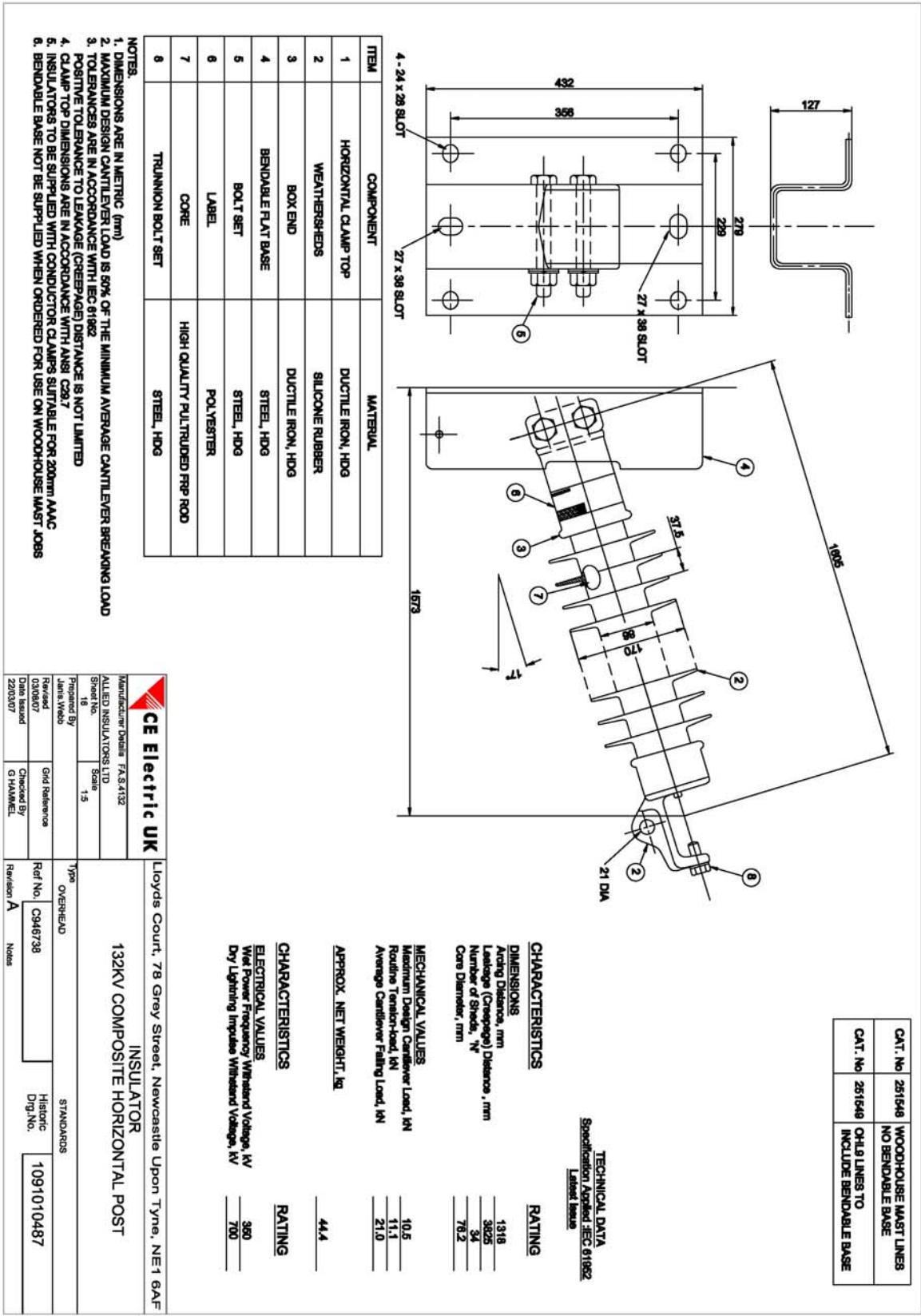
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Appendix D - Detailed Component Drawings

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ITEM	COMPONENT	MATERIAL
1	FLANGE END	DUCTILE IRON, HDG
2	WEATHERSHEDS	SILICONE RUBBER
3	FLANGE END	DUCTILE IRON, HDG
4	LABEL	POLYESTER
5	CORE	HIGH QUALITY PULTRUDED FRP ROD
6	VERTICAL CLAMP ADAPTOR (UNIDSET#2121)	DUCTILE IRON, HDG
7	TRUNNION BOLT SET	STEEL, HDG
8	BOLT SET (LE328-P1-1,2,3)	STAINLESS STEEL OR STEEL, HDG

CHARACTERISTICS

DIMENSIONS

Arching Distance, mm
Leakage (Creepage) Distance, mm
Number of Speds, 'N'
Core Diameter, mm

MECHANICAL VALUES

Maximum Design Cantilever Load, kN (Without Vertical Clamp Adaptor)
Routine Tension Load, kN (Without Vertical Clamp Adaptor)
Average Cantilever Felling Load, kN (Without Vertical Clamp Adaptor)

APPROX. NET WEIGHT, kg

CHARACTERISTICS

ELECTRICAL VALUES

Wet Power Frequency Withstand Voltage, kV
Dry Lightning Impulse Withstand Voltage, kV

RATING

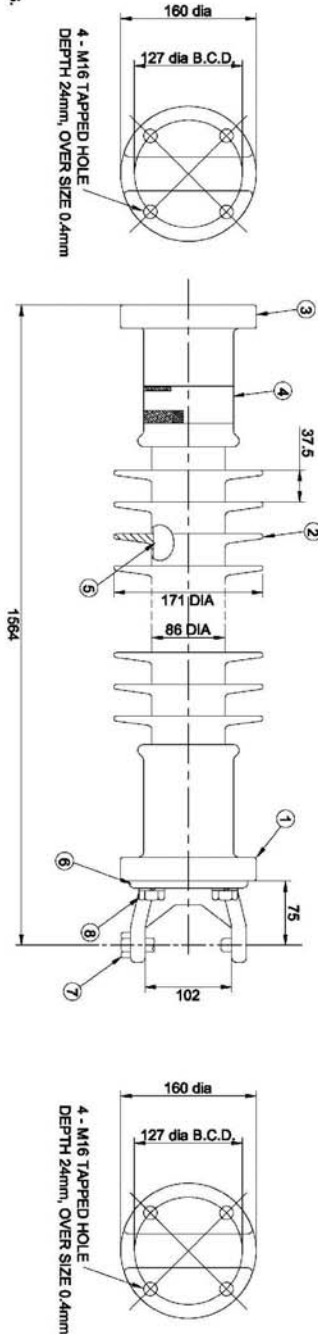
1168
3375
30
76.2

10.5
33.4
21.0

36.74

RATING

350
650



NOTES.

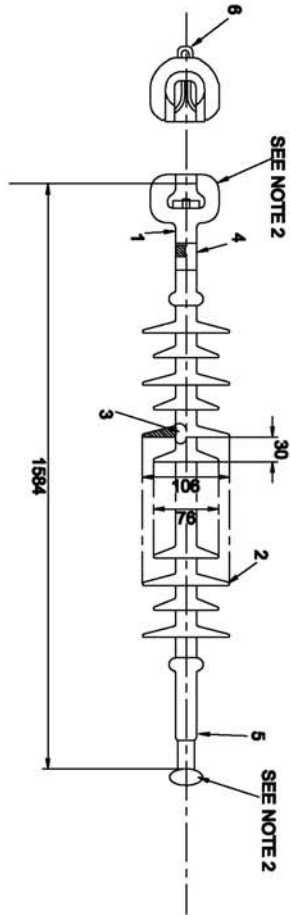
- DIMENSIONS ARE IN METRIC (mm)
 - MAXIMUM DESIGN CANTILEVER LOAD IS 50% OF THE MINIMUM AVERAGE CANTILEVER BREAKING LOAD
 - TOLERANCES ARE IN ACCORDANCE WITH IEC 61952
POSITIVE TOLERANCE TO LEAKAGE (CREEPAGE) DISTANCE IS NOT LIMITED
 - ELECTRICAL TEST : IEC 61952 - 2002
 - BOTTOM FLANGE SHOULD BE INSTALLED BY 4 BOLTS OF THE FOLLOWING MATERIAL IN ORDER TO WITHSTAND AVERAGE CANTILEVER FALLING LOAD
 - WHERE USED AS A PILOT INSULATOR THIS UNIT MUST BE USED WITH A SUPPORT STOOL SEE DRAWING 1091010487 SHEET 34
 - TO BE SUPPLIED COMPLETE WITH CONDUCTOR TRUNNION CLAMPS SUITABLE FOR 200mm AAC
- MATERIAL : SAE GRADE 9 (ISO 898 - 1:1999, CLASS 12.9) OR EQUIVALENT, GALVANISED

CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details F.S.4134		132KV COMPOSITE VERTICAL POST / PILOT INSULATOR (127MM PCD PEDASTEL BASE)	
ALLIED INSULATORS LTD			
Sheet No. 18	Scale 1:5		
Designed By James Webb		Type OVERHEAD	
Reviewed 03/08/07	Grid Reference	Ref No. C946788	
Date Issued 23/03/07	Checked By G HAMMILL	Revision B Notes	
		STANDARDS	
		Historic Dwg No. 1091010487	

TECHNICAL DATA
Specification Applied : IEC 61952
Latest Issue

CAT. No 251551 SUPPLIED WITHOUT STOOL
CAT. No 251550 SUPPLIED WITH SUPPORT STOOL
AND ARMOUR GRIP CLAMPS

Document reference		NSP/004/045			
Version:-	2.0	Date of Issue:-	Sept 2009	Page	80 of 117



ITEM	COMPONENTS	MATERIAL	REQD.
1	SOCKET FITTING	HIGH GRADE F.S. OR D.I. HDG	1
2	WEATHERSHIELDS	SILICONE RUBBER	1
3	CORE	HIGH QUALITY PULTRUDED FRP ROD	1
4	LABEL	POLYESTER	1
5	BALL FITTING	HIGH GRADE FORGED STEEL, HDG	1
6	SPLIT PIN	STAINLESS STEEL	1

TECHNICAL DATA

SPECIFICATION APPLIED : IEC81109, LATEST ISSUE

CHARACTERISTICS

DIMENSIONS

ARROW DISTANCE mm
LEAKAGE (CREEPAGE) DISTANCE mm
NUMBER OF SHEDS 'N'
CORE DIAMETER mm

MECHANICAL VALUES

SPECIFIED MECHANICAL LOAD IN
ROUTINE TENSION LOAD IN

APPROX. NET WEIGHT. kg

ELECTRICAL VALUES

WET POWER FREQUENCY WITHSTAND VOLTAGE IN
DRY LIGHTNING IMPULSE WITHSTAND VOLTAGE IN

RATING

1348
3702
43
17

125
63

6.0

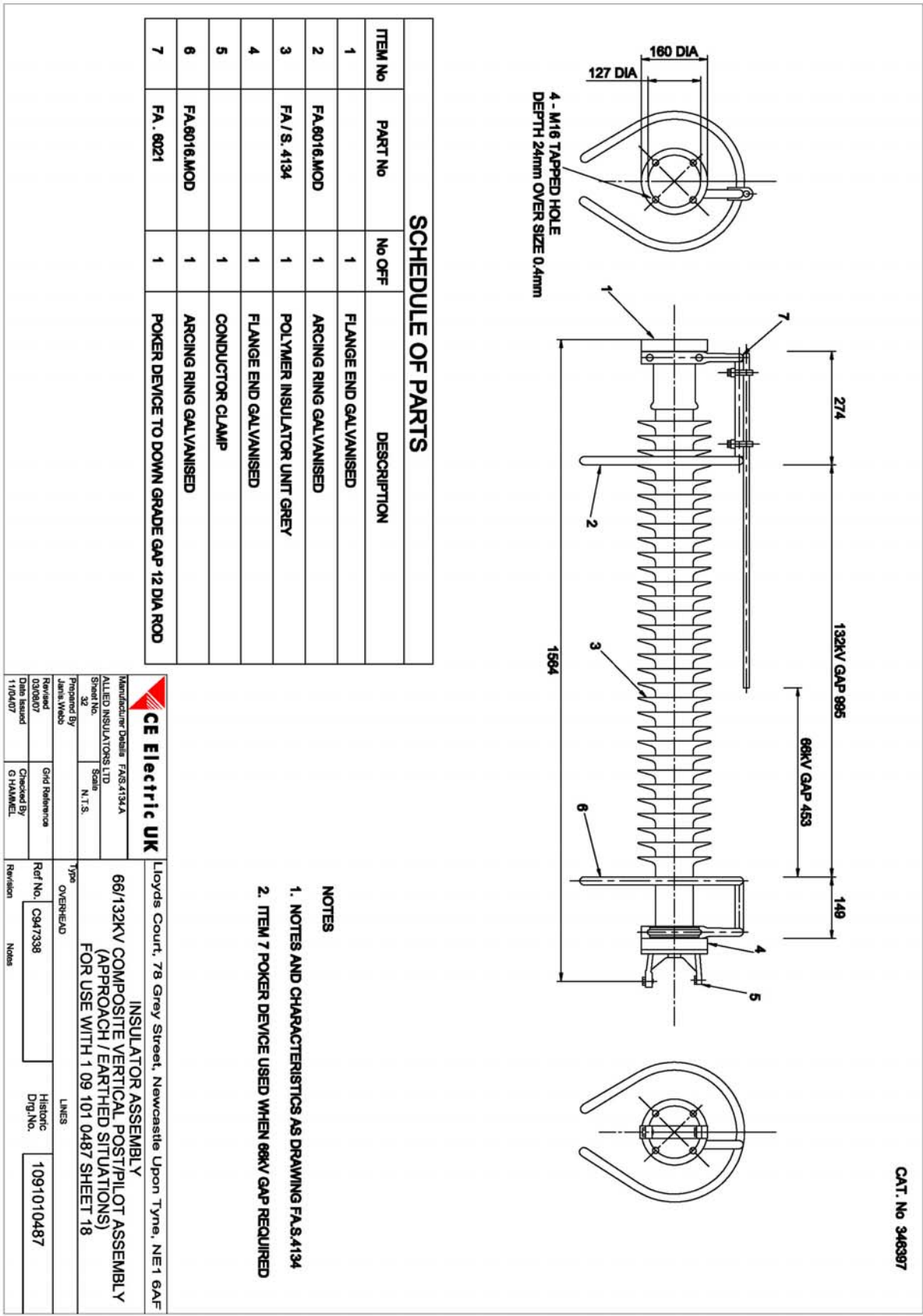
425
765

NOTES

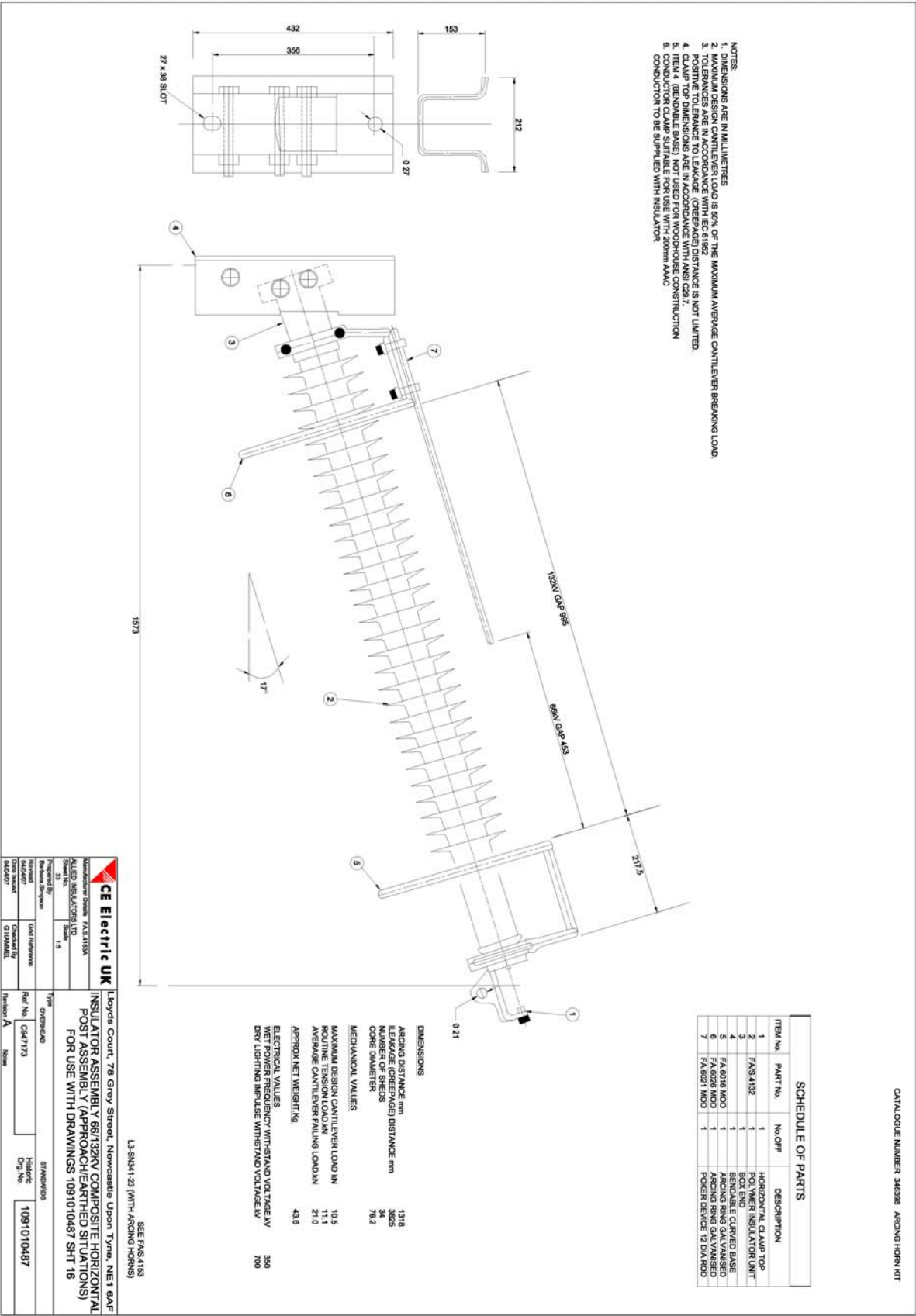
1. DIMENSIONS ARE IN MILLIMETRES
 2. SOCKET AND BALL COUPLINGS CONFORM TO GAUGES FOR IEC PUB80120 / 20
 3. TOLERANCES ARE IN ACCORDANCE WITH IEC 81109, LATEST ISSUE.
- POSITIVE TOLERANCE TO LEAKAGE (CREEPAGE) DISTANCE IS NOT LIMITED

		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details FAS-4117		INSULATOR	
ALLIED INSULATORS LTD		(120KN WITH 20MM BALL / SOCKET COUPLING)	
Sheet No.	Scale	Type OVERHEAD	
29	1:5	Ref No. CM7326	
Proposed By	Grid Reference	Revision	
JAMES WILKES		Notes	
Reviewed	Checked By	Historic	
02/06/07	G HANMEL	Dwg No. 1091010487	
Date Issued			
11/04/07			

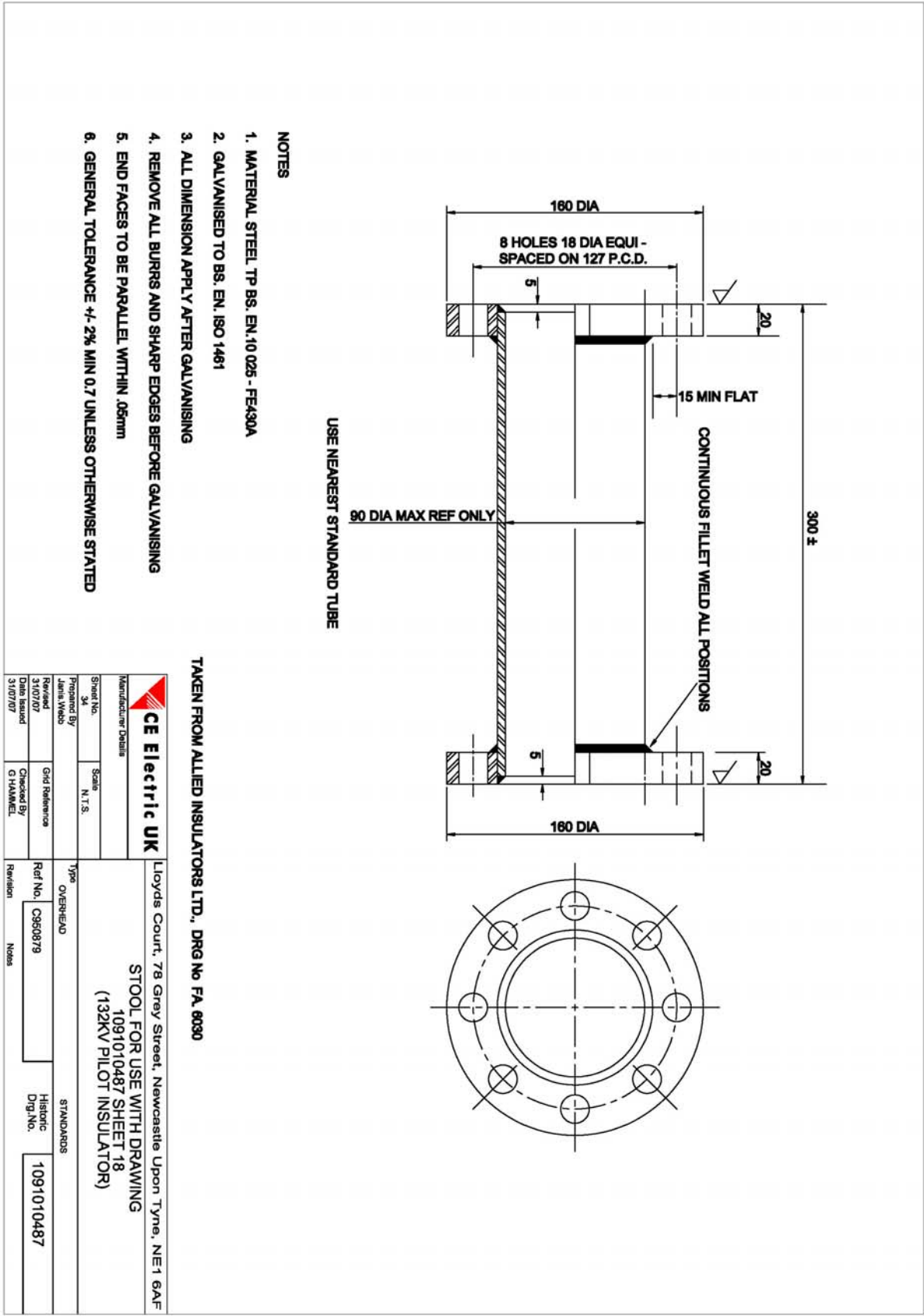
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Version:-	2.0	Date of Issue:-	Sept 2009	Page	81 of 117



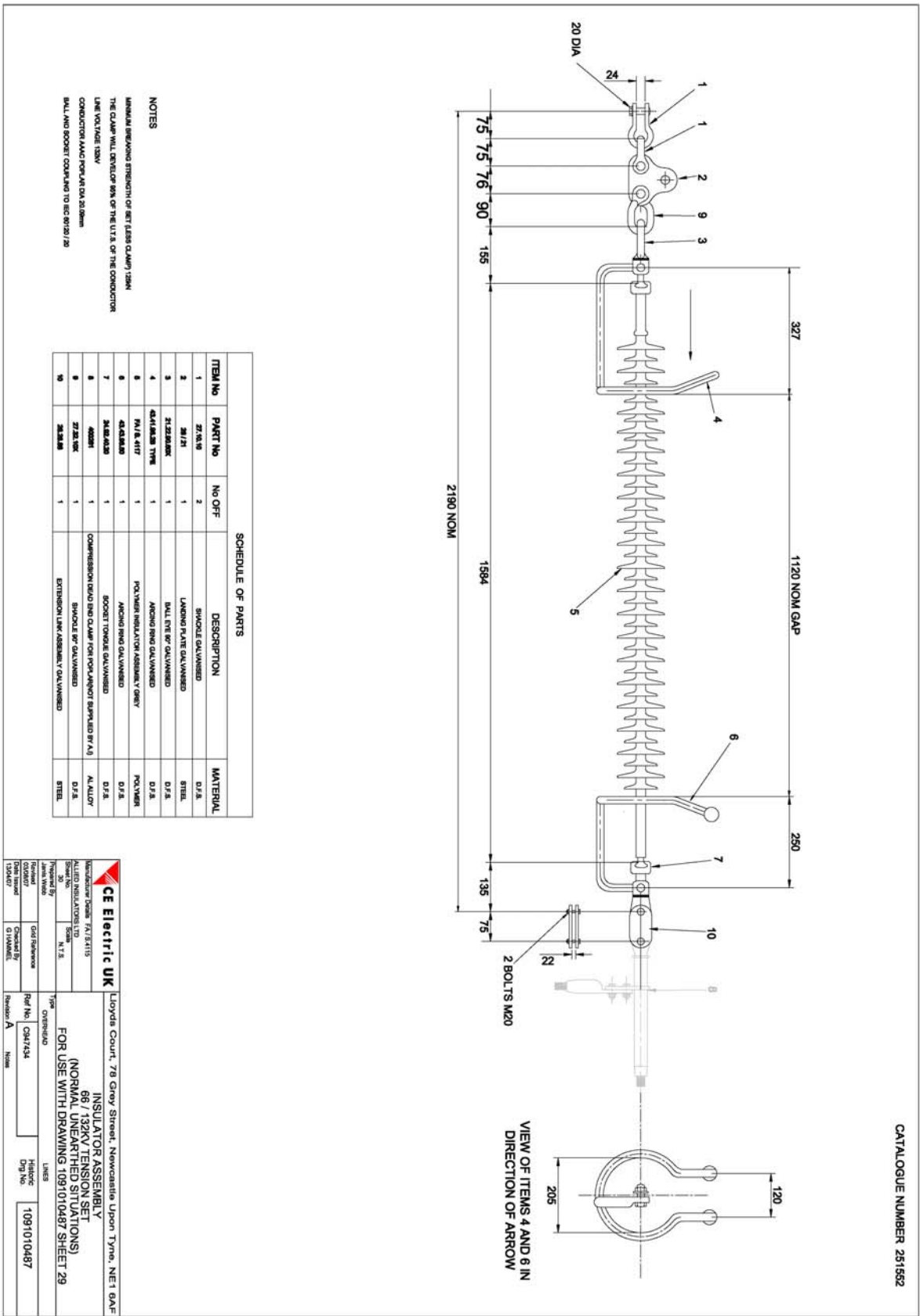
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Version:-	2.0	Date of Issue:-	Sept 2009	Page 82 of 117



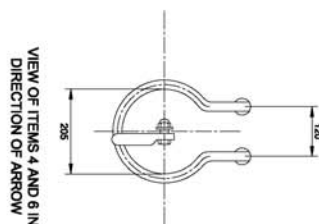
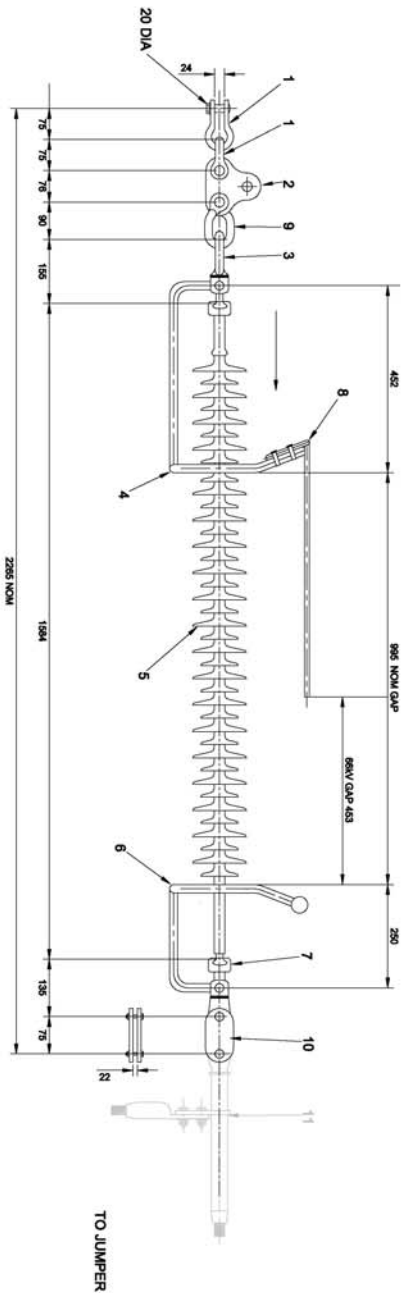
Document reference	NSP/004/045				
Version:-	2.0	Date of Issue:-	Sept 2009	Page	83 of 117



Document reference	NSP/004/045				
Version:-	2.0	Date of Issue:-	Sept 2009	Page	84 of 117



Document reference		NSP/004/045					
Version:-	2.0	Date of Issue:-	Sept 2009	Page	85	of	117



CATALOGUE No. 251553 132KV OPERATION
251554 66/33KV OPERATION

SCHEDULE OF PARTS				
ITEM NO	PART NO	NO OFF	DESCRIPTION	MATERIAL
1	27-15-50	2	SHOCK GALVANIZED	D.I.S
2	28-17	1	LANDING PLATE GALVANIZED	STEEL
3	25-25-20-20-20	1	BALL JOINT Ø7 GALVANIZED	D.I.S
4	42-61-25-20-20	1	ANCHOR RING GALVANIZED	D.I.S
5	PA 1.6-1177	1	POLYMER INSULATION ASSEMBLY Ø600	POLYMER
6	24-16-20-20	1	ANCHOR RING GALVANIZED	D.I.S
7	24-16-20-20	1	ROCKET THROAT GALVANIZED	D.I.S
8	16-20-20	1	POWDER COATING FOR NEW CAP - 400	12 GA STEEL WOOD
9	27-15-100	1	SHOCK Ø7 GALVANIZED	D.I.S
10	25-25-20-20	1	EXTENSION LINK ASSEMBLY GALVANIZED	STEEL
11	42-61-20	1	COMPRESSION RING END FOR POOL (NOT SUPPLIED BY A.I.)	ALUMINUM

NOTES

THE M 10 POWER DEVICE USED WHEN 60KV GAIN IS REQUIRED
MINIMUM BREAKING STRENGTH OF AIR (LESS CLAMP) 150N
THE CLAMP WILL DEVELOP 50% OF THE U.T.S. OF THE CONDUCTOR
LINE VOLTAGE 120KV
CONDUCTOR AAC PDRUM DIA 20.0mm
BALL AND SPOCKET COUPLING TO EC 60150/20

CE Electric UK	
Manufacturer's Design Ref. / Kitemark	
Material	Stainless Steel
Size	N/A
Prepared by	
Checked by	
Drawn by	
DATE	01/01/2018
Checked by	
Drawn by	
DATE	01/01/2018
Lloyd's Court, 78 Gray Street, Newcastle upon Tyne, NE1 5AA INSULATOR ASSEMBLY 66 / 132KV TENSION SET (APPROX / EARTHED SITUATIONS) FOR USE WITH DRAWINGS 1.09.10.0487 SHEET 29	
Type	Overhead
Access	
Ref No.	018774
Drawn by	
Checked by	
DATE	01/01/2018
Ref No.	109.10.0487
Drawn by	
Checked by	
DATE	01/01/2018

Document reference		NSP/004/045			
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CAT. NO. 251547

ITEM No	PART No	No OFF	DESCRIPTION	MATERIAL
1	27.32.10	1	SHACKLE 90 DEGREE GALVANISED	D.F.S.
2	21.22.90	1	BALL EYE 90 DEGREE GALVANISED	D.F.S.
3	FA/S.4117	1	POLYMER INSULATOR ASSEMBLY GREY	POLYMER
4	24.91.40	1	SOCKET TONGUE GALVANISED	D.F.S.
5	43.41.9B.2A	1	ARCING RING GALVANISED	D.F.S.
6	43.41.9B.50	1	ARCING RING GALVANISED	D.F.S.
7	27.C6.10	1	GALVANISED SHACKLE REF 28/29A	D.F.S.

CE Electric UK

Manufacturer Details

Street No. 2

Scale N.T.S.

Revised By J.M.MODDY

Grid Reference

Ref No. C318321

Historic Dwg No. 1091010372

Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF

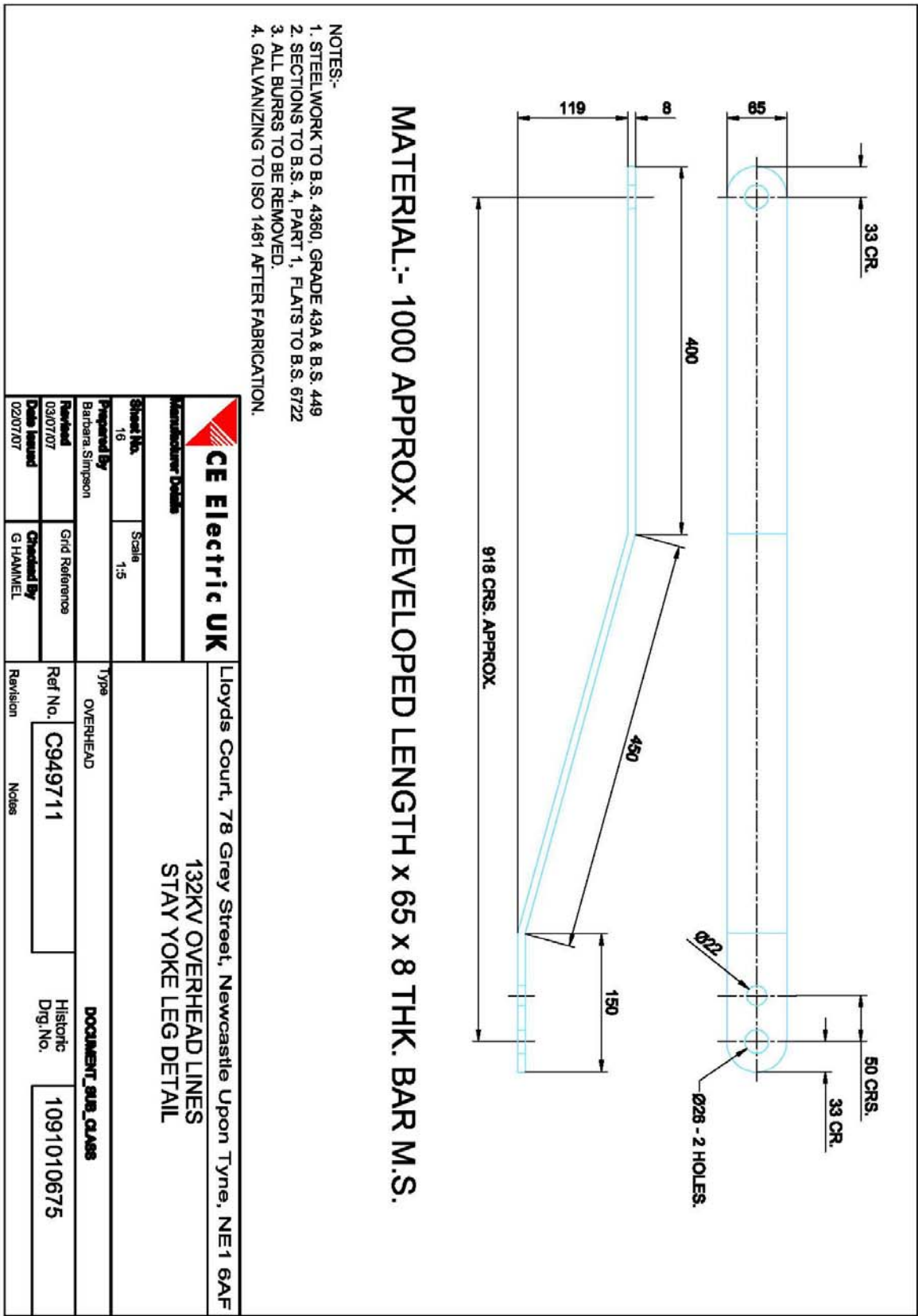
INSULATOR STAY ASSEMBLY (UNEARTHED ONLY) (120kV AND 132kV COMPOSITE)

NOTES

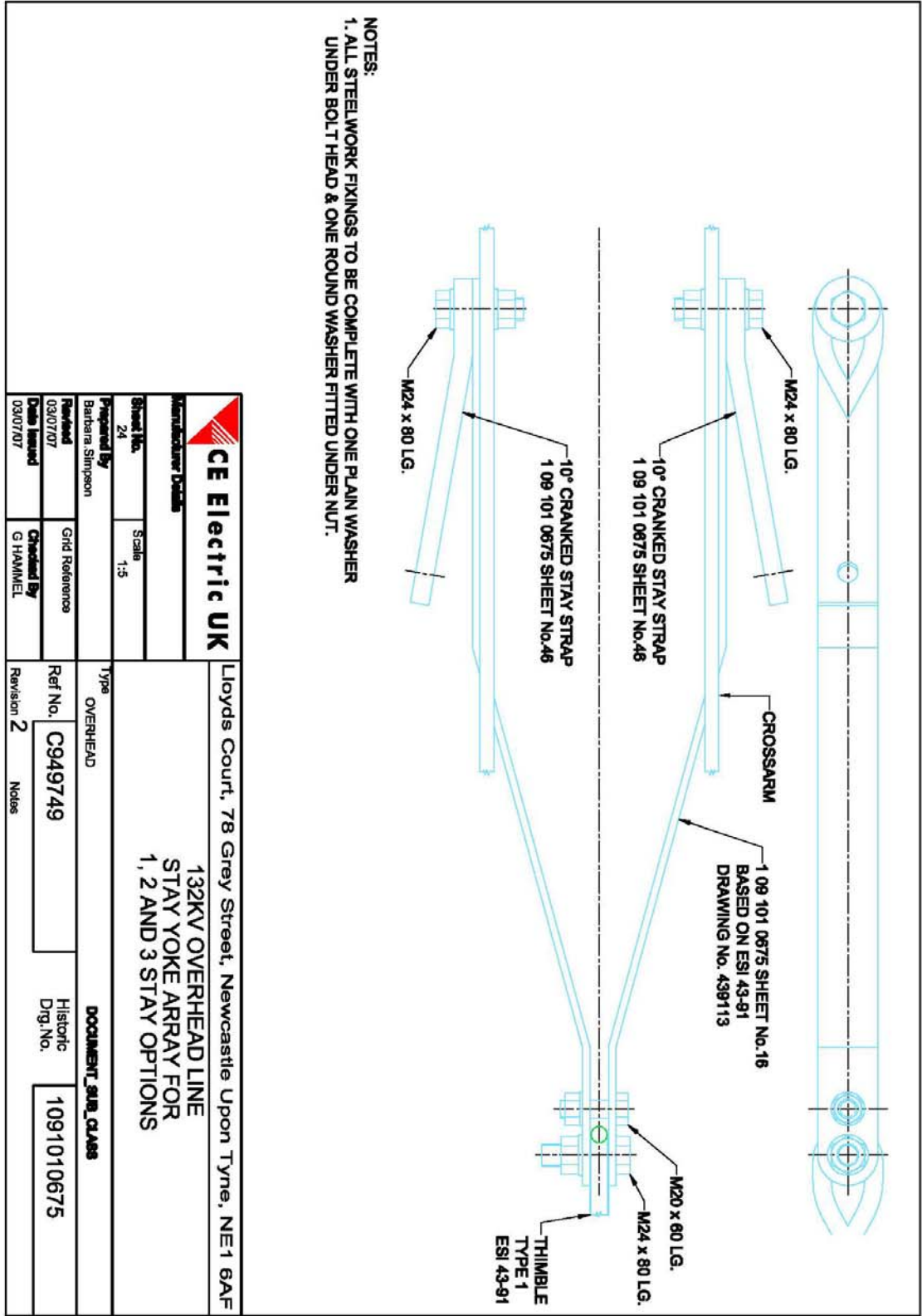
BALL AND SOCKET COUPLING TO IEC 60120 / 20

CAUTION! - This document may be out of date if printed

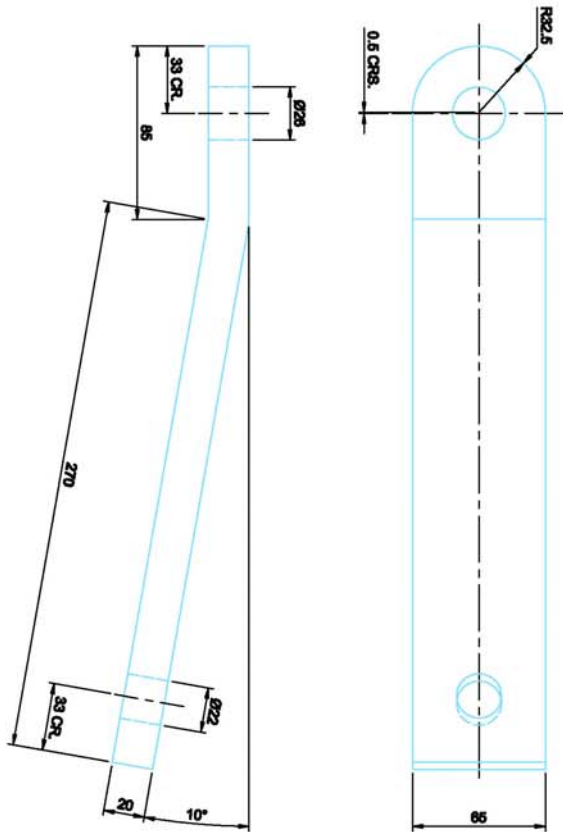
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Document reference		NSP/004/045			
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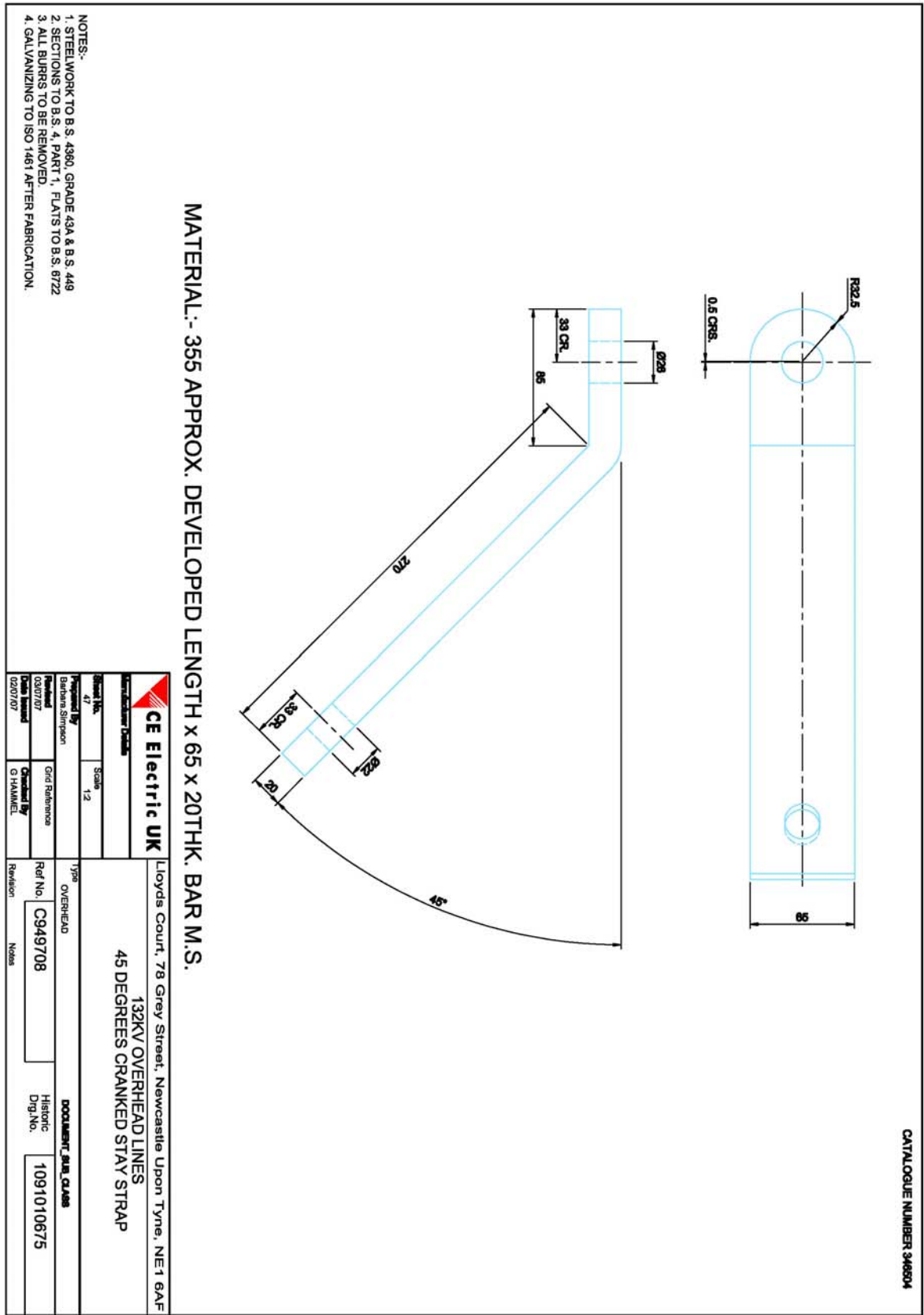
MATERIAL:- 355 APPROX. DEVELOPED LENGTH X 65 X 20THK. BAR M.S.

NOTES:-

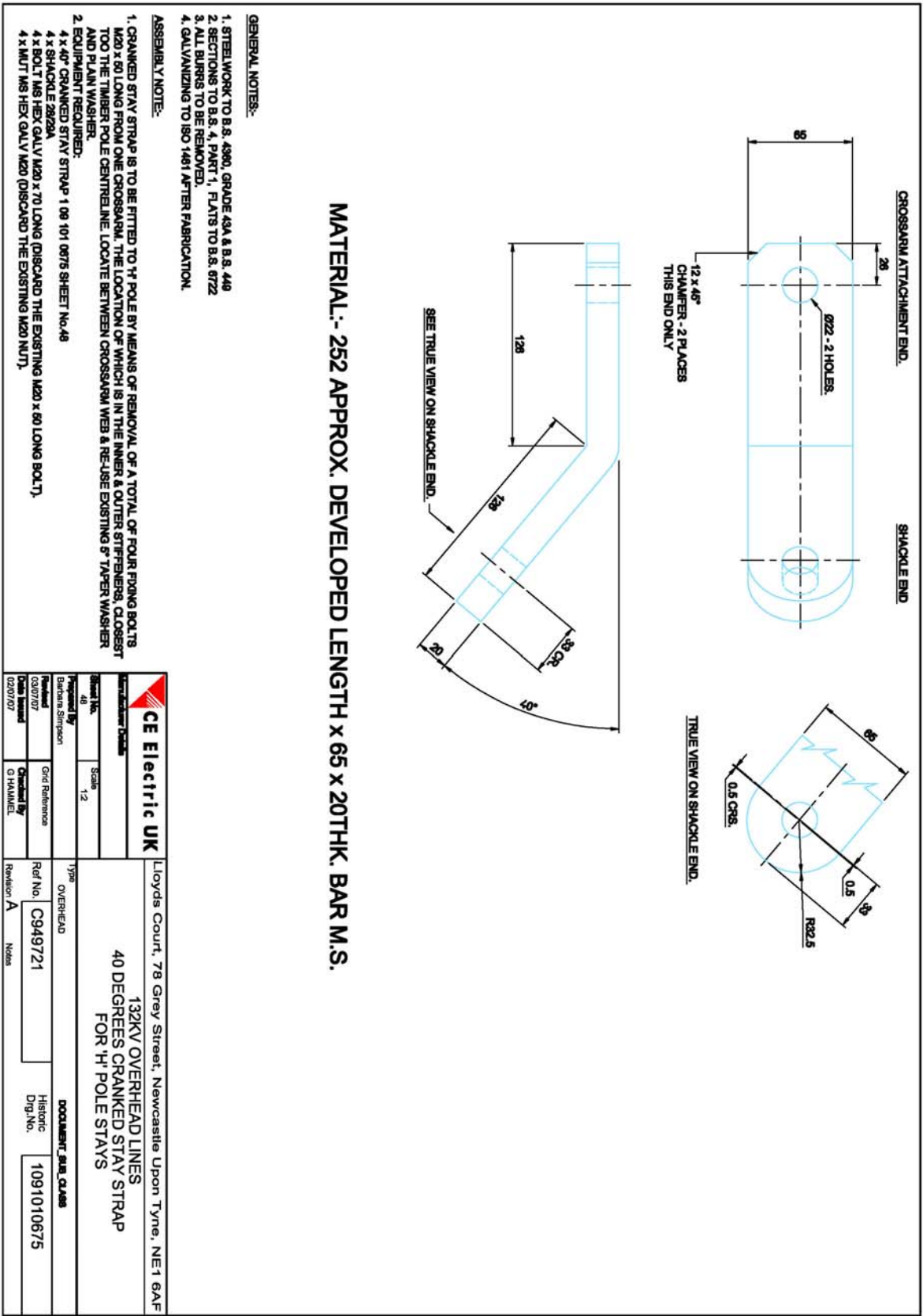
1. STEELWORK TO B.S. 4360, GRADE 43A & B.S. 449
2. SECTIONS TO B.S. 4, PART 1, FLATS TO B.S. 6722
3. ALL BURRS TO BE REMOVED.
4. GALVANIZING TO ISO 1461 AFTER FABRICATION.

 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AR	
Identification Details		132KV OVERHEAD LINES 10 DEGREES CRANKED STAY STRAP	
Serial No. 48	Serial 112		
Requested By Babbar Shrivastava		type OVERHEAD	
Reviewed 03/07/07	Grid Reference	Ref No. C949710	Historic Drg No. 1091010675
Date Issued 02/07/07	Created By G HAMMEL	Revision Notes	

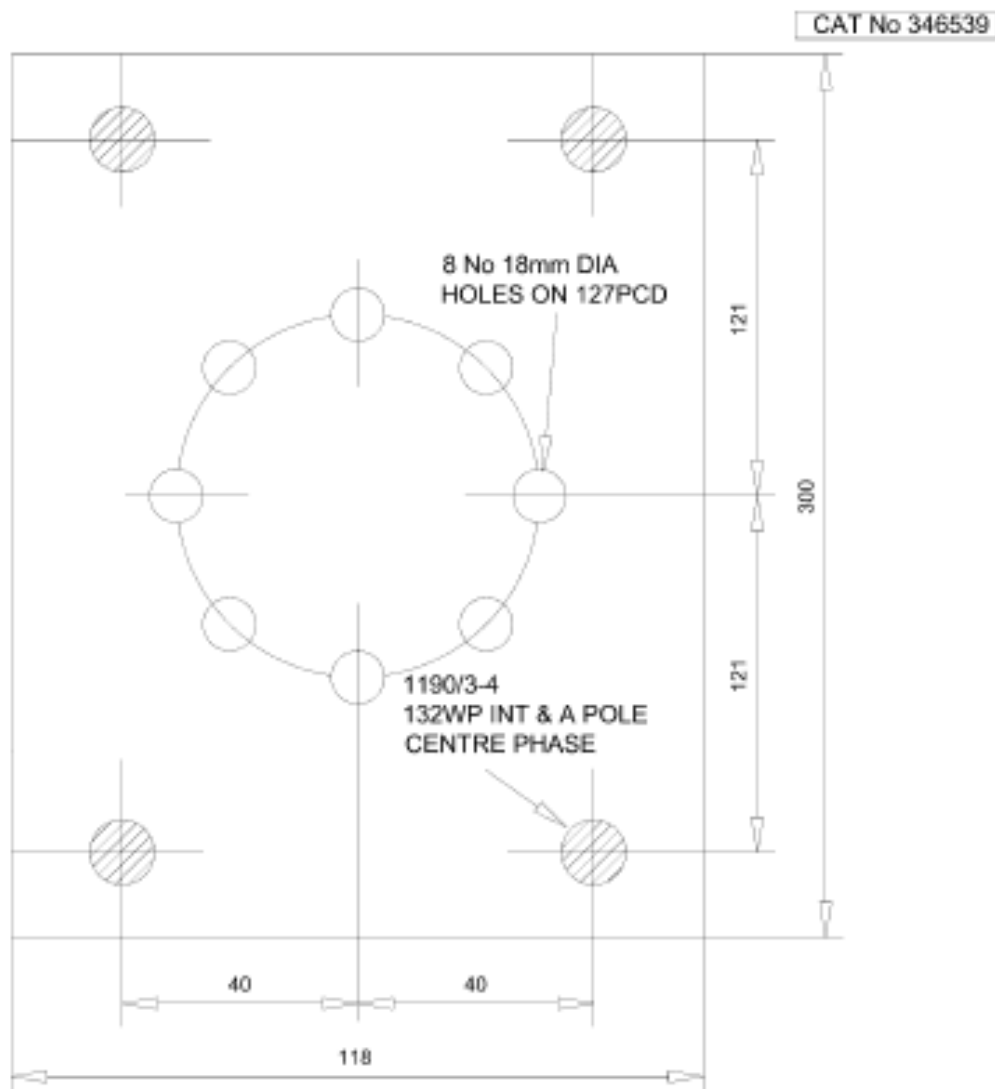
Document reference		NSP/004/045			
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Document reference	NSP/004/045				
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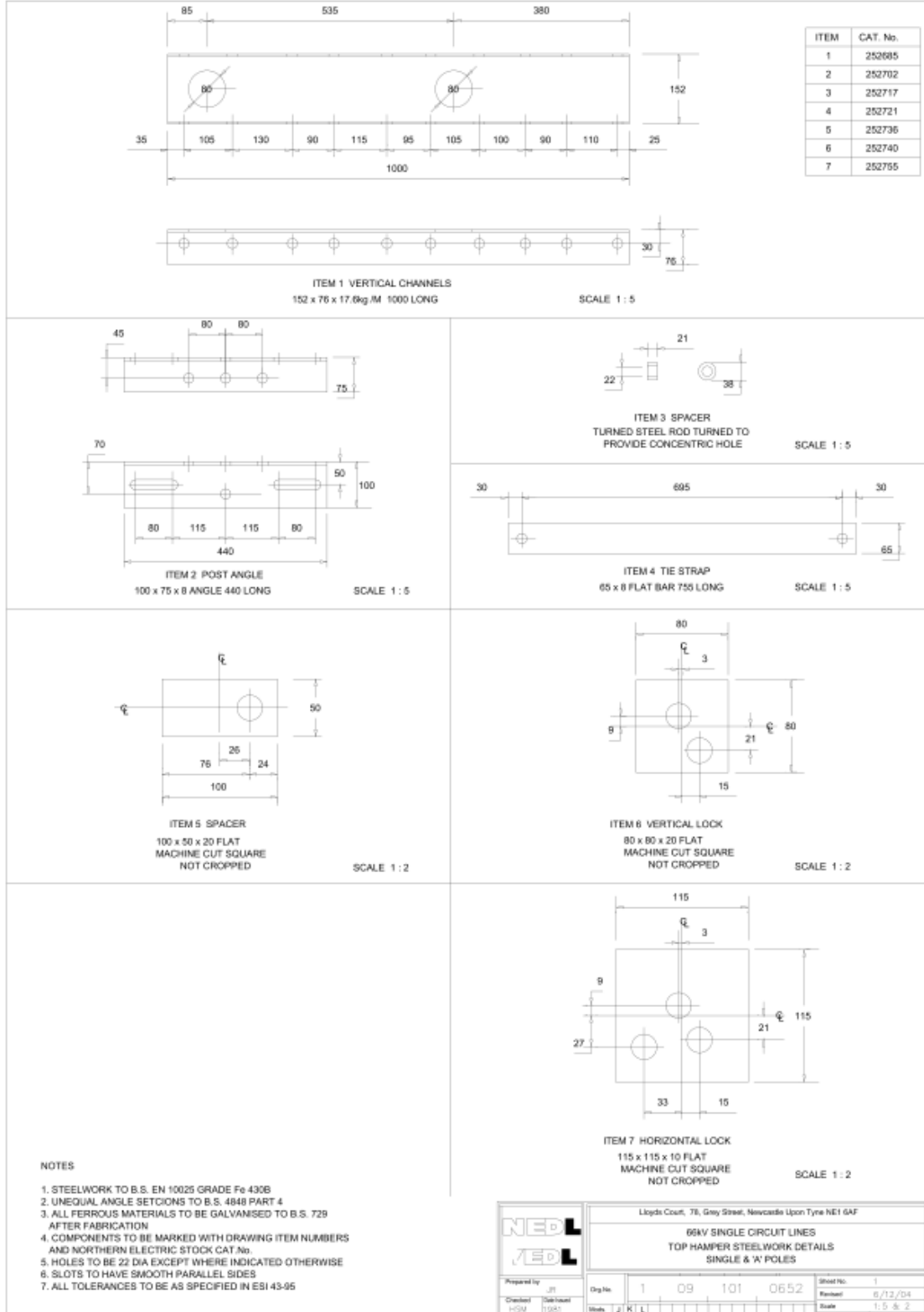
ITEM 6 300 x 235 x 15 STEEL PLATE

NOTES:-

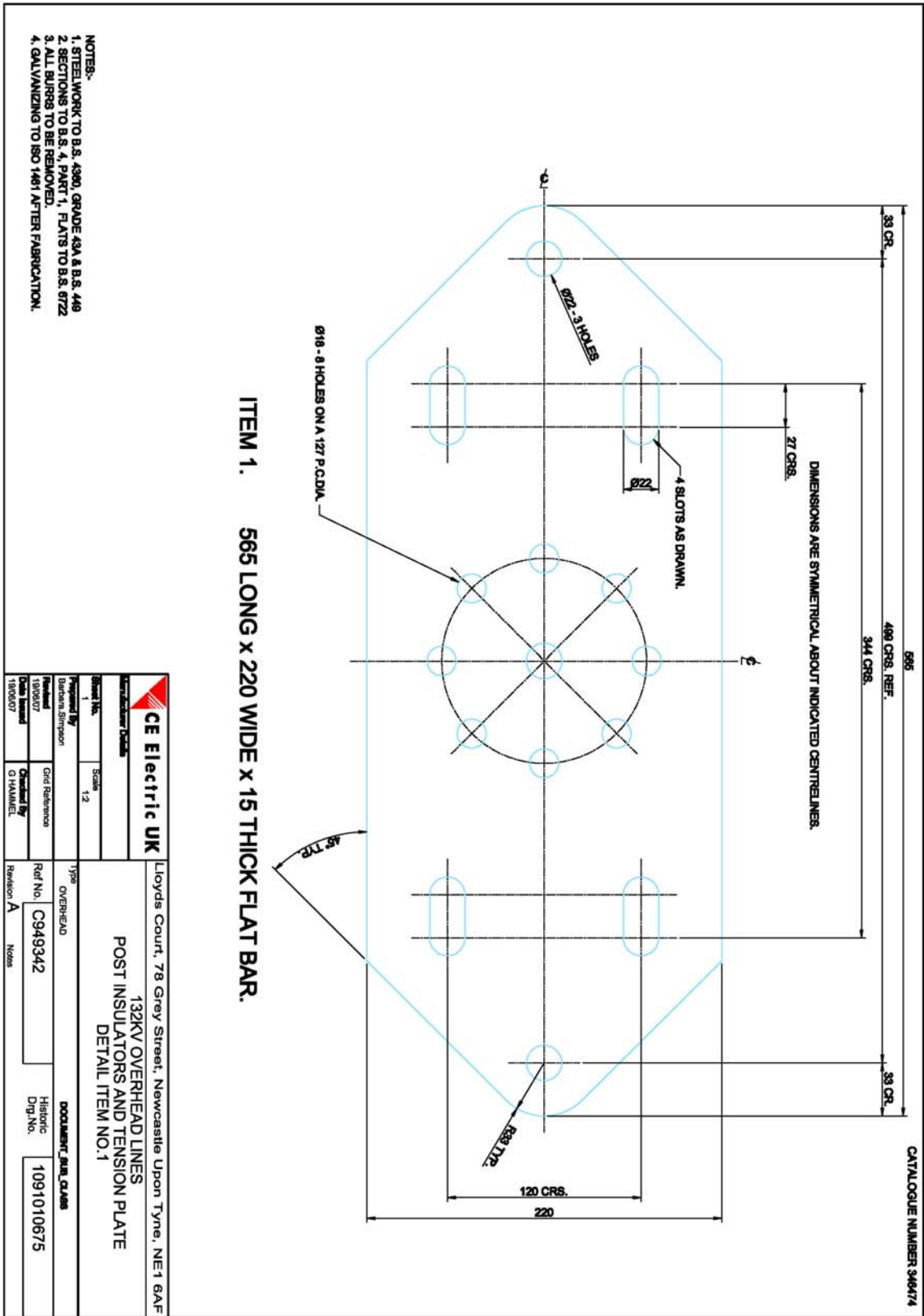
1. STEELWORK TO B.S. 4360, GRADE 43A & B.S. 449
2. SECTIONS TO B.S. 4, PART 1, FLATS TO B.S. 6722
3. GALVANISING TO B.S. 729 AFTER FABRICATION
4. HOLES ARE 22mm DIA. EXCEPT HOLES MARKED OTHERWISE
5. 22mm DIA. SLOTS TO HAVE STRAIGHT PARALLEL SIDES.

		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
		132KV SINGLE CIRCUIT OVERHEAD LINES POLYMERIC INSULATOR - POST INSULATOR BASE PLATE ITEM 6	
Manufacturer Details			
Sheet No. 6	Scale 1:2		
Prepared By Barbara Gordon	Type OVERHEAD	LINES	
Revised 12/08/08	Grid Reference	Ref No. C966011	Historic Org.No. 1091010670
Date Issued 12/08/08	Checked By G HAMMEL	Revision B	Notes

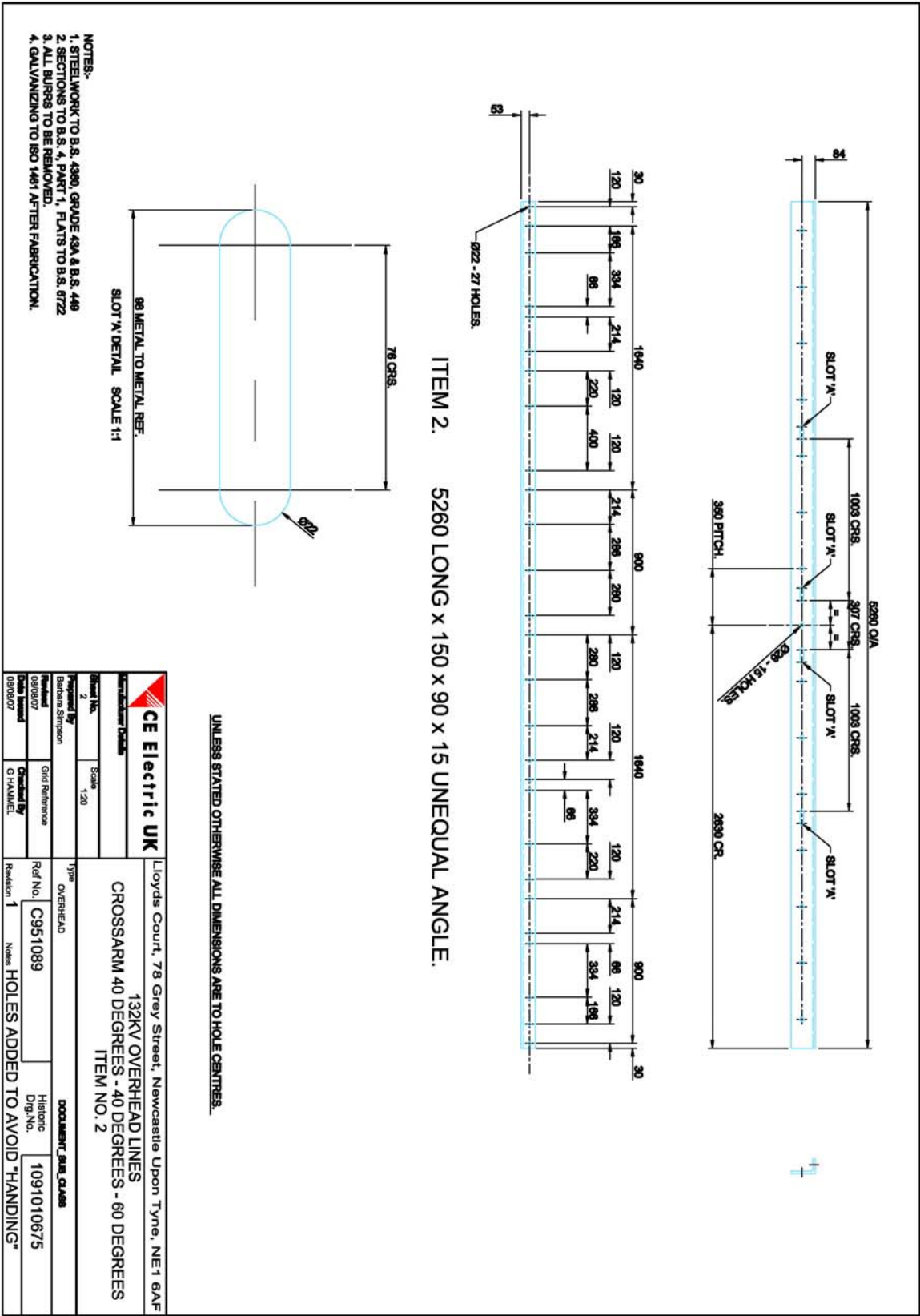
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Document reference	NSP/004/045			
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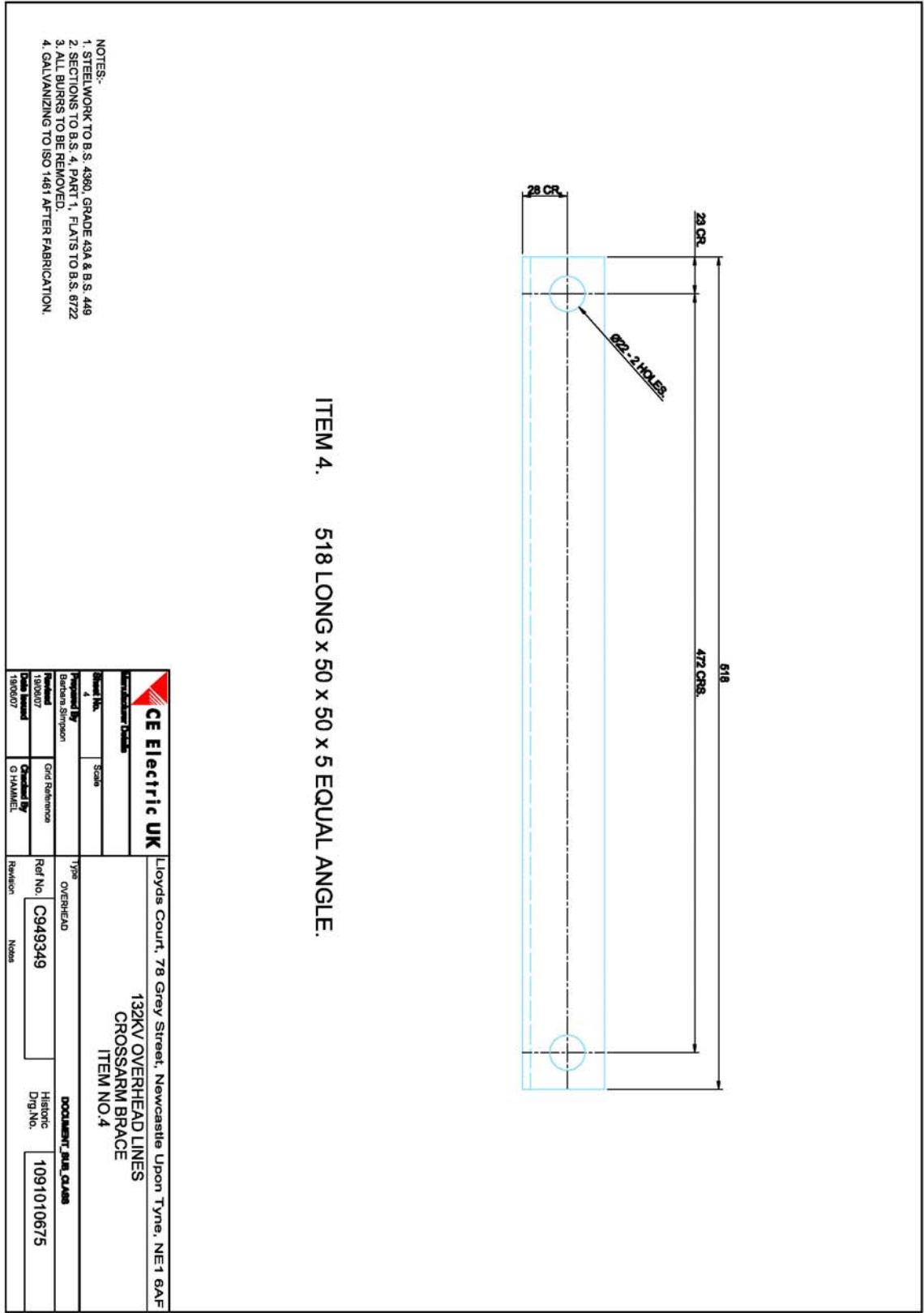
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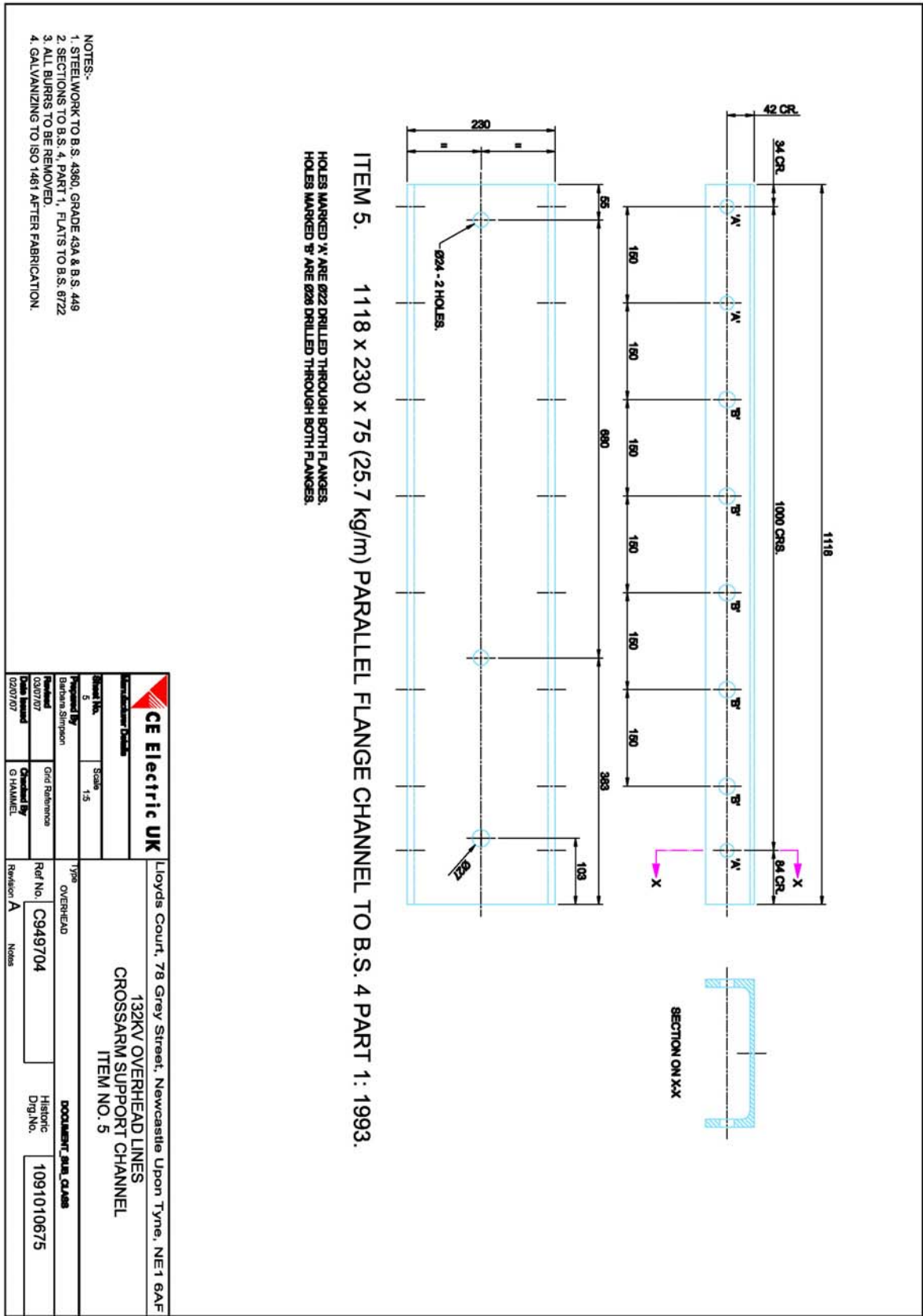
CE Electric UK

Project Name	Scale	Type	Document Ref. Class
2	1:20	OVERHEAD	132KV OVERHEAD LINES
Barbara Simpson	Grid Reference	Ref No.	C951089
080807	Checked By	Historic	Dwg No. 1091010675
080807	G HAMMILL	Revision	1
Holes Added to Avoid "Handing"			

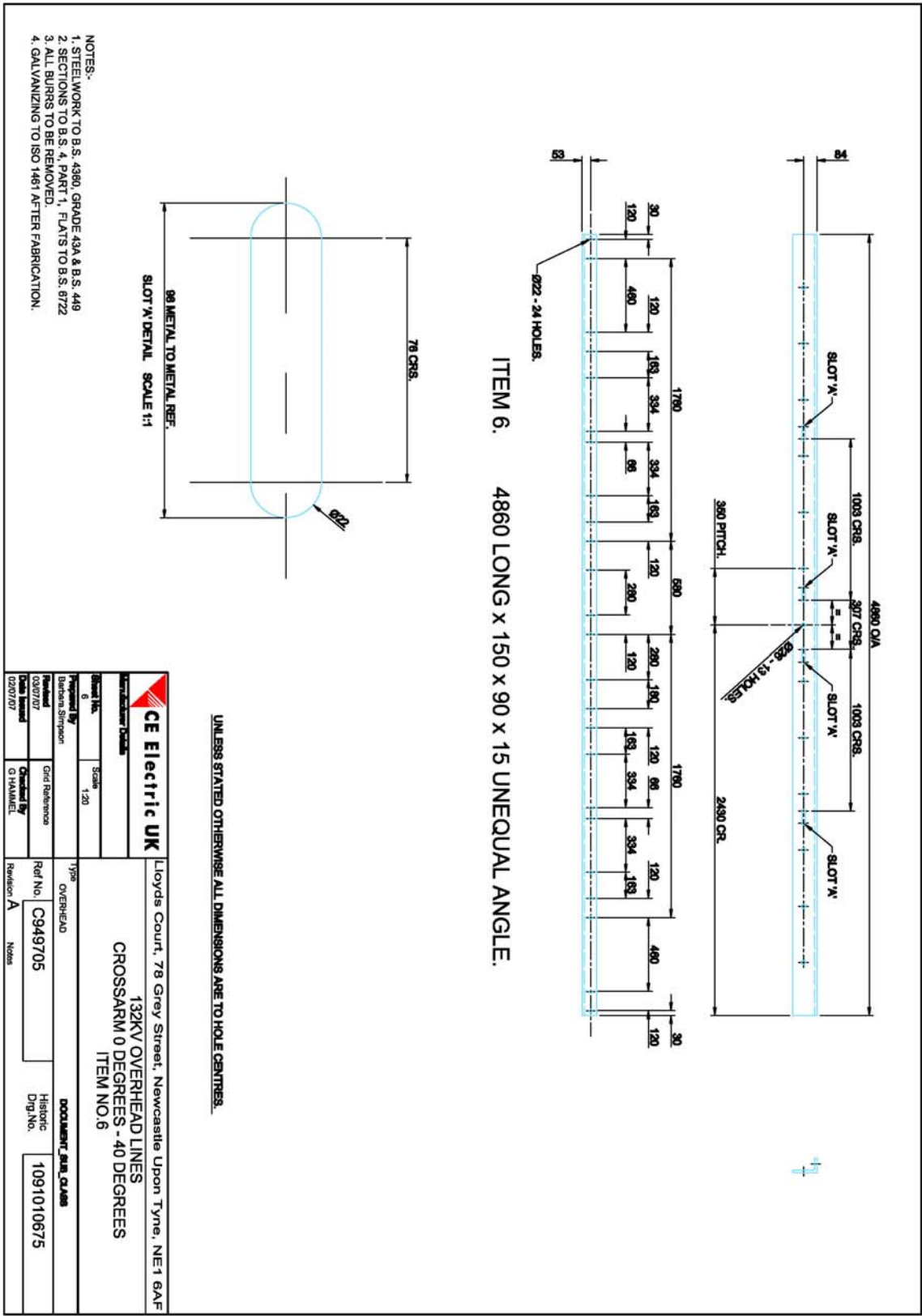
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Version:-	2.0	Date of Issue:-	Sept 2009	Page	96 of 117



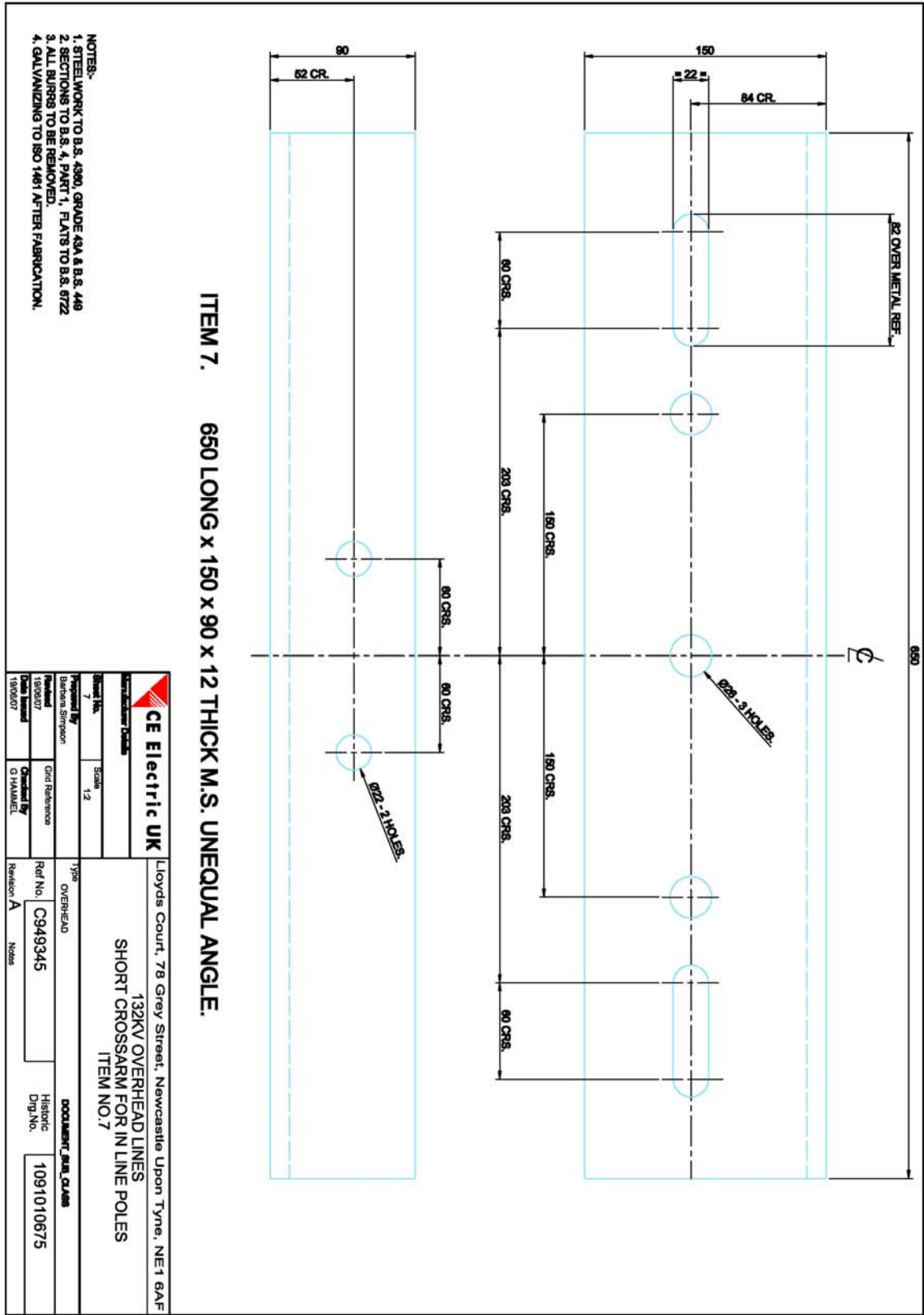
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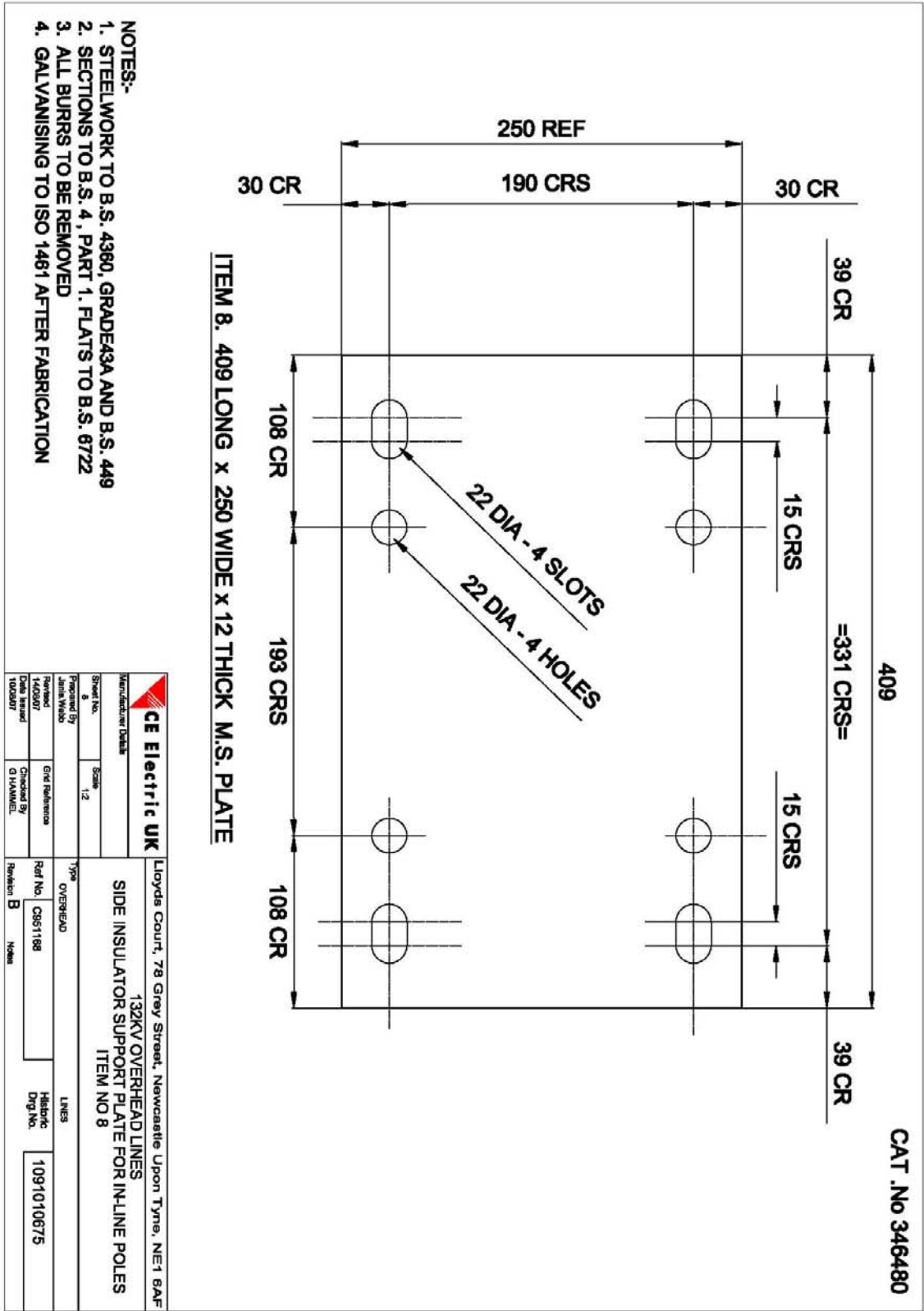
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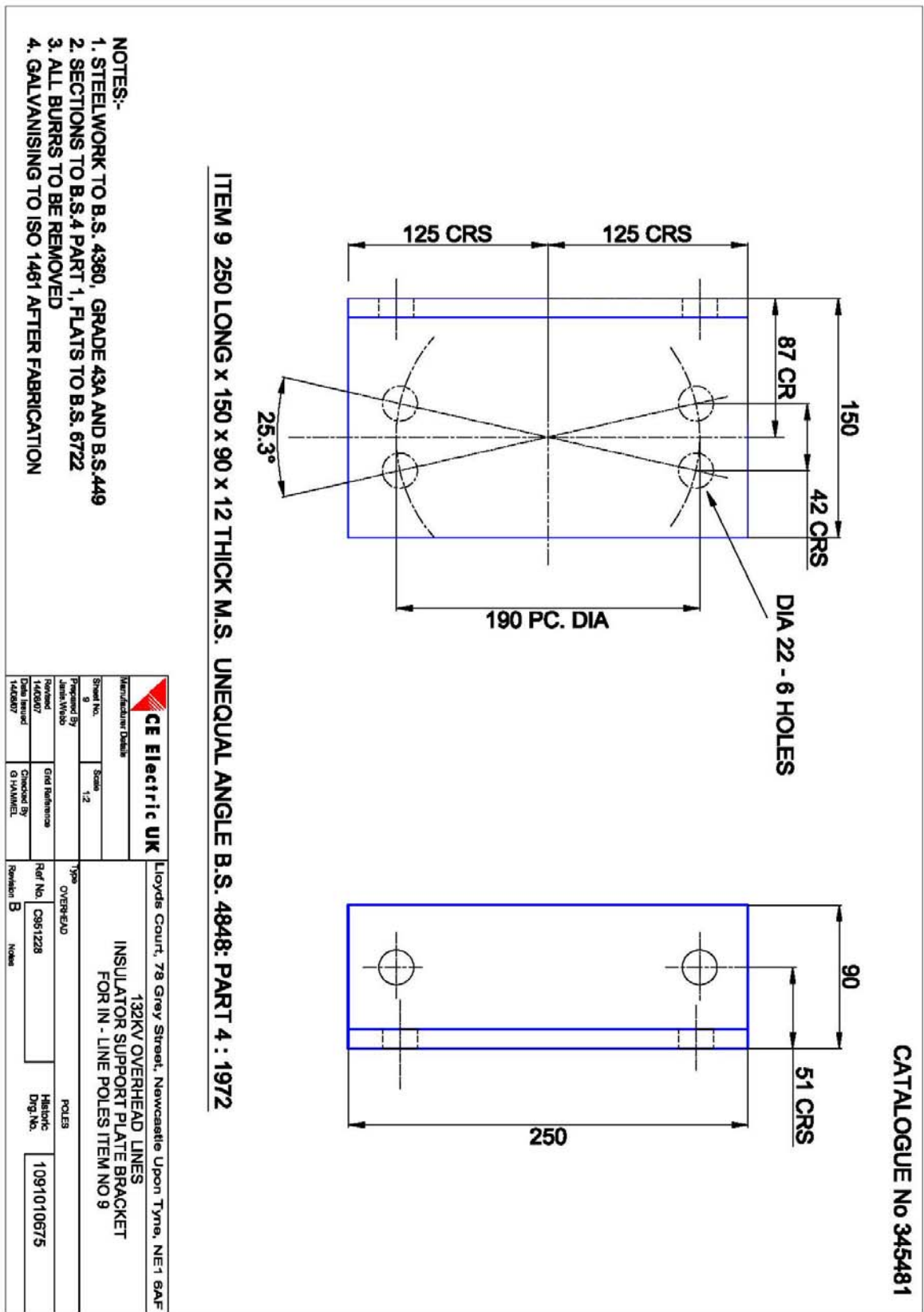
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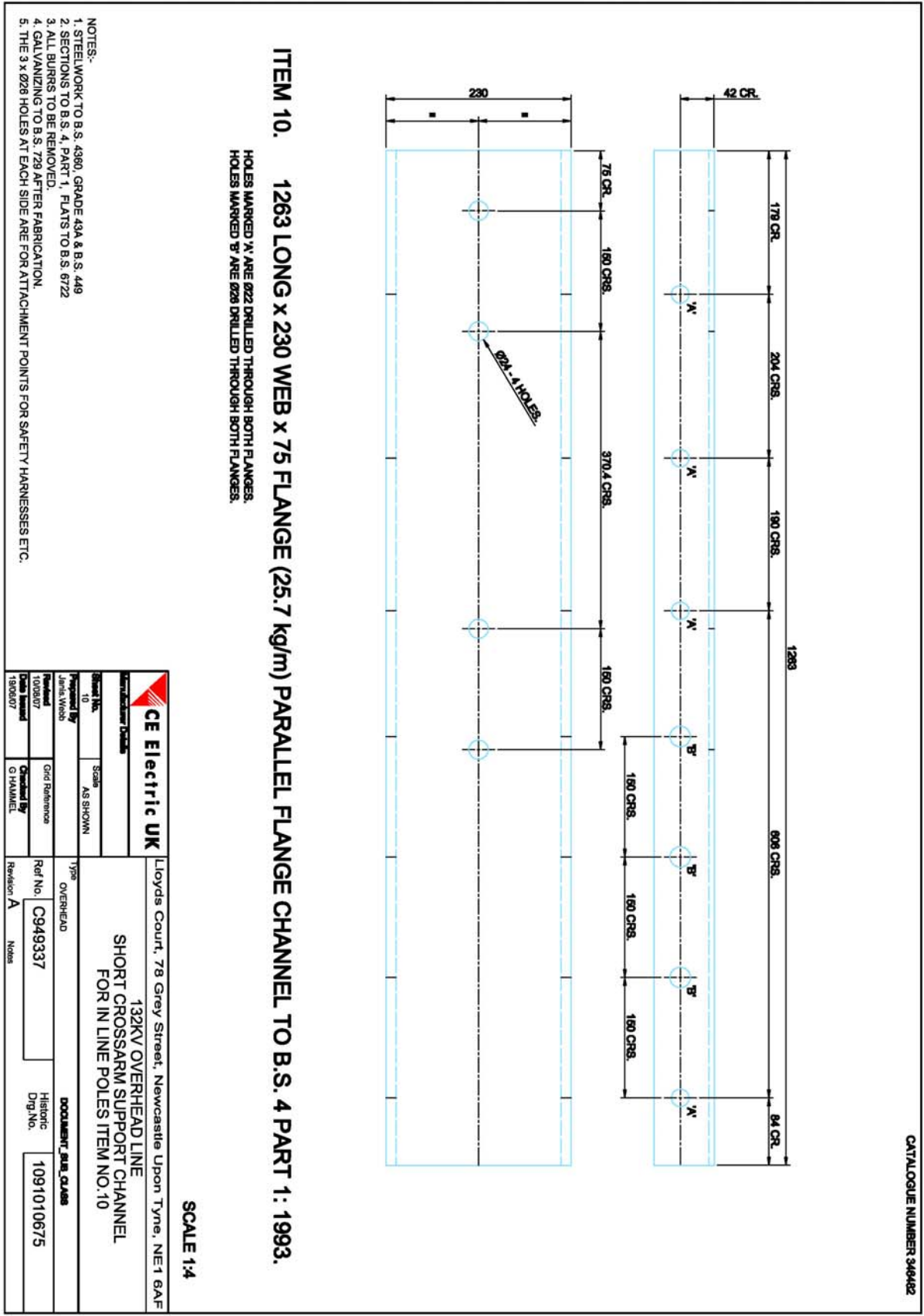
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ITEM NO.	LENGTH 'A'	HEIGHT 'B'	THICKNESS 'C'	'D' CENTRES	MATERIAL
1	175	300	100	150	ANGELIM PEDRA
2	175	250	100	86	ANGELIM PEDRA
3	1300	250	125	86	PINUS SYLVESTRIS
4	2500	250	125	86	PINUS SYLVESTRIS
5	1300	250	125	150	PINUS SYLVESTRIS
6	2500	250	125	150	PINUS SYLVESTRIS
7	240	300	130	150	ANGELIM PEDRA
8	210	300	130	150	ANGELIM PEDRA
9	2500	250	125	150	PINUS SYLVESTRIS
10	1300	250	125	150	PINUS SYLVESTRIS
*	*	*	*	*	*
*	*	*	*	*	*
*	*	*	*	*	*

NOTES

1. SEE MATERIAL COLUMN IN TABLE. A SUITABLE ALTERNATIVE MAY BE USED PROVIDED IT IS OF SUITABLE STRENGTH & DURABILITY.
2. BLOCKS TO BE CUT FROM SAWN TIMBER.
3. TIMBER GRAIN TO BE LONGITUDINALLY ALONG LENGTH 'A'.
4. LONGITUDINALLY CUT SURFACES TO BE STRAIGHT & CLEAN & ANNUAL GROWTH RINGS SHALL BE APPROX. PARALLEL.

Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF

132KV OVERHEAD LINES

ROUTER TYPE POLE

SHEAR BLOCK SCHEDULE

Drawn By: BURGESS SIMPSON

Checked By: G HAMMILL

Rev No: C949240

Doc No: 1091010675

Historic Dwg No: 1091010675

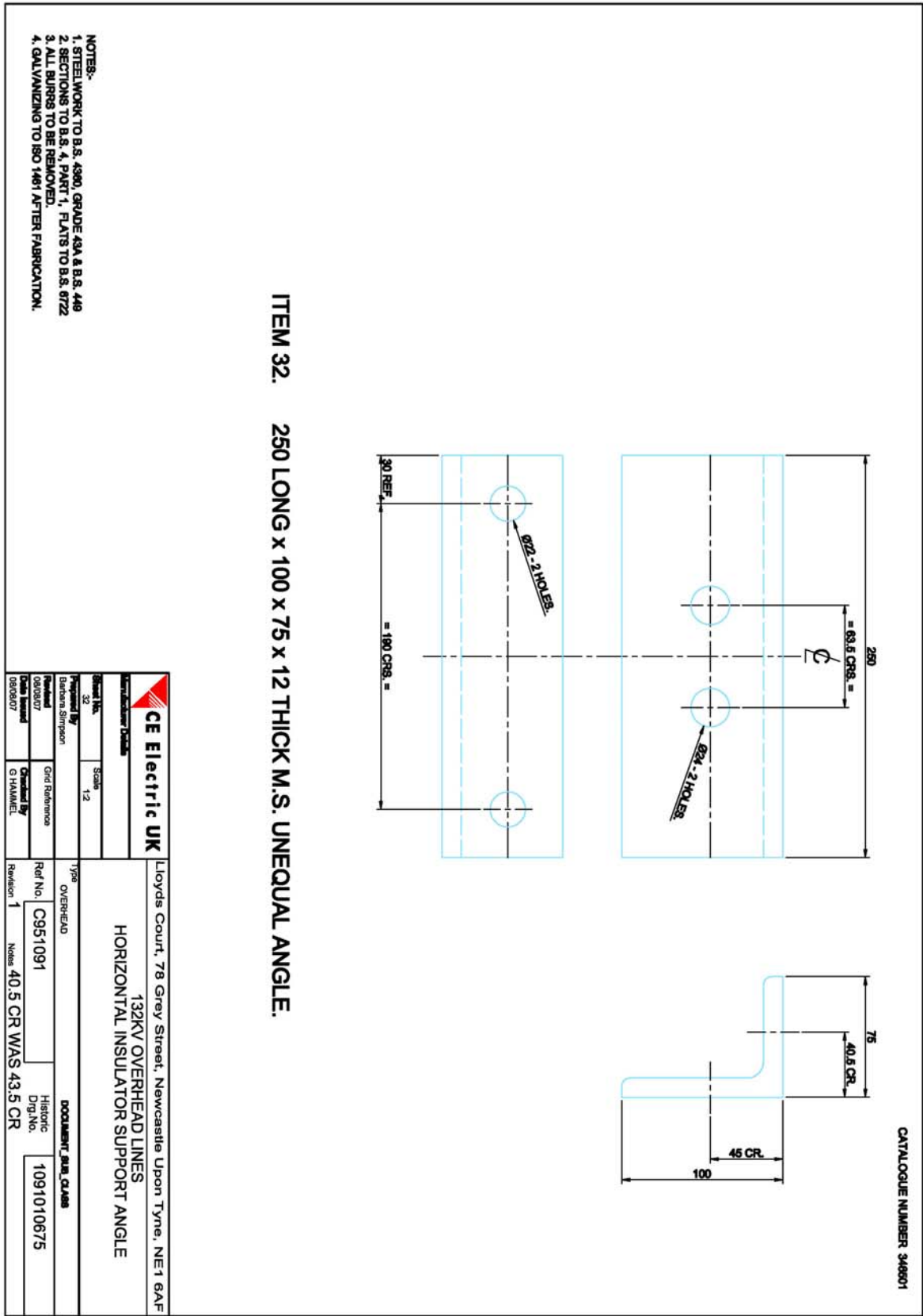
Revision 4

Notes ITEMS 7,8,9 AND 10 ADDED

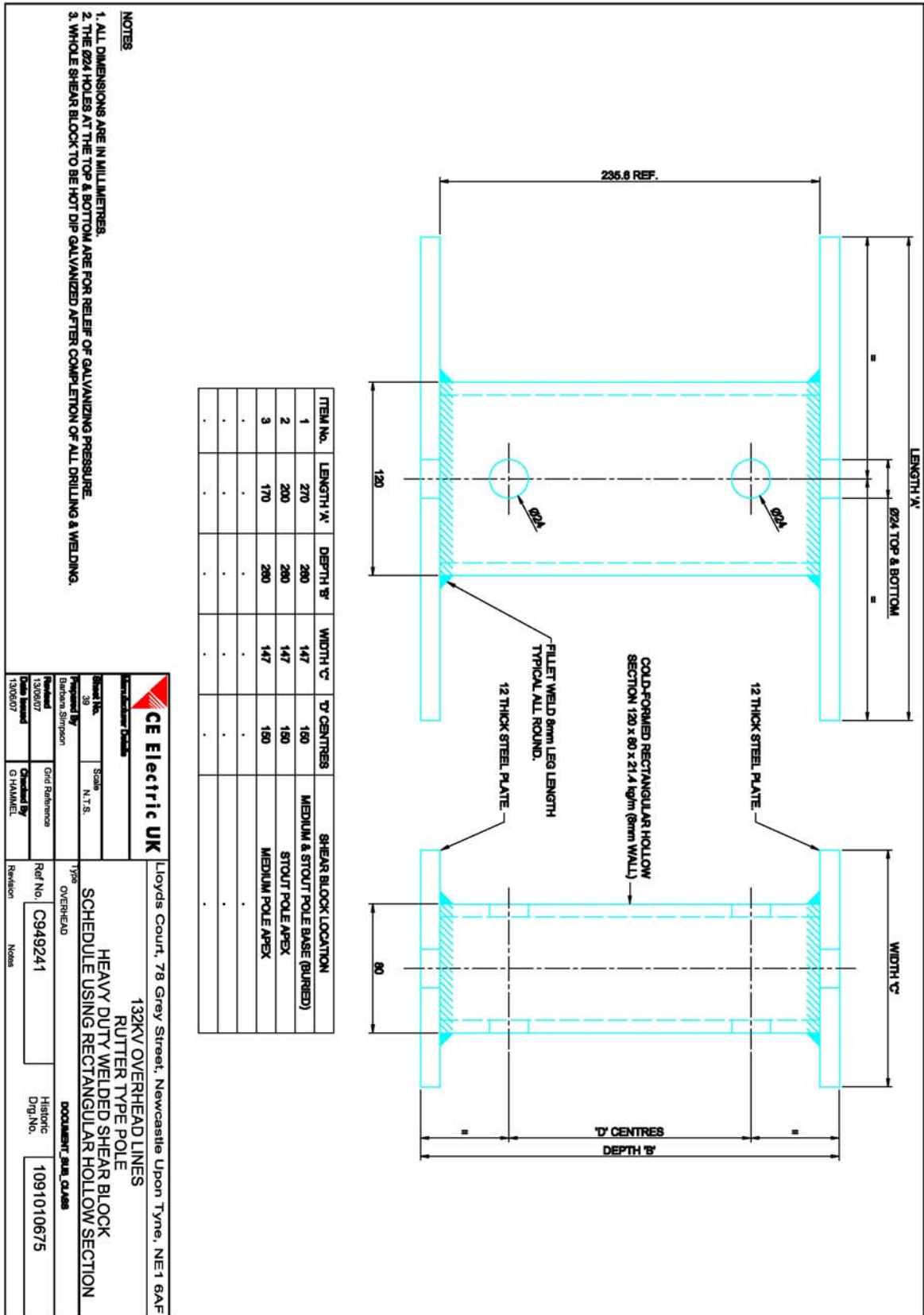


NOTES:-
1. STEELWORK TO B.S. 4360, GRADE 43A & B.S. 449
2. SECTIONS TO B.S. 4, PART 1, FLATS TO B.S. 6722
3. ALL BURRS TO BE REMOVED.
4. GALVANIZING TO ISO 1461 AFTER FABRICATION.

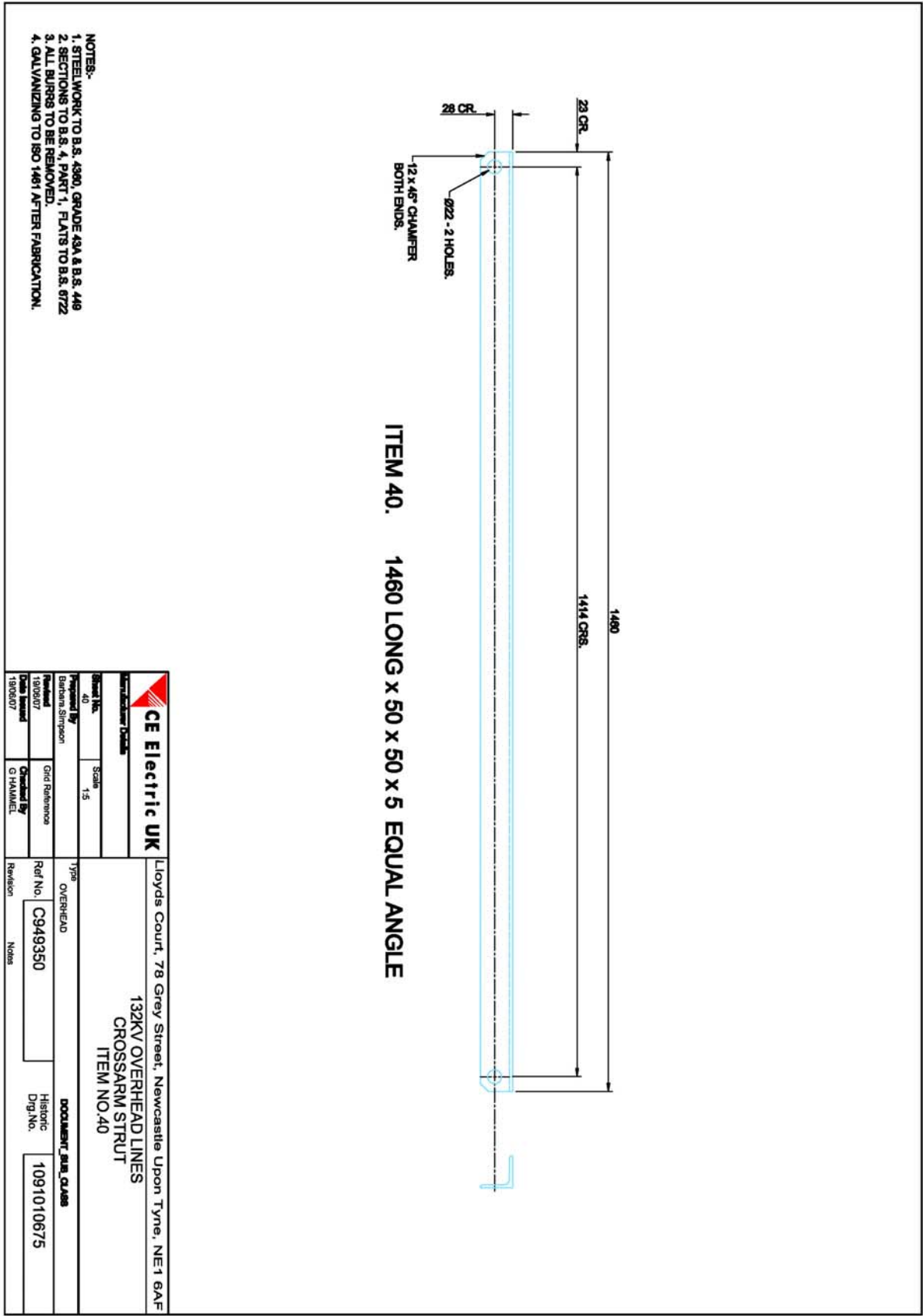
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Version:-	2.0	Date of Issue:-	Sept 2009	Page 105 of 117



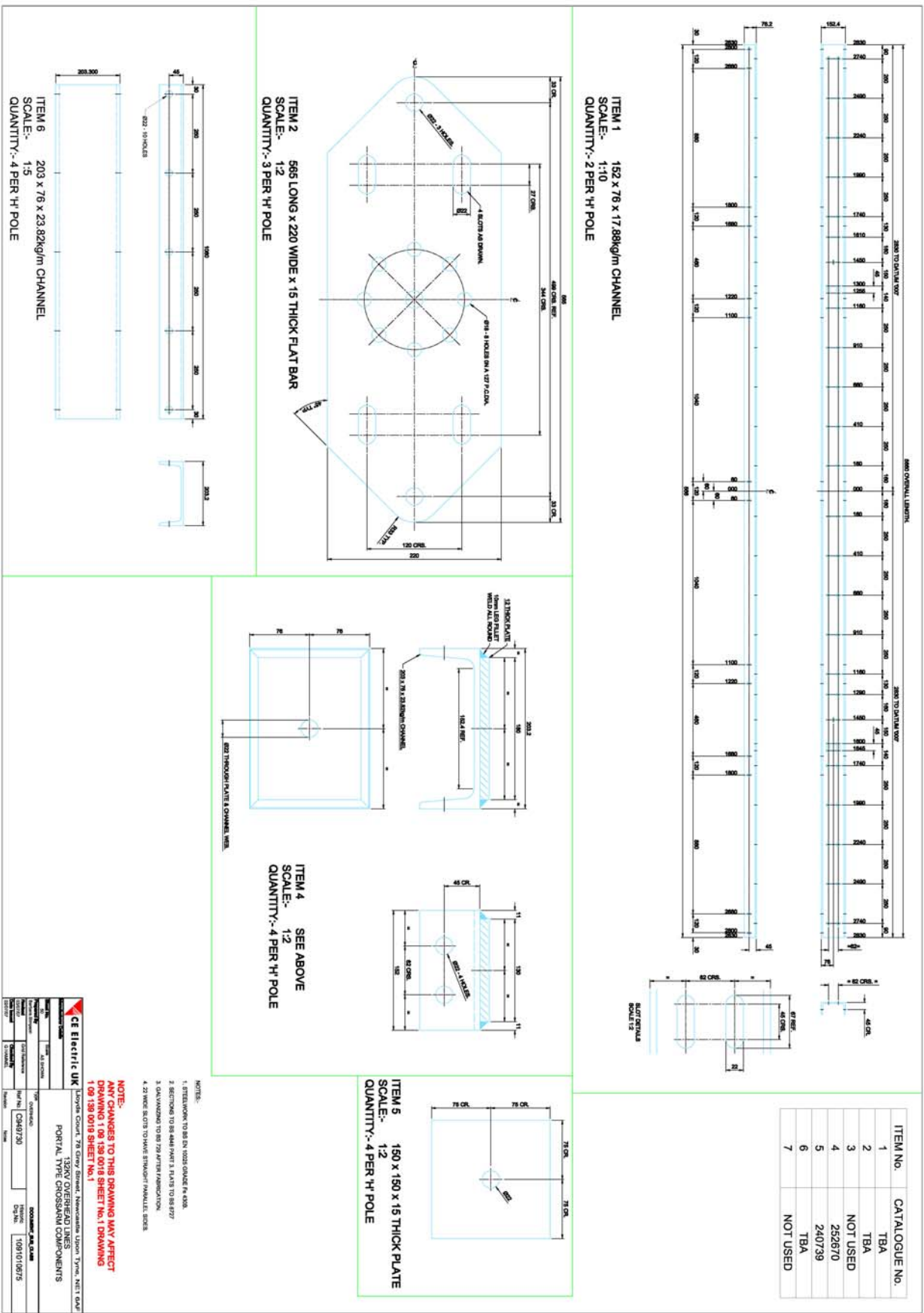
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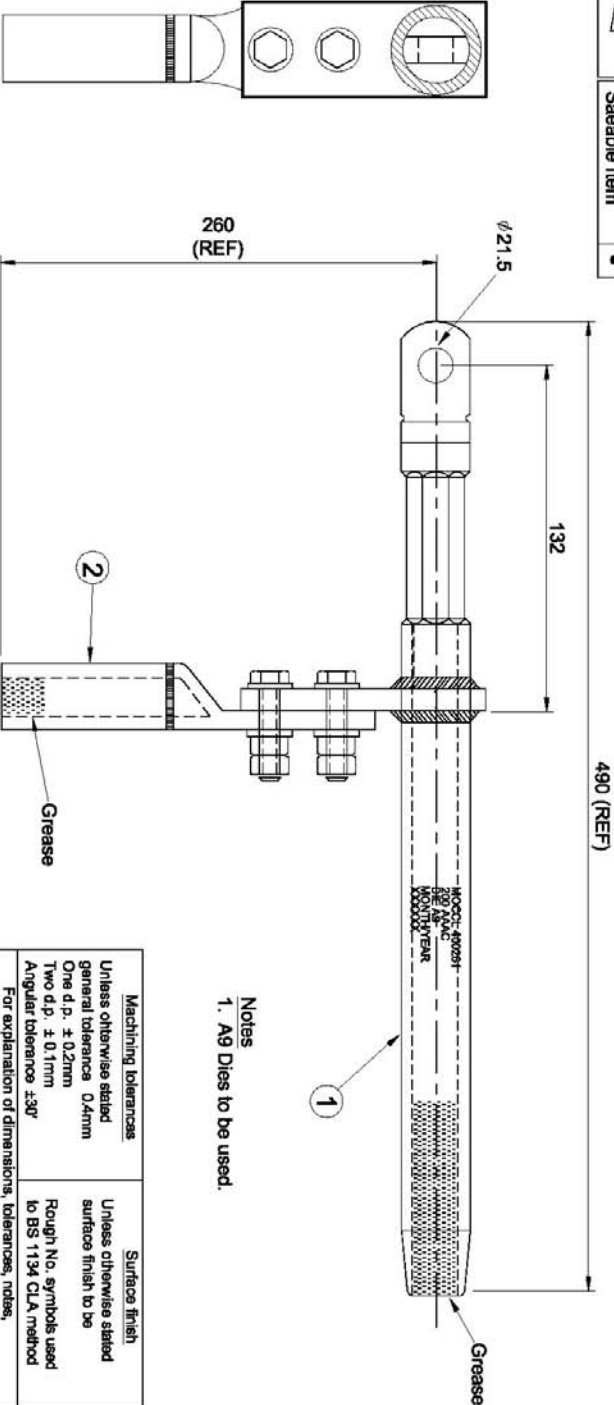


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Projection	Raw Material
	Component
	Seable Item



260 (REF)

132

490 (REF)

21.5

MOORE-400262

200MM AAC COMPRESSION ANCHOR CLAMP

Grease

Notes

1. A9 Dies to be used.

Item	Description	Quantity	Part Number
1	Anchor Clamp	1	400262
2	Straight Jumper Lug c/w Bolt Kit	1	226361

BASED ON HPAA 2.20 (E) 20 POPLAR

CE Electric UK	
Manufacturer Details	400261
MOSDORFER CCL SYSTEMS	
Sheet No.	Scale
3	N.T.S.
Prepared By	Grid Reference
Barbara Simpson	
Revised	Date Issued
26/06/07	18/06/07
Checked By	
G HAMMEL	

Type	Revision	Notes
OVERHEAD	A	

LINES	Historic Dwg.No.
	1091010653

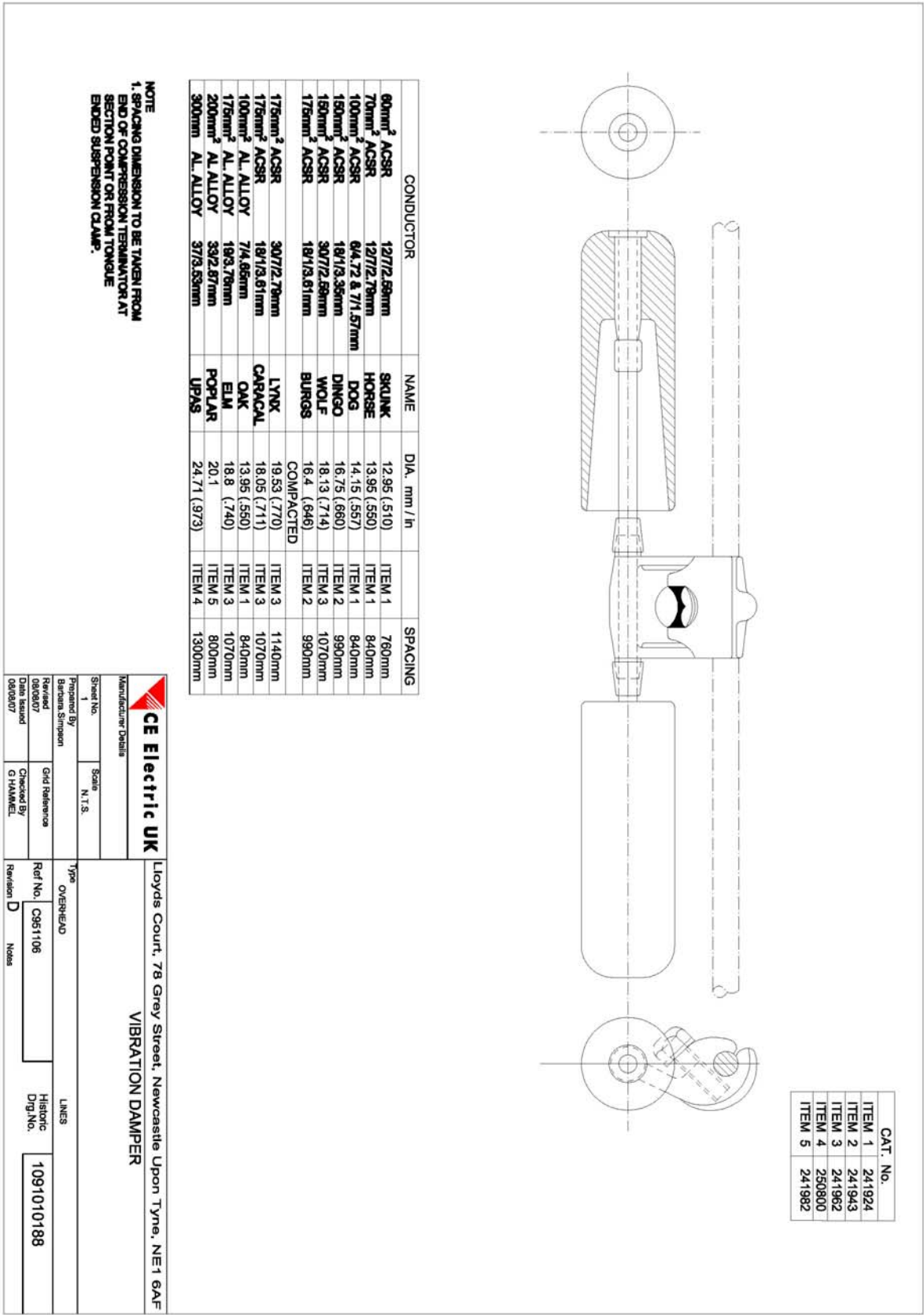
Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF

POPLAR 200MM AAC COMPRESSION ANCHOR CLAMP

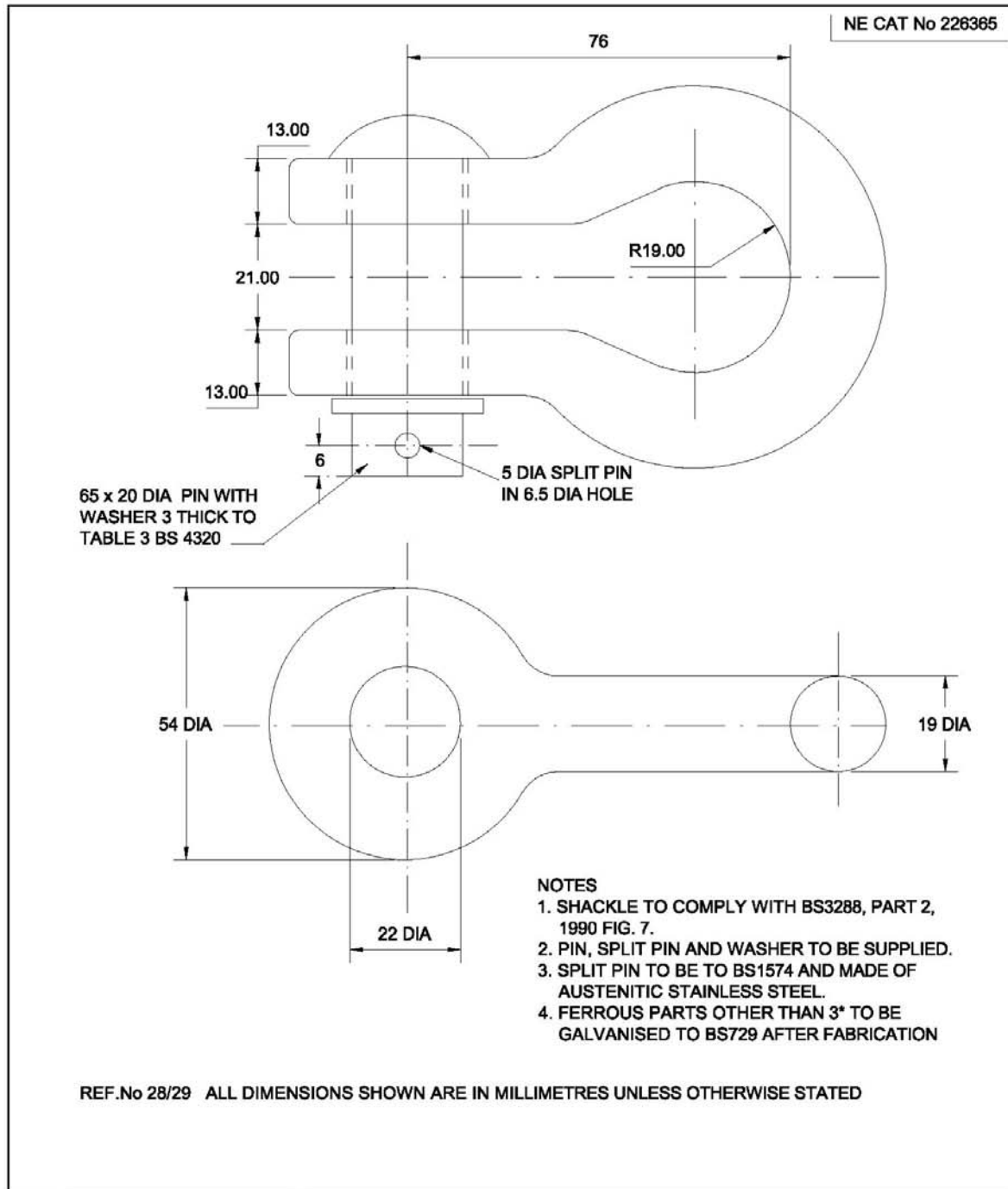
FOR USE WITH A 28/31 SOCKET CLEVIS

Machining tolerances		Surface finish											
Unless otherwise stated general tolerance 0.4mm	One d.p. $\pm 0.2\text{mm}$	Unless otherwise stated surface finish to be	Rough No. symbols used to BS 1134 CLA method										
Two d.p. $\pm 0.1\text{mm}$	Angular tolerance $\pm 30^\circ$												
For explanation of dimensions, tolerances, notes, geometric symbols etc. see BS 308													
Normal	micrometres	50	25	12.5	6.3	3.15	1.6	0.8	0.4	0.2	0.1	0.05	0.025
Value	Ra	micrometres	250	125	63	31.5	16	8	4	2	1	0.5	0.25
Roughness number	N12	N11	N10	N9	N8	N7	N6	N5	N4	N3	N2	N1	

Document reference	NSP/004/045			
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Version:- 2.0					



 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		SHACKLE MINIMUM FAILING LOAD 125KN			
Sheet No. 1	Scale 1:1				
Prepared By Barbara.Simpson		Type OVERHEAD	LINES		
Revised 08/08/07	Grid Reference	Ref No.	C951103	Historic Drg.No.	1091010461
Date Issued 08/08/07	Checked By G HAMMEL	Revision D	Notes		

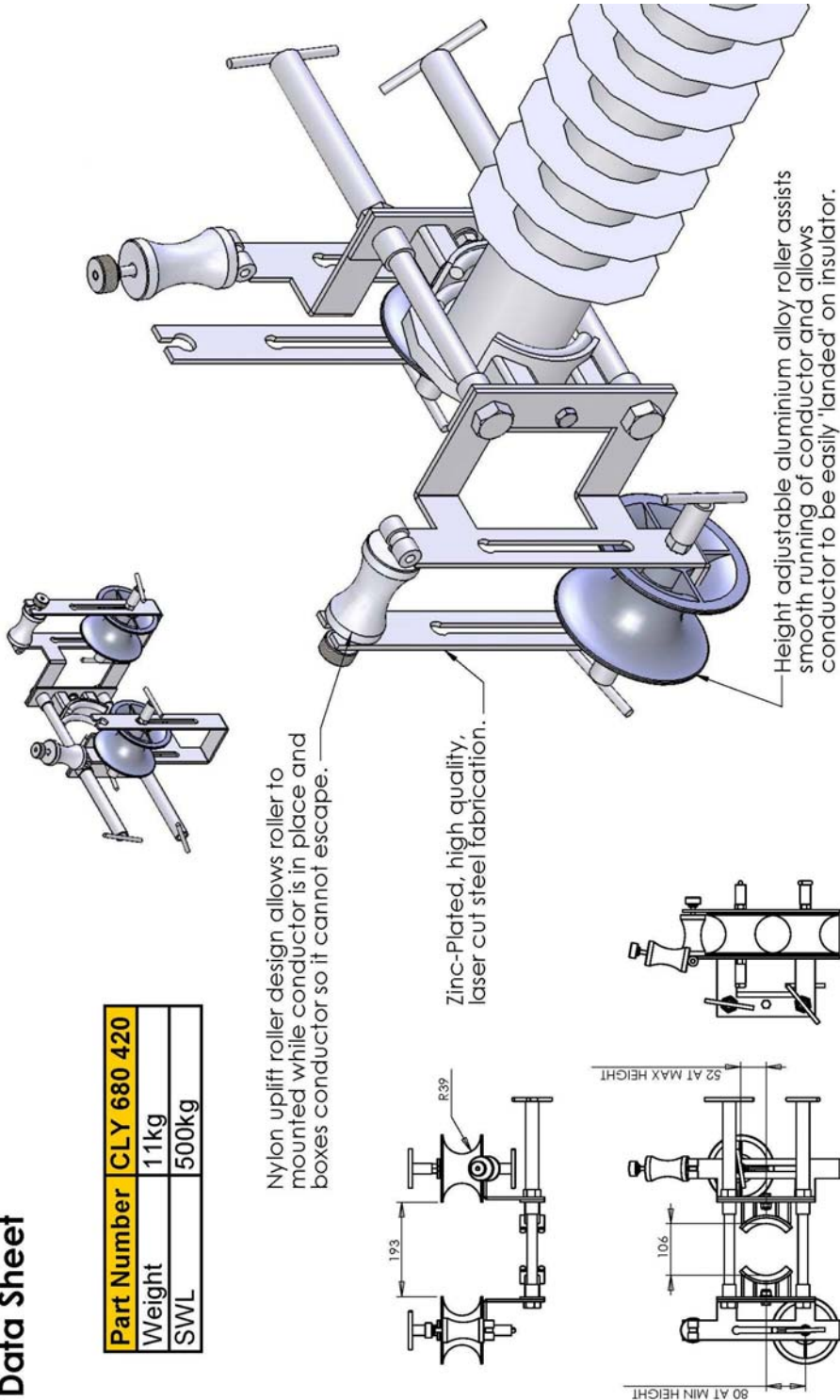
Document reference		NSP/004/045			
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Appendix E – Specialist Pole Top Equipment

E1 Horizontal Insulator Mounted Roller

CLY 680 420: Horizontal Post Insulator Mounted Roller Data Sheet

Part Number	CLY 680 420
Weight	11kg
SWL	500kg



DS CLY 680 420 14/09/07 REV0 CB

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E2 Vertical Insulator Mounted Roller

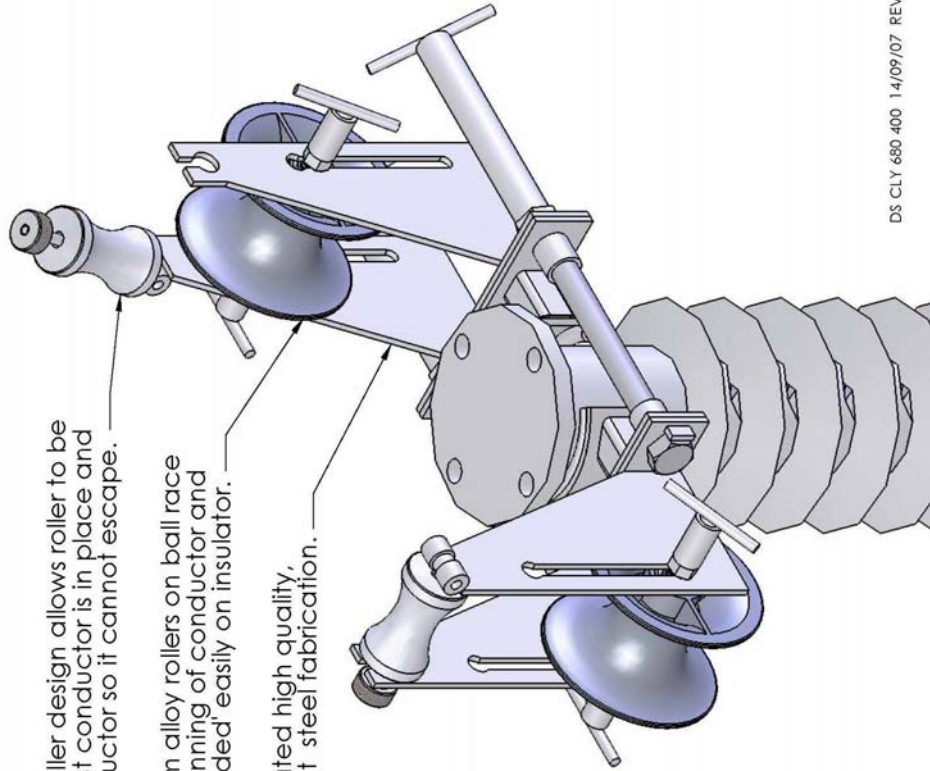
CLY 680 400: Vertical Post Insulator Mounted Roller Data Sheet

Part Number	CLY 680 400
Weight	9.5kg
SWL	500kg

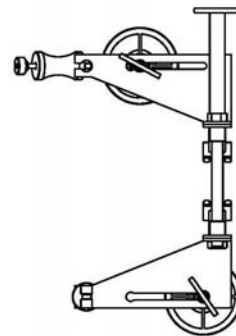
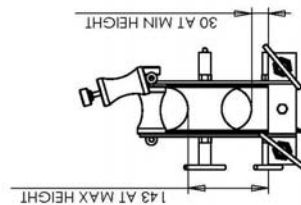
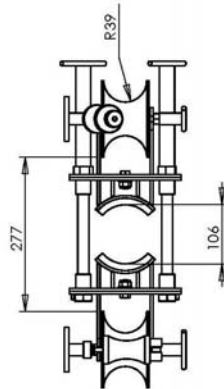
Nylon Uplift Roller design allows roller to be mounted whilst conductor is in place and boxes in conductor so it cannot escape.

Height adjustable aluminium alloy rollers on ball race suspension assists smooth running of conductor and allows conductor to be 'landed' easily on insulator.

Zinc-plated high quality, laser cut steel fabrication.

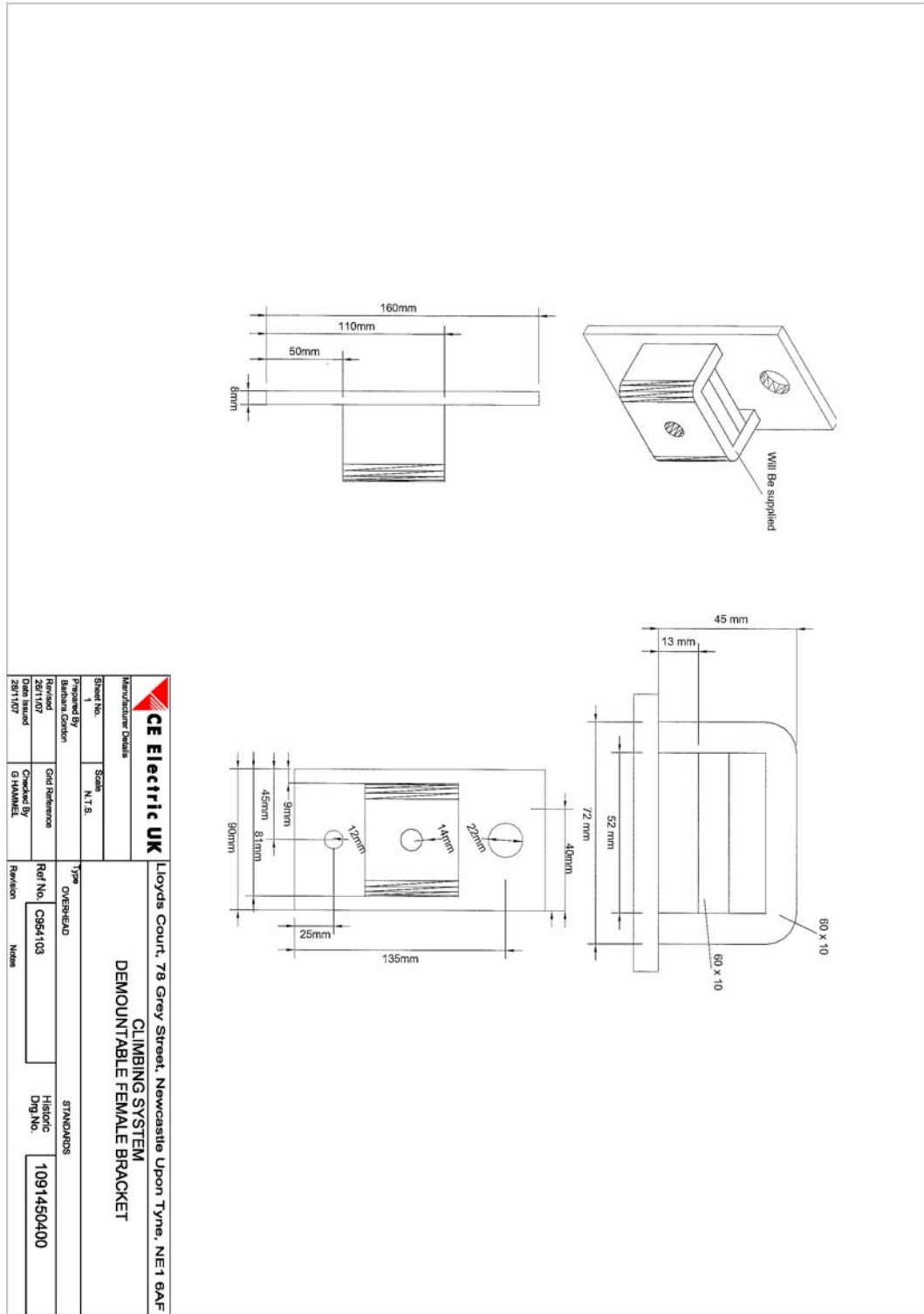


DS CLY 680 400 14/09/07 REV0 CB



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E3 Fixed Bracket for Demountable Ladders



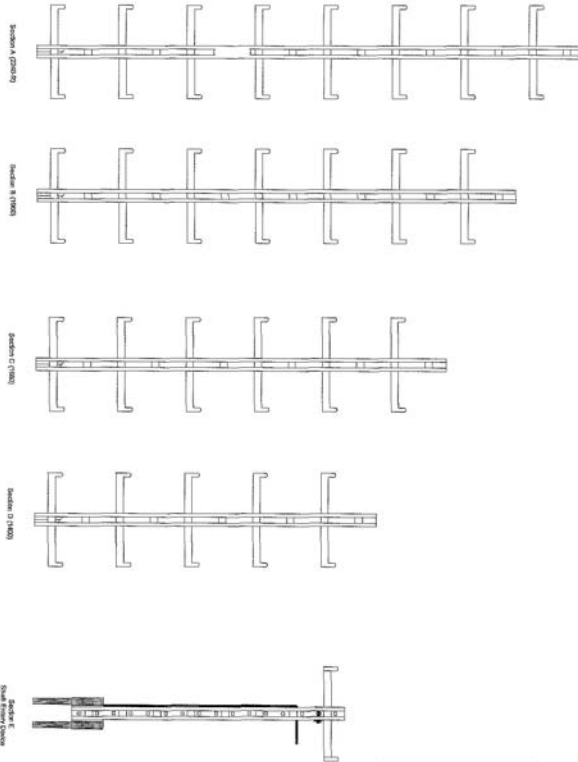
Document reference		NSP/004/045					
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E4 Demountable Ladder System

TYPE	DESCRIPTION	CAT NUMBER	No RINGS	WEIGHT in Kgs
A	Common base ladder section 2240mm in length	313030	9	9.2
B	Intermediate ladder section 1800mm in length	313031	6	6.3
C	Intermediate ladder section 1800mm in length	313032	6	7.4
D	Intermediate ladder section 1400mm in length	313033	5	6.5
E	Small Entry Top section 1120mm in length	313034	1	4.5
F	Permanent Female Mounting Brackets	313035	-	-
G	Galalock Tiesaver	313036	-	-

LADDER SELECTION TABLE

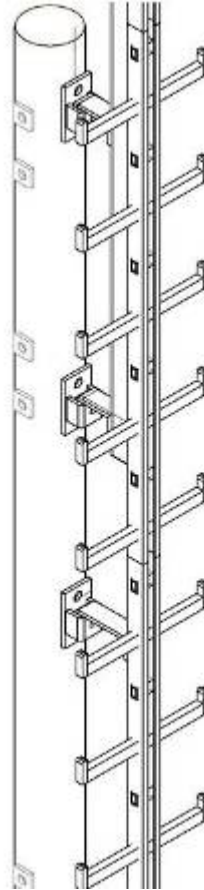
STRUCTURE TYPE	(MULTI POLE)	SINGLE SECTION POLES & SECTION / TERMINAL POLES
12m	(1A) + (1B) + (2E)	(1A) + (1B)
13m	(1A) + (1C) + (1D) + (2E)	(1A) + (1B) + (1C) + (1D)
14m	(1A) + (2C) + (2E)	(1A) + (1C) + (2B)
15m	(1A) + (2C) + (2E)	(1A) + (1B)
16m	(1A) + (2C) + (2E)	(1A) + (2C) + (2D)
17m	(1A) + (1B) + (2E)	(1A) + (2C) + (2D)



 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AA	
Manufacturing Details			
Sheet No.	Scale		
	N.T.S.		
Prepared By	Type		
Barbara Gordon	OVERHEAD		
Reviewed	Grid Reference	STANDARDS	
27/11/07		Ref No.	Hydroc Dig No.
Date Issued	Checked By	C954131	1091450402
27/11/07	G HAMMILL	Revision	Notes

Document reference		NSP/004/045				
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E4.1 Demountable Ladder System



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E5 Traveller Attachment Device

